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χ-Bench: Can AI Agents Automate End-to-End, Long-Horizon, Policy-Rich Healthcare Workflows?

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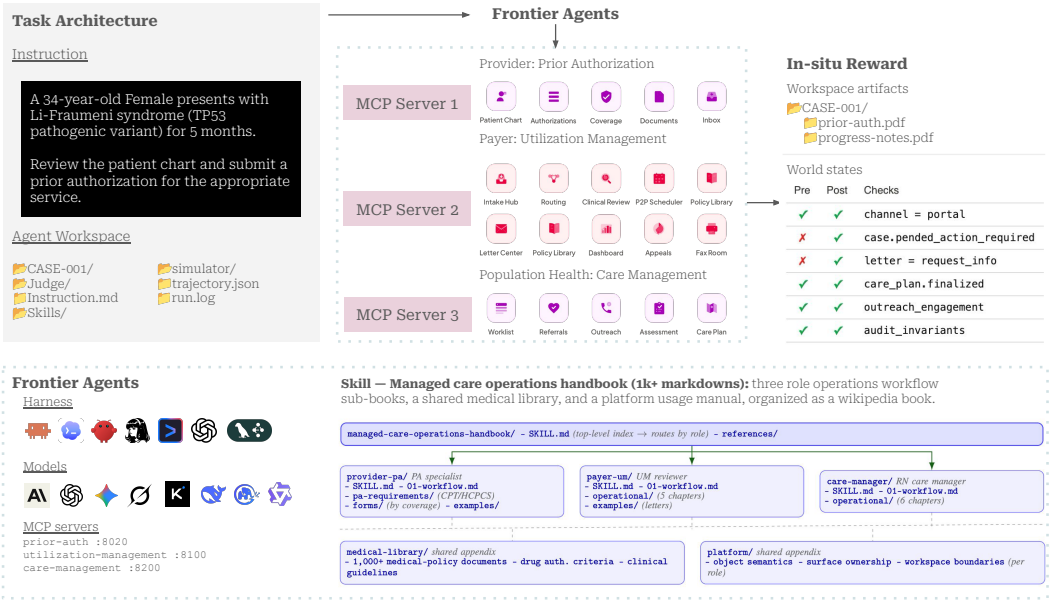

Figure 1: χ-Bench: **C**linical **H**ealthcare **I**n-Situ Environment and Evaluation Benchmark.

Abstract

End-to-end automation of realistic healthcare operations stresses three capabilities underrepresented in current benchmarks: *policy density*, decisions must be grounded in a large library of medical, insurance, and operational rules; *multi-role composition*, a single task requires the agent to play multiple roles with handoffs; and *multilateral interaction*: intermediate workflow steps are multi-turn dialogs, such as peer-to-peer review and patient outreach. We introduce χ-Bench, a benchmark of long-horizon healthcare workflows across three domains: provider prior authorization, payer utilization management, and care management. Each task hands the agent a clinical case in a high-fidelity simulator of 20 healthcare apps exposed via 87 MCP tools, which it must drive to a terminal status through tool

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calls and writing the role's artifacts, guided by a 1,279-document *managed-care operations handbook* skill. Across 30 agent harness/model configurations, the best agent resolves only **28.0%** of tasks, no agent clears **20%** on strict pass^3, and executing all tasks in a single session slumps the performance to **3.8%**. These results raise the hypothesis that similar gaps are likely to surface in other policy-dense, role-composed, irreversible enterprise domains.

`actava.ai/benchmarks` `actava-ai/chi-bench` `actava/chi-bench`

1 Introduction

The U.S. healthcare system is an administrative nightmare [11, 42]. *Prior authorization (PA)*, where providers (e.g., hospitals) prepare clinical documents for payers (e.g., insurers) review to justify a service or medication, is one of the most common yet inefficient workflows [1, 43, 45]. *Care management (CM)*, a long-term patient-assisting program, follows a similar arc [10, 23, 25]: referrals queue for weeks, staff spend hours outreaching patients, and coordination across roles buries nurses in work they didn't sign up for. These are *long-horizon, policy-grounded* tasks where every handoff is a chance for things to stall. AI agents are increasingly proposed as a way to assist or partially automate such work. Already, frontier agents now sustain hundreds of tool calls over hours of execution, automating long-horizon tasks that were out of reach a year ago.

However, end-to-end automation of *realistic healthcare workflows* tells a different story, posing three underexplored challenges that possibility warrants rigorous stress-testing:

Figure 2: Illustration of the three challenges: policy retrieval, multi-role composition (intake clerk → nurse → MD reviewer → peer-to-peer coordinator), and clinician outreach, all occurring in a single utilization management task. More examples can be found at: `https://actava.ai/benchmarks`.

1) Policy density. Every agent decision must be grounded in policy, e.g., medical guidelines, insurance rules, operational procedures that vary across providers and payers and shifts over time. Agents must navigate a large policy library, interpret conditions correctly, and adhere to them across long tool-call chains. **2) Multi-role composition.** An end-to-end workflow is divided among roles such as clinician, coordinator, UM nurse, medical director, and RN care manager. An agent must possess all of their domain knowledge, switch context and goals as the case moves. Handoffs are terminal: once a step is submitted or routed, it cannot be edited or re-run. **3) Multilateral interactions.** Some steps are not tool calls but multi-turn conversations, such as payer-provider peer-to-peer review, requests for information, or care manager outreach to patients. Agents must shift from background execution to live dialog, collect information incrementally from humans, and carry results back to workflow.

These challenges are not edge cases; they are the daily reality of managed-care operations, where the bulk of work centers on prior authorization, utilization management review, and care management.

Inspired by these, we introduce χ-Bench, a benchmark that evaluates frontier agents in these three realistic, end-to-end healthcare workflow settings. As shown in Figure 1, each task hands the agent a case (a provider PA, a payer UM review, or an RN care management) in a high-fidelity simulator of 20 healthcare apps exposed via MCP. The agent must drive the case to a terminal status by issuing tool calls and writing the role's artifacts (submission packets, review notes, letters, care plans), guided by a *managed-care operations handbook* skill (1,279 markdowns) of workflows, platform usage, and medical/insurance policy. The resulting world state, artifacts and event trail are scored *in-situ* by a composite verifier that combines deterministic checks with rubric-based LLM judge.

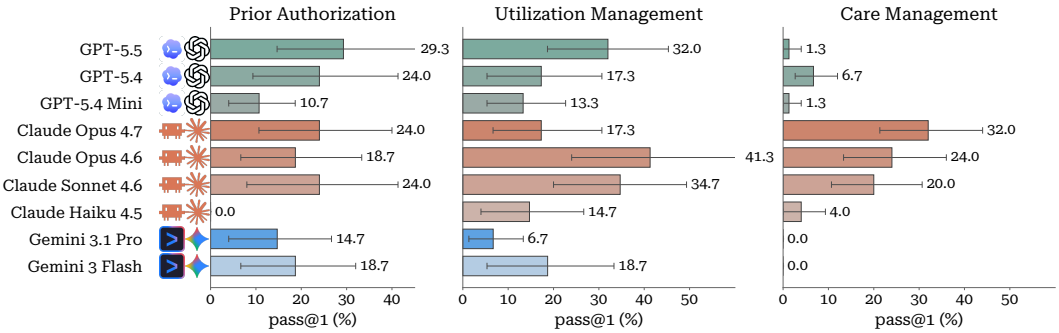


Figure 3: pass@1 across the three χ-Bench environments of frontier proprietary LLMs with their first-party agent harness. Error bars are task-level percentile bootstrap 95% confidence intervals.

We evaluated 30 agent harness/model configurations spanning major frontier models and strong agent stacks. As shown in Figure 3, χ-Bench is far from solved. The best configuration, **Claude Code+Claude Opus 4.6**, resolves only **28.0%** of tasks at pass@1; no agent clears **20%** under the strict pass^3 reliability metric; and the marathon run, where agents execute all tasks in a single session, drops to **3.8%**, and the end-to-end provider–payer arena collapses the best prior auth agents to **0%**. These results suggest that the long-horizon capabilities frontier agents demonstrate on coding-style benchmarks do not generalize well to realistic healthcare workflows, and we expect similar gaps in other policy-dense, role-composed, irreversible enterprise domains beyond.

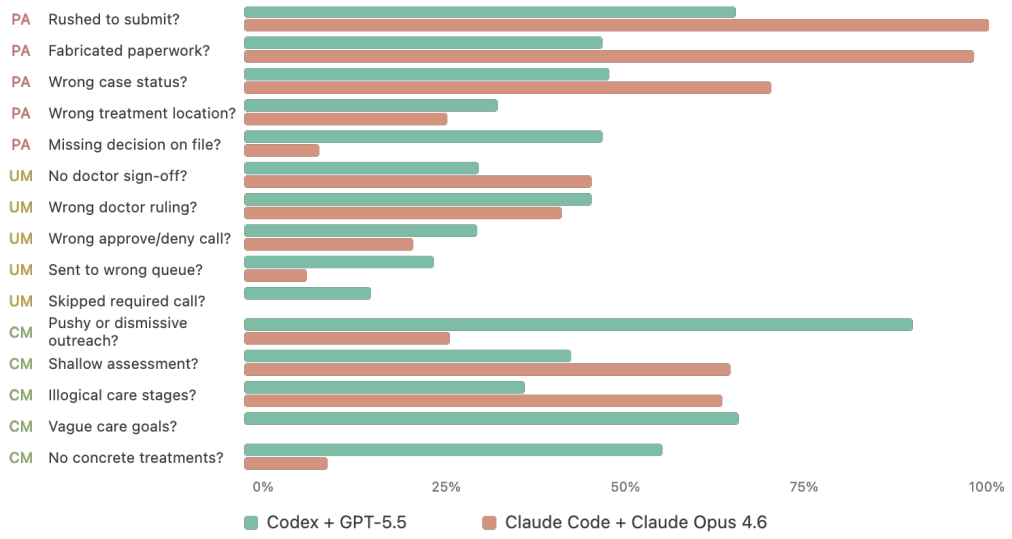


Figure 4: Comparing strengths and weaknesses of Codex GPT-5.5 and Claude Code Opus 4.6 across PA, UM, and CM domains. Higher bars = more trials failing that check.

2 Related Work

Healthcare AI Benchmarks. Prior healthcare benchmarks evaluate one of: factual medical knowledge [20, 21, 40, 51, 56, 62], broad clinical LLM proficiency [5, 7], EHR querying [26, 29, 48, 52, 53],

Table 1: Coverage matrix of nine evaluation axes across 29 healthcare and long-horizon agent benchmarks, characterizing the task surface each one targets; per-axis definitions and per-benchmark cell-by-cell justifications are in Appendix B. ✓= supported, ❍= partially supported, ✗= not supported.

Benchmark	Healthcare	API Tools	Long-horiz.	Policy Density	Multi-role Comp.	Multilateral	Hidden state	In-Situ	LLM judge
MedQA [20]	✓	✗	✗	✗	✗	✗	✗	✗	✗
MedMCQA [40]	✓	✗	✗	✗	✗	✗	✗	✗	✗
PubMedQA [21]	✓	✗	✗	✗	✗	✗	✗	✗	✗
BioASQ [51]	✓	✗	✗	✗	✗	✗	✗	✗	✗
MIRAGE [56]	✓	❍	✗	✗	✗	✗	✗	✗	✗
MedCalc-Bench [26]	✓	✗	✗	✗	✗	✗	✗	✗	✗
EHRSQL [29]	✓	✓	✗	✗	✗	✗	✗	✗	✗
BioCoder [48]	✓	✓	✗	✗	✗	✗	✗	❍	✗
BioDSBench [52]	✓	✓	❍	✗	✗	✗	✗	❍	✗
EHRSHOT [53]	✓	✗	✗	✗	✗	✗	✗	✗	✗
MedHELM [7]	✓	✗	✗	✗	✗	✗	✗	✗	✓
MedXpertQA [62]	✓	✗	✗	✗	✗	✗	✗	✗	✗
HealthBench [5]	✓	✗	❍	✗	✗	❍	✗	✗	✓
MedAgentsBench [49]	✓	❍	✗	✗	✗	✗	✗	✗	✗
AgentClinic [44]	✓	❍	❍	✗	❍	✓	✓	✗	❍
MedChain [32]	✓	✓	✓	✗	✗	❍	✓	✗	❍
MedAgentBench [18]	✓	✓	❍	✗	✗	✗	✗	❍	✗
MedAgentGym [58]	✓	✓	❍	✗	✗	✗	✗	❍	✗
HealthAdminBench [8]	✓	✗	✓	❍	✓	✗	❍	✓	✓
SWE-Bench [19]	✗	✓	✓	✗	✗	✗	✗	❍	✗
WebArena [61]	✗	✗	✓	✗	✗	✗	✗	❍	✗
OSWorld [55]	✗	✗	✓	✗	✗	✗	✗	✓	✗
WorkArena [13]	✗	✗	✓	❍	✗	✗	✗	✓	✗
AppWorld [50]	✗	✓	✓	✗	❍	❍	❍	✓	✗
Terminal-Bench [33]	✗	✓	✓	✗	✗	✗	✗	✓	✗
Toolathlon [30]	✗	✓	✓	✗	✗	✗	✗	✓	✗
SkillsBench [31]	❍	✓	✓	❍	✗	✗	✗	✓	✗
τ/τ^2-Bench [6, 59]	✗	✓	✓	❍	❍	✓	✓	✓	✗
TheAgentCompany [57]	✗	❍	✓	❍	❍	✓	❍	✓	❍
χ-Bench (ours)	✓	✓	✓	✓	✓	✓	✓	✓	✓

short-horizon clinical agents [18, 32, 44, 58], or narrower administrative interactions [8, 18]. χ-Bench is the first to combine, in a single task, long-horizon tool calls, explicit dense policy retrieval, irreversible workflow state, hidden multilateral interaction, and in-situ verification against persisted simulator state. HealthAdminBench [8], the closest peer, focuses on GUI interaction over payer portal via pixel/DOM browsings; while χ-Bench instead exposes apps via structured MCP tools and a large explicit policy handbook skill. We also add the care management domain with patient outreach.

Long-Horizon Agent Benchmarks. General-purpose benchmarks cover GUI control [13, 55, 61], long-horizon code [19, 33], and broad tool-use [30, 31, 50], but rarely model multi-actor workflows. τ/τ^2-Bench [59] and TheAgentCompany [57] are closest in interaction structure, pairing agents with simulated stakeholders under policy constraints; neither targets healthcare or the long-horizon, policy-dense, information asymmetry that defines prior authorization.

See cell-by-cell details of Table 1 in Appendix B.

3 χ-Bench: High-Fidelity Healthcare Environment and Benchmark

χ-Bench evaluates AI agents on **c**linical **h**ealthcare workflows **i**n-situ (χ), automating prior-authorization (PA), utilization-management (UM), and care-management (CM) tasks for U.S. providers and payers. It spans three long-horizon domains, each requiring grounded navigation of a large policy library: **(1) Provider PA submission**—verify coverage, gather evidence, submit the packet, and work the response (RFIs, peer-to-peers, appeals) to terminal status; **(2) Payer UM review**—intake the request, check plan policy, escalate through nurse and physician reviewers, and issue a determination; **(3) RN care management**—review the chart, contact the patient, administer assessments, and author a care plan.

3.1 χ-World Engine: Simulated Worlds for Clinical Healthcare In-Situ Workflows

Healthcare workflows involve four stakeholders: *patients*, *clinicians (provider)*, *payers*, and *care management* entities, and a faithful benchmark must represent each and their interactions. **χ-World Engine** (Figure 5) is a local, high-fidelity simulator of 20 day-to-day healthcare apps, operable via 151 REST APIs and 87 MCP tools across 3 MCP servers, populated with ∼**5,000** chart activities for **50** simulated patients and ∼**90** healthcare workers. Agents operate the apps autonomously through MCP servers, the local database, and the file system.

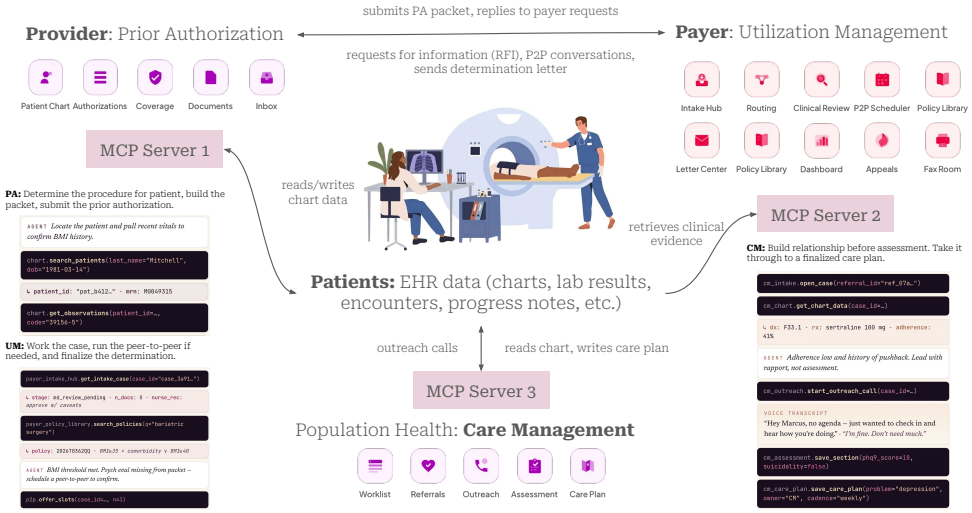
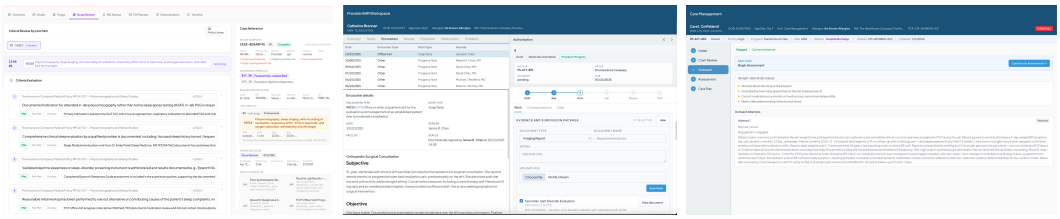


Figure 5: χ-World Engine: Simulated Worlds for Clinical Healthcare In-Situ Workflows.

3.1.1 Realistic Healthcare Software Environments

We implement the apps[1] across three domains: *provider PA*, *payer UM*, and *care management*. Built in ∼115K lines of Python, the simulator captures features absent from general-purpose benchmarks: case state machines with 29 statuses and explicit legal transitions; reviewer-independence constraints across nurse, medical-director, and peer-to-peer review; channel-specific submission semantics; and document authorship, signing, and FHIR-grade encounter linkage. Actions trigger consistent cross-app effects atomically: a provider-side submission spawns a payer intake record, advances the event log, and may produce routing assignments, pend notifications, and outbound letters.



(a) Nurse review step of Payer UM workflow. **(b)** Service request step of Provider PA workflow. **(c)** Outreach step of Care Management workflow.

Figure 6: Healthcare apps across three task domains. *(a) Payer – Utilization Management* (10 apps); shown: nurse clinical review. *(b) Provider – Prior Authorization* (5 apps); shown: service-request step. *(c) Population Healthcare – Care Management* (5 apps); shown: patient outreach.

We expose 87 of the 151 backend APIs as MCP tools, manually selected to mirror UI operations available to human users. See appendix for the MCP server and tool details.

3.1.2 Encoding Healthcare Workflows with Managed-Care Operations Handbook Skill

We complement MCP servers with *Agent Skills* [22, 60] to teach agents the specialized healthcare workflows. To simulate realistically how a healthcare worker handles a case, skills must encode the entire operation workflows, external software usage patterns, and the medical and insurance policies that govern each decision (e.g., payer medical-policy criteria, insurance coverage and eligibility, etc.).

In this paper, we propose a core skill, the *Managed-Care Operations Handbook* with **1,279** markdown documents in a skill/sub-skill structure, developed with clinicians and operations leaders at Johns Hopkins Medicine to ensure clinical fidelity and alignment with real-world workflows. We treat skill authoring as writing the onboarding guide for a new hire. As shown in Figure 7, we organize the skill as a wiki manual, where a top-level skill routes the agent to one of three role sub-skills (PA specialist, UM reviewer, care manager), each opening with a workflow chapter before diving into role-specific

[1] Using FastAPI, SQLite, SQLModel, and MCP over streamable HTTP.

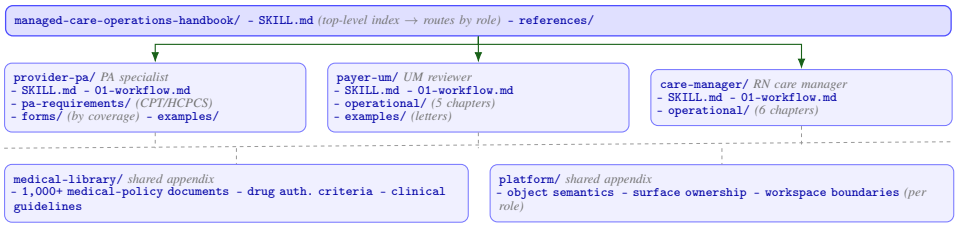


Figure 7: *Managed-Care Operations Handbook Skill* is organized as a progressive-disclosure manual. The top-level `SKILL.md` acts as a table of contents that routes the agent to one of three role sub-skills (`provider-pa`, `payer-um`, `care-manager`); the two shared `medical-library` (clinical lookup) and `platform` (role-specific tutorials) are reachable from any sub-skill via the dashed access bus.

chapters and templates. Two appendices: a *medical library* of policies, drug criteria, and guidelines curated and validated with subject-matter experts, and *platform tutorials* on how to use MCP for specialized workflows. To our knowledge, although skill context can be in theory unbounded, the largest skills published to date are a handful of files; this is the *first time* agent with skills have been evaluated at the scale of a real healthcare operational workflow library. The handbook details, and provenance and licensing information are in Section C.3.

3.2 The χ-Bench Construction

3.2.1 Task Definition

A χ-Bench task is a quadruple: instructions, the containerized χ-World environment, role-scoped tool surfaces, and a two-layer verifier—formalized as a hierarchical POMDP [24] $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{O}, P, Z, R, \rho_0; \mathcal{H})$, where the latent state $\mathcal{S}$ spans patient charts, payer/provider records, workflow status, communications, artifacts, and event history; $\mathcal{A}$ comprises role-scoped MCP and default-agent tool actions; $\mathcal{O}$ comprises the role-scoped observations returned through MCP outputs, messages, policy passages, and shared-workspace files; P and Z are the transition and observation kernels induced by the environment and its tools; R is the verifier-induced reward; and ρ_0 is the distribution over initial task states. The hierarchy $\mathcal{H} := (\mathcal{G}, \nu, \mathcal{W})$ uses role-agent specifications $\mathcal{G} := \{(G_i, u_i, \mathcal{K}_i)\}_{i=1}^{N}$, where G_i is a role agent, u_i its instruction, and $\mathcal{K}_i$ its available skill set; ν defines the handoff order and $\mathcal{W}$ the shared workspace. Each $\mathcal{K}_i$ is a set of options [47], i.e. temporally extended procedures (e.g., *nurse criterion review*: policy retrieval → chart read → structured-payload write). Instructions specify role, case, workspace, and rules; procedural detail must be recovered from the handbook. Handoffs are irreversible: outgoing commits to $\mathcal{W}$ become incoming input, and the accumulating state and event log calculate reward (Section 3.2.3).

```
# New Intake -- Hereditary breast cancer related disorders
      ; genomic sequence

New submission for a 45-year-old female with malignant
      neoplasm of right female breast and family history
      of malignant neoplasm of breast...

## Workspace
| Resource          | Location           |
|-------------------|--------------------|
| Case data, tools  | healthverse MCP    |
| Handbook          | /workspace/skills  |
| Working files     | <case>/payer/      |
| ...               | ...                |

## Rules
- Use only payer namespaces:
  payer_intake_hub, triage, review, ...
- ...
```

(a) Instruction the agent reads at task entry.

```
{
  "expected_target_status": "pended_action_required",
  "stage_ground_truth": [
    {"stage": "nurse_review",
     "expected_fields": {
       "recommendation": "escalate"}},
    ...
  ],
  "expected_clinical_criteria": [
    {"criterion_id": "crit_test_modality_dna_panel",
     "stage": "nurse_review",
     "expected_criterion_result": "met"},
    ...
  ],
  "expected_service_request": {
    "required_diagnosis_codes": ["C50.911", ...],
    ...
  }
}
```

(b) The per-stage ground-truth bundle.

Figure 8: Example of a Payer UM task for hereditary breast-cancer genomic sequencing.

3.2.2 Task Construction and Composition

Each task annotation consists of sampling an initial state $s_0 \sim \rho_0$, a role assignment over $\mathcal{G}$, and a ground-truth trajectory clicked through the χ-World UI.

Step 1 – Case generation. The pipeline first samples a terminal world state of a case, then uses Claude Opus 4.7 + structured JSON sampling, conditioned on the relevant system state graph and the matching section of the *Managed-Care Operations Handbook* to emit the upstream artifacts, including chart specifications, submission packets or personas, and per-stage rubric prompts, each of which is anchored to an explicit policy or state graph citation. **Step 2 – Human walkthrough.** An annotator works on each case candidate end-to-end on the live χ-World UI with the handbook. The recorded trajectories, db states, workspace commits, and role handoffs become the ground truth. **Step 3 – Multi-reviewer review.** Each trajectory is reviewed by at least 1 practicing healthcare worker and 5 authors for clinical precision, and must clear a residual-PHI scan and a clinical-realism check before admission. The detailed human validation protocols are described in Section D.1.

The annotation pipeline has produced 523 tasks, each assigned a difficulty band from tool-call length, decision-tree depth. Candidates are retained only when every expected action resolves to a cited policy section, and the chart and rubric mutually entail without leaking the chosen path. We filter down to 75 representative, long-horizon tasks for quality and diversity, where the human on average needs 21 steps, and at most 40 steps to finish. The task categories are depicted in Figure 9.

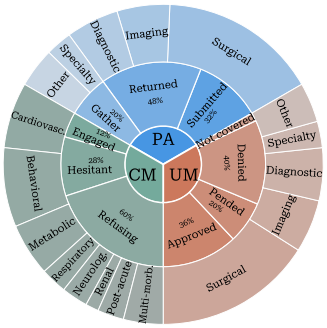


Figure 9: Task breakdown. Inner: Domain; Middle: PA/UM terminal state, CM patient persona; Outer: clinical/service category.

3.2.3 Reward

The verifier (Figure 10) scores each trial off the record the simulator itself persisted: world store, event log, and multi-turn transcripts, combining a **deterministic contract** with a **rubric LLM judge** into a binary reward $R = \text{DeterministicPass} \wedge \text{JudgePass}$, with a fractional scorecard for diagnostics.

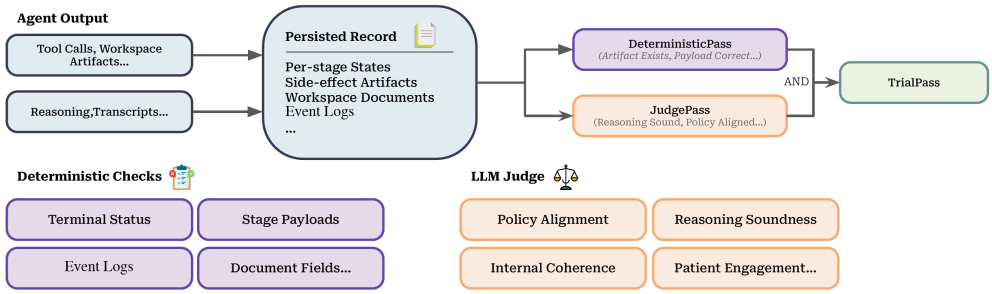


Figure 10: **Verification pipeline.** Each trial emits a persisted record to *deterministic contract* verifier and *rubric-based LLM judge* under strict-majority vote. A trial passes only when both layers pass.

4 Experiments

4.1 Experiment Setup

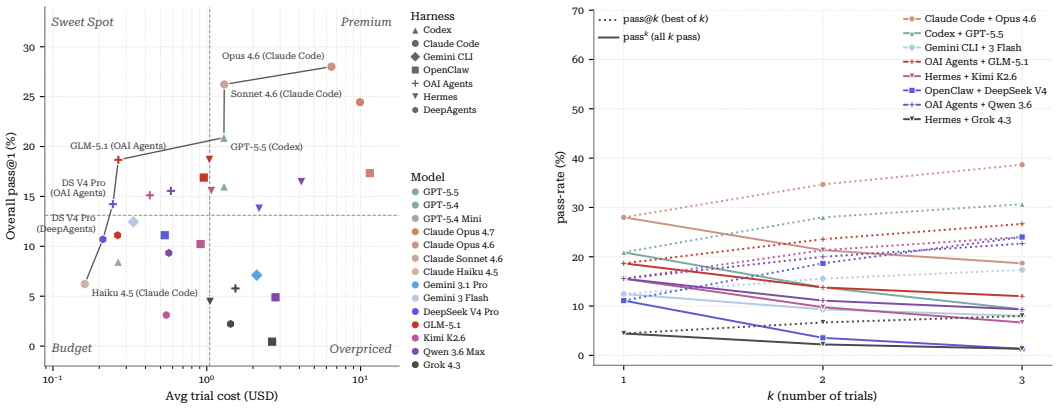
We evaluate 30 agent harness/model configurations across two stacks: a *proprietary stack* pairing each frontier lab's first-party CLI (*Claude Code* [3], *OpenAI Codex* [37], *Gemini CLI* [15]) with that lab's closed-weight models [4, 16, 38], plus an *open-source stack* sweeping four agent frameworks (*OpenClaw* [39], *Hermes* [35], *OpenAI Agents SDK* [36] (*OAI Agents*), and *DeepAgents* [28]) over five OpenRouter-served open-weight models [12, 14, 27, 41, 54], plus an additional OpenClaw + Claude Opus 4.7 reference cell. For each task we run 3 independent trials and report pass@1 [9], pass@3, and pass^3 [59]. The evaluation protocol is shown in Figure 10. Detail configurations like sandbox, judge, and runtime are deferred to Appendix F.

4.2 χ-Bench Results

Table 2 summarizes benchmark performance across agent harnesses and frontier models.

Table 2: χ-Bench results across agent harnesses and frontier models. Per-column maxima are bolded. The three **Overall** columns show task-level bootstrap 95% CIs in $value^{+\mathrm{hi}}_{-\mathrm{lo}}$ form; per-domain pass cells show mean only and the two *Efficiency* columns are averaged over all 225 trials per row.

Agent Harness	Model	Overall			Prior Authorization			Utilization Management			Care Management			Efficiency	
		pass@1	pass@3	pass^3	pass@1	pass@3	pass^3	pass@1	pass@3	pass^3	pass@1	pass@3	pass^3	Steps	Cost ($)
Codex	GPT-5.5	$20.9^{+8.4}_{-7.6}$	$30.7^{+10.7}_{-10.7}$	$9.3^{+8.0}_{-5.3}$	**29.3**	**40.0**	16.0	32.0	48.0	12.0	1.3	4.0	0.0	54	$1.29
Codex	GPT-5.4	$16.0^{+7.1}_{-6.7}$	$25.3^{+9.3}_{-9.3}$	$8.0^{+6.7}_{-5.3}$	24.0	32.0	16.0	17.3	24.0	8.0	6.7	20.0	0.0	58	$1.30
Codex	GPT-5.4 Mini	$8.4^{+4.4}_{-4.0}$	$20.0^{+10.7}_{-9.3}$	$0.0^{+0.0}_{-0.0}$	10.7	24.0	0.0	13.3	32.0	0.0	1.3	4.0	0.0	58	$0.27
Claude Code	Claude Opus 4.7	$24.4^{+8.4}_{-8.0}$	**41.3**$^{+12.0}_{-12.0}$	$10.7^{+8.0}_{-6.7}$	24.0	32.0	16.0	17.3	28.0	8.0	**32.0**	**64.0**	**8.0**	68	$9.91
Claude Code	Claude Opus 4.6	**28.0**$^{+8.9}_{-8.4}$	$38.7^{+10.7}_{-10.7}$	**18.7**$^{+9.3}_{-8.0}$	18.7	24.0	12.0	**41.3**	44.0	**40.0**	24.0	48.0	4.0	76	$6.47
Claude Code	Claude Sonnet 4.6	$26.2^{+7.6}_{-8.0}$	**41.3**$^{+10.7}_{-10.7}$	$12.0^{+8.0}_{-6.7}$	24.0	28.0	**20.0**	34.7	**52.0**	16.0	20.0	44.0	0.0	82	$1.30
Claude Code	Claude Haiku 4.5	$6.2^{+5.3}_{-4.0}$	$10.7^{+8.0}_{-6.7}$	$2.7^{+4.0}_{-2.7}$	0.0	0.0	0.0	14.7	24.0	8.0	4.0	8.0	0.0	41	$0.16
Gemini CLI	Gemini 3.1 Pro	$7.1^{+4.9}_{-4.0}$	$13.3^{+8.0}_{-6.7}$	$1.3^{+2.7}_{-1.3}$	14.7	24.0	4.0	6.7	16.0	0.0	0.0	0.0	0.0	82	$2.11
Gemini CLI	Gemini 3 Flash	$12.4^{+7.1}_{-6.2}$	$17.3^{+9.3}_{-8.0}$	$8.0^{+6.7}_{-5.3}$	18.7	28.0	8.0	18.7	24.0	16.0	0.0	0.0	0.0	142	$0.33
OpenClaw	Claude Opus 4.7	$17.3^{+6.2}_{-5.8}$	$37.3^{+10.7}_{-12.0}$	$4.0^{+5.3}_{-4.0}$	18.7	28.0	8.0	13.3	32.0	4.0	20.0	52.0	0.0	41	$11.48
OpenClaw	Kimi K2.6	$10.2^{+5.3}_{-4.9}$	$18.7^{+9.3}_{-8.0}$	$2.7^{+4.0}_{-2.7}$	12.0	20.0	4.0	18.7	36.0	4.0	0.0	0.0	0.0	72	$0.91
OpenClaw	DeepSeek V4 Pro	$11.1^{+5.3}_{-4.9}$	$24.0^{+9.3}_{-9.3}$	$1.3^{+2.7}_{-1.3}$	14.7	28.0	4.0	12.0	28.0	0.0	6.7	16.0	0.0	42	$0.53
OpenClaw	GLM-5.1	$16.9^{+7.1}_{-6.7}$	$30.7^{+10.7}_{-10.7}$	$6.7^{+6.7}_{-5.3}$	13.3	24.0	4.0	26.7	36.0	16.0	10.7	32.0	0.0	116	$0.96
OpenClaw	Qwen 3.6 Max	$4.9^{+4.0}_{-3.1}$	$10.7^{+8.0}_{-6.7}$	$0.0^{+0.0}_{-0.0}$	10.7	24.0	0.0	4.0	8.0	0.0	0.0	0.0	0.0	79	$2.80
OpenClaw	Grok 4.3	$0.4^{+0.9}_{-0.4}$	$1.3^{+2.7}_{-1.3}$	$0.0^{+0.0}_{-0.0}$	1.3	4.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	65	$2.66
OAI Agents	Kimi K2.6	$15.1^{+7.1}_{-6.2}$	$22.7^{+9.3}_{-9.3}$	$8.0^{+6.7}_{-5.3}$	17.3	28.0	12.0	25.3	36.0	12.0	2.7	4.0	0.0	60	$0.43
OAI Agents	DeepSeek V4 Pro	$14.2^{+7.1}_{-6.2}$	$22.7^{+9.3}_{-9.3}$	$9.3^{+6.7}_{-5.3}$	10.7	16.0	8.0	28.0	40.0	20.0	4.0	12.0	0.0	52	$0.25
OAI Agents	GLM-5.1	$18.7^{+8.4}_{-8.0}$	$26.7^{+10.7}_{-9.3}$	$12.0^{+8.0}_{-6.7}$	18.7	24.0	12.0	33.3	44.0	24.0	4.0	12.0	0.0	58	$0.27
OAI Agents	Qwen 3.6 Max	$15.6^{+8.0}_{-6.7}$	$22.7^{+10.7}_{-9.3}$	$9.3^{+6.7}_{-5.3}$	16.0	20.0	12.0	26.7	36.0	16.0	4.0	12.0	0.0	48	$0.58
OAI Agents	Grok 4.3	$5.8^{+4.4}_{-3.6}$	$10.7^{+8.0}_{-6.7}$	$1.3^{+2.7}_{-1.3}$	0.0	0.0	0.0	16.0	28.0	4.0	1.3	4.0	0.0	32	$1.54
Hermes	Kimi K2.6	$15.6^{+7.6}_{-6.7}$	$24.0^{+10.7}_{-9.3}$	$6.7^{+6.7}_{-5.3}$	18.7	24.0	12.0	21.3	36.0	8.0	6.7	12.0	0.0	31	$1.07
Hermes	DeepSeek V4 Pro	$13.8^{+7.1}_{-6.2}$	$22.7^{+9.3}_{-9.3}$	$8.0^{+6.7}_{-5.3}$	8.0	16.0	4.0	25.3	32.0	20.0	8.0	20.0	0.0	26	$2.19
Hermes	GLM-5.1	$18.7^{+8.9}_{-7.1}$	$28.0^{+10.7}_{-9.3}$	$10.7^{+8.0}_{-6.7}$	10.7	16.0	8.0	34.7	44.0	24.0	10.7	24.0	0.0	30	$1.04
Hermes	Qwen 3.6 Max	$16.4^{+6.7}_{-6.7}$	$28.0^{+10.7}_{-10.7}$	$5.3^{+5.3}_{-4.0}$	9.3	16.0	4.0	26.7	36.0	12.0	13.3	32.0	0.0	29	$4.12
Hermes	Grok 4.3	$4.4^{+4.4}_{-3.1}$	$8.0^{+6.7}_{-5.3}$	$1.3^{+2.7}_{-1.3}$	0.0	0.0	0.0	13.3	24.0	4.0	0.0	0.0	0.0	32	$1.05
DeepAgents	Kimi K2.6	$3.1^{+3.1}_{-2.2}$	$8.0^{+6.7}_{-5.3}$	$0.0^{+0.0}_{-0.0}$	8.0	20.0	0.0	1.3	4.0	0.0	0.0	0.0	0.0	39	$0.55
DeepAgents	DeepSeek V4 Pro	$10.7^{+5.8}_{-4.9}$	$18.7^{+9.3}_{-8.0}$	$2.7^{+4.0}_{-2.7}$	14.7	24.0	4.0	10.7	20.0	4.0	6.7	12.0	0.0	15	$0.21
DeepAgents	GLM-5.1	$11.1^{+6.2}_{-5.8}$	$17.3^{+8.0}_{-8.0}$	$5.3^{+5.3}_{-4.0}$	17.3	24.0	12.0	10.7	16.0	4.0	5.3	12.0	0.0	21	$0.26
DeepAgents	Qwen 3.6 Max	$9.3^{+5.8}_{-4.9}$	$16.0^{+9.3}_{-8.0}$	$4.0^{+5.3}_{-4.0}$	12.0	16.0	8.0	10.7	16.0	4.0	5.3	16.0	0.0	18	$0.57
DeepAgents	Grok 4.3	$2.2^{+2.7}_{-1.8}$	$5.3^{+5.3}_{-4.0}$	$0.0^{+0.0}_{-0.0}$	0.0	0.0	0.0	5.3	12.0	0.0	1.3	4.0	0.0	21	$1.43



(a) ROI quadrants.

(b) Reliability degradation.

Figure 11: (a) Each marker is one row of Table 2: x = mean per-trial cost in USD (log scale), y = Overall pass@1. Dashed cross-hairs at the median cost and median pass@1 split the plane into four quadrants (*Sweet Spot*, *Premium*, *Budget*, *Overpriced*); the Pareto-optimal frontier is connected with a dark line. (b) pass@k (dotted) and pass^k (solid) for $k \in \{1, 2, 3\}$ pooled across all 75 tasks.

Performance, Reliability and ROI. Claude Code paired with Claude Opus 4.6 tops Overall pass@1 at 28.0%, with Sonnet 4.6 (26.2%), Opus 4.7 (24.4%), and Codex + GPT-5.5 (20.9%) close behind; the best domain-level rows are split across Opus 4.6 for UM (41.3%), Opus 4.7 for CM (32.0%), and Codex + GPT-5.5 for PA (29.3%). Reliability further collapses on repeat trials (Figure 11b): pass^3 sits well below pass@1 for the main cells (Opus 4.6 28.0→18.7, GPT-5.5 20.9→9.3, OAI Agents + GLM-5.1 18.7→12.0, Hermes + Grok 4.3 4.4→1.3), exposing run-to-run inconsistency that any production deployment would need to close.

The ROI quadrants in Figure 11a separate absolute capability from cost-normalized value: high-performing configurations (e.g. Claude Code + Opus 4.6) sit in *Premium*, while OAI Agents + GLM-5.1 stands out as a strong cost-normalized point, anchoring the *Sweet Spot* and the low-cost end of the Pareto frontier. The *Overpriced* quadrant collects all Grok 4.3 cells, OpenClaw + Qwen 3.6 Max, and Gemini 3.1 Pro + Gemini CLI; the *Budget* quadrant contains low-cost rows whose savings come with below-median completion rates.

4.3 χ-Bench-Arena: Can Prior Authorization Workflows be Automated End-to-End?

The arena runs a provider agent and a payer agent, both running Codex + GPT-5.5 (our best PA configuration) as a two-player game end-to-end on 23 PA tasks.[2] Each holds its own role-scoped MCPs and state, and they exchange information only through MCP tools. Each side is scored independently; a trial passes only when every check on both sides passes. Pass@1 collapses from **30.4%** to **0%** once the payer agent and cross-role checks join: 2 tasks did not get submitted; 18 did not finish MD decision, and 5 failed the final judge. P2P tasks fail in both sides: 0 P2P request on 5 P2P-required tasks appears and 2 spontaneous P2Ps happen.

Configuration	pass@1
PA provider-only (23 tasks)	30.4
E2E two-agent	0.0

Table 3: E2E two-agent PA vs. same-tasks single-agent baseline.

4.4 χ-Bench-Marathon: Can Long-Running Agents Stay on Track Across All 25 Tasks?

χ-Bench-Marathon stress-tests long-horizon capabilities by loading all 25 tasks of a domain into a shared χ-World. The agent is instructed to finish all tasks, lists them via MCP tools and attempts in any order, in one agent run. Context compaction follows the harness's default setting. Each case is scored individually after the agent reports completion. We evaluate Claude Code + Opus 4.7 and Codex + GPT-5.5. Pass@1 slumps for both configurations regardless of baseline (Table 4). On PA, neither agent submits a single authorization across any of the 25 queued cases, despite touching most cases via write-side tool calls. On UM and CM, agents reach a finalized determination or care plan on only 3-8 of 25 cases per session. Codex + GPT-5.5 reaches its context window and auto-compacts 4-6 times per PA session and 1-2 times on UM; Claude Code + Opus 4.7, with a 1M-token context, never compacts yet completes a similar number of cases. However both agents fan out across the queue, save partial work, and fail to drive most cases to a terminal action.

Table 4: χ-Bench-Marathon pass@1 vs. the per-task baseline. Marathon = all 25 tasks queued in a single agent session, pass@1 averaged over 3 independent sessions; Per-task = isolated single-task trials from Table 2. Δ = Marathon − Per-task (percentage points).

Agent Harness	Model	Prior Authorization			Utilization Management			Care Management		
		Marathon	Per-task	Δ	Marathon	Per-task	Δ	Marathon	Per-task	Δ
Codex	GPT-5.5	8.0	29.3	−21.3	2.7	32.0	−29.3	0.0	1.3	−1.3
Claude Code	Claude Opus 4.7	8.0	24.0	−16.0	1.3	17.3	−16.0	2.7	32.0	−29.3

4.5 Effects of Handbook Skills Components

We trimmed the 1,279-document *Managed-Care Operations Handbook Skill* three ways (−*Domain* drops the domain handbook, −*Medical* drops the medical library, −*Both* drops both), ran all tasks with Codex + GPT-5.5, and found that the handbook's effect is domain-dependent (Figure 12). **UM is handbook-bound**: −*Domain* collapses pass@1 from 32.0 to 17.3, while −*Medical* barely moves it. **PA inverts**: −*Both* modestly beats the other two trimming settings because, with one handbook present, the agent enters an exhaustive verification mode and refuses to submit when uncertain; with no handbook, it commits and the verifier accepts the packet. **CM stays near the**

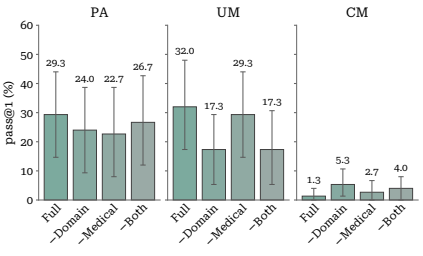


Figure 12: Pass@1 under trimmed skills.

[2]Two tasks not applicable to the two-agent setting are excluded.

floor regardless: the complexity is conversation driving, not policy. The finding is that large skills can help policy-heavy reviews, but can also induce over-verification, refusal, or cognitive overload.

4.6 MCP vs. CLI for Healthcare Agent Workflows

As an exploratory probe, we re-surface every MCP tool as a CLI bash command via MCPorter [46] and re-run Codex + GPT-5.5, on the 75-task suite with 3 trials per task. Table 5 shows a small PA regression, a clear UM drop, and a small CM gain. On this configuration, MCPorter-style CLI re-surfacing is neutral-to-worse rather than uniformly beneficial. We hypothesize that the effect of tool surface format is neutral for OOD tasks like healthcare workflows.

Domain	MCP	CLI	Δ
PA	29.3	28.0	−1.3
UM	32.0	25.3	−6.7
CM	1.3	4.0	+2.7

Table 5: pass@1 of MCP vs. CLI.

4.7 Failure Mode Analysis

We analyze all 5,886 failed trials with the two-layer taxonomy defined in Section G.2: first-level categories capture the broad failure source, while second-level modes specify how the failure occurred. Figure 13 reports the first-level distribution, separating non-agent *Harness-Fault* (1.0%) from agent-side failures: *Clinical-Reasoning* (35.4%, medical or protocol judgment errors), *Workflow-Completion* (23.3%, a required terminal action was never invoked), *Abstain-or-Stuck* (15.6%, wall-clock timeouts, looping, premature closes, and explicit refusal to act), *Policy-Compliance* (13.2%, dominantly literal misreading of cited criterion text), *Tool-Use-Error* (10.7%, concentrated in DeepAgents, where a single malformed tool call escalates into a trial-fatal exit), and *Hallucination* (0.8%).

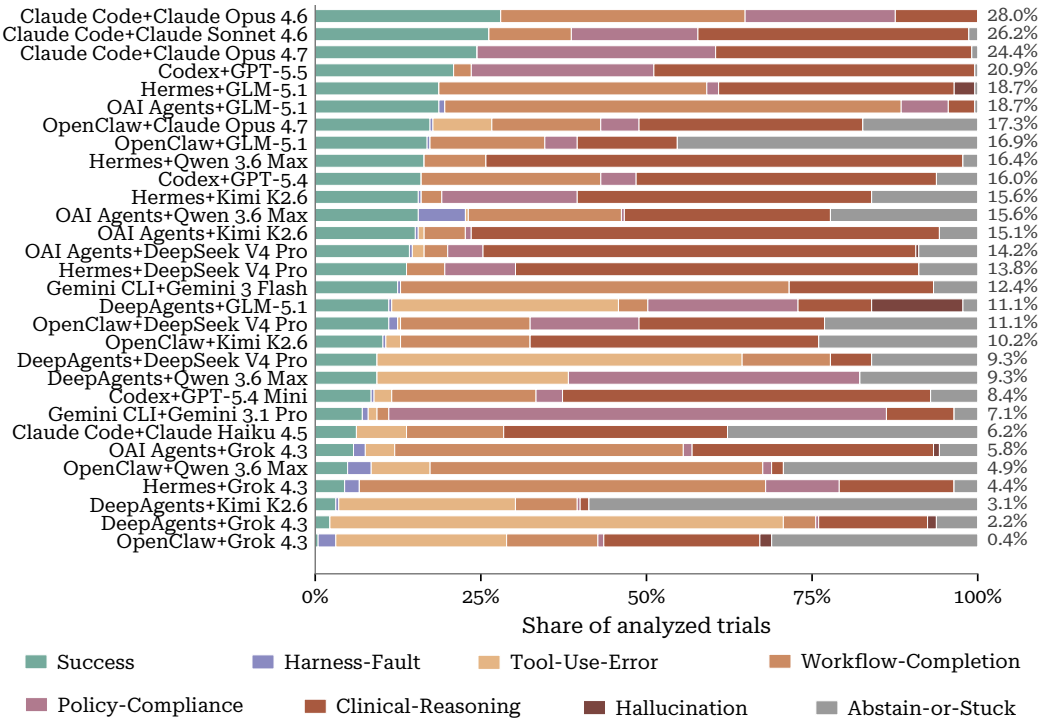


Figure 13: Failure-mode distribution sorted by overall pass@1.

Abstain-or-Stuck concentrates in PA/CM and in DeepAgents + Kimi K2.6 and OpenClaw-based configurations. Nearly half simply exhaust the 1800 s wall-clock cap, and the rest are loops, premature closes, or refusals to act. We therefore read this category as a reliability and termination problem, whereas *Policy-Compliance* captures completed decisions based on misread criteria.

Figure 14 shows that the dominant second-level modes are *criteria misapplication*, where agents see the relevant evidence but make the wrong medical or protocol judgment, *skipped required steps*

(18.7%), and *policy criteria misreading* (13.2%). We distinguish policy criteria misreading from criteria misapplication by the locus of error: the former misreads the rule text itself, while the latter applies the correct rule or evidence to the case incorrectly. A separate CM-specific mode, *illegitimate consent* (337 failures, 5.7%), captures concern-mining: the agent repeatedly reframes and expands care program scopes until an initially refusing member says "yes," instead of using autonomy-first engagement. Detailed failure-mode definitions, analysis, and case examples are in Appendix G.

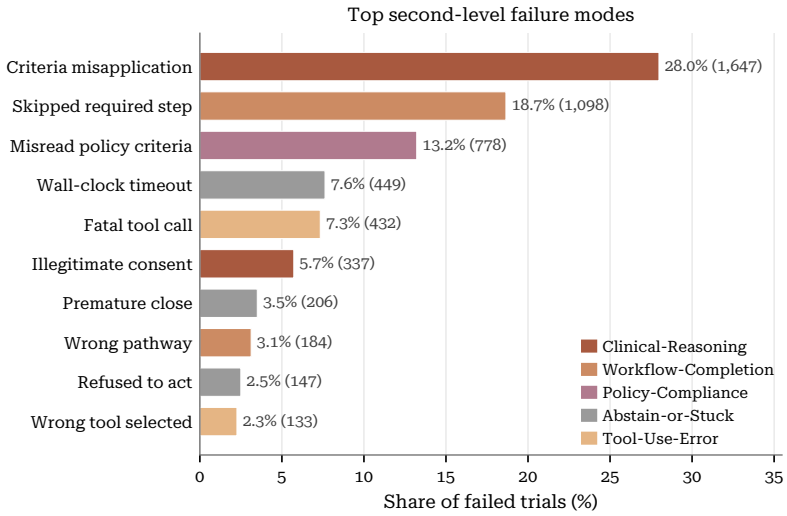


Figure 14: Second-level failure modes. % is over failed trials; colors show first-level categories.

5 Conclusion

We developed χ-Bench, a high-fidelity benchmark that evaluates agents on long-horizon healthcare operations: prior authorization, utilization management, and care management, grounded in a 1,279-document managed-care operations handbook. The strongest agent (Claude Code + Opus 4.6) resolves only **28.0%** of tasks at pass@1, no agent exceeds **20%** at pass^3. Our analysis attributes most failures to three first-level categories: *Clinical-Reasoning* (35.4%), *Workflow-Completion* (23.3%), and *Policy-Compliance* (13.2%). Second level modes, e.g. *criteria misapplication*, *skipped required steps*, and *policy criteria misreading* show that failures arise from distinct bottlenecks. The CM-specific *illegitimate consent* mode further shows that an agent can advance the workflow while violating autonomy-first engagement, so completion alone is not an adequate safety criterion.

Limitations. χ-Bench evaluates language-only agents; real-world healthcare operations often require multimodal reasoning over imaging and speech. Additionally, while χ-World workflows are high-impact, the healthcare industry encompasses hundreds of long-tail workflows with empirical values. Extending coverage along both axes is our immediate next step. Besides, Opus 4.7 is the only judge model, and the effects of using different judge models are yet to be studied.

Broader Impacts. χ-Bench is intentionally a stress test: 28% pass@1 on a static benchmark might be risky for live patient care. The failures our analysis surfaces translate directly into clinical, financial, and regulatory harm if left unchecked. We release χ-Bench to expose these gaps and to encourage caution before agents are deployed on irreversible workflows where the affected party is a patient.

References

[1] American Medical Association. 2024 AMA prior authorization physician survey. Presented at the Annual Meeting of the American Medical Association, Chicago, IL, 2024. URL `https://www.ama-assn.org/system/files/prior-authorization-survey.pdf`.

[2] Anthropic. Introducing the Model Context Protocol. `https://www.anthropic.com/news/model-context-protocol`, 2024. Accessed: 2026-04-30.

[3] Anthropic. Claude Code. https://github.com/anthropics/claude-code, 2025. Accessed: 2026-04-30.

[4] Anthropic. Claude Opus 4.7 system card. https://www.anthropic.com/system-cards, 2026. Accessed: 2026-04-30. Covers Claude Opus 4.7, Sonnet 4.6, and Haiku 4.5.

[5] R. K. Arora, J. Wei, R. S. Hicks, P. Bowman, J. Quiñonero-Candela, F. Tsimpourlas, M. Sharman, M. Shah, A. Vallone, A. Beutel, J. Heidecke, and K. Singhal. Healthbench: Evaluating large language models towards improved human health, 2025. URL https://arxiv.org/abs/2505.08775.

[6] V. Barres, H. Dong, S. Ray, X. Si, and K. Narasimhan. τ^2-bench: Evaluating conversational agents in a dual-control environment, 2025. URL https://arxiv.org/abs/2506.07982.

[7] S. Bedi, H. Cui, M. Fuentes, A. Unell, M. Wornow, J. M. Banda, N. Kotecha, T. Keyes, Y. Mai, M. Oez, et al. Medhelm: Holistic evaluation of large language models for medical tasks. *arXiv preprint arXiv:2505.23802*, 2025.

[8] S. Bedi, R. Welch, E. Steinberg, M. Wornow, T. M. Kim, H. Ahmed, P. Sterling, B. Purohit, Q. Akram, A. Acosta, et al. Healthadminbench: Evaluating computer-use agents on healthcare administration tasks. *arXiv preprint arXiv:2604.09937*, 2026.

[9] M. Chen, J. Tworek, H. Jun, Q. Yuan, H. P. d. O. Pinto, J. Kaplan, H. Edwards, Y. Burda, N. Joseph, G. Brockman, et al. Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374*, 2021.

[10] A. Cuellar, A. H. Krist, L. M. Nichols, and A. J. Kuzel. Facilitators and barriers to care coordination in patient-centered medical homes (PCMHs) from coordinators' perspectives. *Journal of the American Board of Family Medicine*, 31(1):90–101, 2018. doi: 10.3122/jabfm.2018.01.170133. PMC4809054.

[11] D. Cutler, E. Wikler, and P. Basch. Reducing administrative costs and improving the health care system. *New England Journal of Medicine*, 367(20):1875–1878, 2012. doi: 10.1056/NEJMp1209711.

[12] DeepSeek-AI. DeepSeek-V4 Pro model card. https://huggingface.co/deepseek-ai/DeepSeek-V4-Pro, 2026. Accessed: 2026-04-30.

[13] A. Drouin, M. Gasse, M. Caccia, I. H. Laradji, M. D. Verme, T. Marty, L. Boisvert, M. Thakkar, Q. Cappart, D. Vazquez, N. Chapados, and A. Lacoste. WorkArena: How capable are web agents at solving common knowledge work tasks? In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024. URL https://arxiv.org/abs/2403.07718.

[14] GLM-5 Team. GLM-5: From vibe coding to agentic engineering. *arXiv preprint arXiv:2602.15763*, 2026.

[15] Google. Gemini CLI. https://github.com/google-gemini/gemini-cli, 2025. Accessed: 2026-04-30.

[16] Google DeepMind. Gemini 3.1 Pro model card. https://deepmind.google/models/model-cards/gemini-3-1-pro/, 2026. Accessed: 2026-04-30. Covers Gemini 3.1 Pro and Gemini 3 Flash.

[17] Harbor Framework. Harbor: A framework for agent evaluations and RL environments. https://github.com/harbor-framework/harbor, 2026. Accessed: 2026-04-30.

[18] Y. Jiang, K. C. Black, G. Geng, D. Park, J. Zou, A. Y. Ng, and J. H. Chen. Medagentbench: a virtual ehr environment to benchmark medical llm agents. *Nejm Ai*, 2(9):AIdbp2500144, 2025.

[19] C. E. Jimenez, J. Yang, A. Wettig, S. Yao, K. Pei, O. Press, and K. Narasimhan. Swe-bench: Can language models resolve real-world github issues?, 2024. URL https://arxiv.org/abs/2310.06770.

[20] D. Jin, E. Pan, N. Oufattole, W.-H. Weng, H. Fang, and P. Szolovits. What disease does this patient have? a large-scale open domain question answering dataset from medical exams. *Applied Sciences*, 11(14):6421, 2021. URL https://arxiv.org/abs/2009.13081.

[21] Q. Jin, B. Dhingra, Z. Liu, W. W. Cohen, and X. Lu. PubMedQA: A dataset for biomedical research question answering. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2567–2577. Association for Computational Linguistics, 2019. URL https://arxiv.org/abs/1909.06146.

[22] A. Jones and C. Kelly. Code execution with mcp: Building more efficient agents, 2025.

[23] H.-H. Ju. Improving care coordination of patients with chronic diseases. *The Journal for Nurse Practitioners*, 18(9):926–929, 2022. doi: 10.1016/j.nurpra.2022.07.005.

[24] L. P. Kaelbling, M. L. Littman, and A. R. Cassandra. Planning and acting in partially observable stochastic domains. *Artificial intelligence*, 101(1-2):99–134, 1998.

[25] M. Karam, M.-C. Chouinard, M. Kevork, R. Fleming, and A. Duhoux. Nurses' and patients' perspectives on care coordination across health care and social services sectors: A qualitative study. *SAGE Open Nursing*, 2026. doi: 10.1177/08445621251395347.

[26] N. Khandekar, Q. Jin, G. Xiong, S. Dunn, S. S. Applebaum, Z. Anwar, M. Sarfo-Gyamfi, C. W. Safranek, A. A. Anwar, A. Zhang, A. Gilson, M. B. Singer, A. Dave, A. Taylor, A. Zhang, Q. Chen, and Z. Lu. MedCalc-Bench: Evaluating large language models for medical calculations. In *Advances in Neural Information Processing Systems 37: Datasets and Benchmarks Track*, 2024. URL https://arxiv.org/abs/2406.12036.

[27] Kimi Team. Kimi K2: Open agentic intelligence. *arXiv preprint arXiv:2507.20534*, 2025.

[28] LangChain. DeepAgents. https://github.com/langchain-ai/deepagents, 2025. Accessed: 2026-04-30.

[29] G. Lee, H. Hwang, S. Bae, Y. Kwon, W. Shin, S. Yang, M. Seo, J.-Y. Kim, and E. Choi. EHRSQL: A practical text-to-SQL benchmark for electronic health records. In *Advances in Neural Information Processing Systems 35: Datasets and Benchmarks Track*, 2022. URL https://arxiv.org/abs/2301.07695.

[30] J. Li, W. Zhao, J. Zhao, W. Zeng, H. Wu, X. Wang, R. Ge, Y. Cao, Y. Huang, W. Liu, et al. The tool decathlon: Benchmarking language agents for diverse, realistic, and long-horizon task execution. *arXiv preprint arXiv:2510.25726*, 2025.

[31] X. Li, W. Chen, Y. Liu, S. Zheng, X. Chen, Y. He, Y. Li, B. You, H. Shen, J. Sun, et al. Skillsbench: Benchmarking how well agent skills work across diverse tasks. *arXiv preprint arXiv:2602.12670*, 2026.

[32] J. Liu, W. Wang, Z. Ma, G. Huang, Y. Su, K.-J. Chang, W. Chen, H. Li, L. Shen, and M. R. Lyu. MedChain: Bridging the gap between LLM agents and clinical practice through interactive sequential benchmarking. In *Advances in Neural Information Processing Systems 38: Datasets and Benchmarks Track*, 2025. URL https://arxiv.org/abs/2412.01605.

[33] M. A. Merrill, A. G. Shaw, N. Carlini, B. Li, H. Raj, I. Bercovich, L. Shi, J. Y. Shin, T. Walshe, E. K. Buchanan, et al. Terminal-bench: Benchmarking agents on hard, realistic tasks in command line interfaces. *arXiv preprint arXiv:2601.11868*, 2026.

[34] Modal Labs. Modal: High-performance serverless infrastructure for AI and data. https://modal.com, 2025. Accessed: 2026-04-30.

[35] Nous Research. Hermes Agent: The agent that grows with you. https://github.com/NousResearch/hermes-agent, 2026. Accessed: 2026-04-30.

[36] OpenAI. OpenAI Agents SDK (python). https://github.com/openai/openai-agents-python, 2025. Accessed: 2026-04-30.

[37] OpenAI. OpenAI Codex CLI. https://github.com/openai/codex, 2025. Accessed: 2026-04-30.

[38] OpenAI. GPT-5.5 system card. https://openai.com/index/gpt-5-5-system-card/, 2026. Accessed: 2026-04-30. Covers the GPT-5.5, GPT-5.4, and GPT-5.4 Mini family.

[39] OpenClaw. OpenClaw: Your own personal ai assistant. https://github.com/openclaw/openclaw, 2025. Accessed: 2026-04-30.

[40] A. Pal, L. K. Umapathi, and M. Sankarasubbu. MedMCQA: A large-scale multi-subject multi-choice dataset for medical domain question answering. In *Proceedings of the Conference on Health, Inference, and Learning (CHIL)*, volume 174 of *Proceedings of Machine Learning Research*, pages 248–260. PMLR, 2022. URL https://arxiv.org/abs/2203.14371.

[41] Qwen Team. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.

[42] N. R. Sahni, P. Gupta, M. Peterson, and D. M. Cutler. Active steps to reduce administrative spending associated with financial transactions in US health care. *Health Affairs Scholar*, 1(5): qxad053, 2023. doi: 10.1093/haschl/qxad053.

[43] N. R. Sahni, B. Istvan, and D. M. Cutler. Perceptions of prior authorization burden and solutions. *Health Affairs Scholar*, 2(9):qxae096, 2024. doi: 10.1093/haschl/qxae096.

[44] S. Schmidgall, R. Ziaei, C. Harris, E. Reis, J. Jopling, and M. Moor. AgentClinic: a multimodal agent benchmark to evaluate AI in simulated clinical environments, 2024. URL https://arxiv.org/abs/2405.07960.

[45] C. A. Sinsky, L. Colligan, L. Li, M. Prgomet, S. Reynolds, L. Goeders, J. Westbrook, M. Tutty, and G. Blike. Allocation of physician time in ambulatory practice: A time and motion study in 4 specialties. *Annals of Internal Medicine*, 165(11):753–760, 2016. doi: 10.7326/M16-0961.

[46] P. Steinberger. MCPorter: TypeScript runtime and CLI for connecting to MCP servers. https://github.com/steipete/mcporter, 2025. npm package mcporter; accessed 2026-05-03.

[47] R. S. Sutton, D. Precup, and S. Singh. Between mdps and semi-mdps: A framework for temporal abstraction in reinforcement learning. *Artificial intelligence*, 112(1-2):181–211, 1999.

[48] X. Tang, B. Qian, R. Gao, J. Chen, X. Chen, and M. Gerstein. BioCoder: a benchmark for bioinformatics code generation with large language models. *Bioinformatics*, 40(Supplement_1): i266–i276, 2024. doi: 10.1093/bioinformatics/btae230. URL https://arxiv.org/abs/2308.16458.

[49] X. Tang, D. Shao, J. Sohn, J. Chen, J. Zhang, J. Xiang, F. Wu, Y. Zhao, C. Wu, W. Shi, A. Cohan, and M. Gerstein. MedAgentsBench: Benchmarking thinking models and agent frameworks for complex medical reasoning, 2025. URL https://arxiv.org/abs/2503.07459.

[50] H. Trivedi, T. Khot, M. Hartmann, R. Manku, V. Dong, E. Li, S. Gupta, A. Sabharwal, and N. Balasubramanian. AppWorld: A controllable world of apps and people for benchmarking interactive coding agents. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*. Association for Computational Linguistics, 2024. URL https://arxiv.org/abs/2407.18901.

[51] G. Tsatsaronis, G. Balikas, P. Malakasiotis, I. Partalas, M. Zschunke, M. R. Alvers, D. Weissenborn, A. Krithara, S. Petridis, D. Polychronopoulos, Y. Almirantis, J. Pavlopoulos, N. Baskiotis, P. Gallinari, T. Artières, A.-C. N. Ngomo, N. Heino, E. Gaussier, L. Barrio-Alvers, M. Schroeder, I. Androutsopoulos, and G. Paliouras. An overview of the BIOASQ large-scale biomedical semantic indexing and question answering competition. *BMC Bioinformatics*, 16(1):138, 2015. doi: 10.1186/s12859-015-0564-6. URL https://bmcbioinformatics.biomedcentral.com/articles/10.1186/s12859-015-0564-6.

[52] Z. Wang, B. Danek, Z. Yang, Z. Chen, and J. Sun. Can large language models replace data scientists in biomedical research?, 2024. URL https://arxiv.org/abs/2410.21591.

[53] M. Wornow, R. Thapa, E. Steinberg, J. A. Fries, and N. H. Shah. EHRSHOT: An EHR benchmark for few-shot evaluation of foundation models. In *Advances in Neural Information Processing Systems 36: Datasets and Benchmarks Track*, 2023. URL https://arxiv.org/abs/2307.02028.

[54] xAI. Grok 4 model card. https://data.x.ai/2025-08-20-grok-4-model-card.pdf, 2025. Accessed: 2026-04-30.

[55] T. Xie, D. Zhang, J. Chen, X. Li, S. Zhao, R. Cao, T. J. Hua, Z. Cheng, D. Shin, F. Lei, Y. Liu, Y. Xu, S. Zhou, S. Savarese, C. Xiong, V. Zhong, and T. Yu. OSWorld: Benchmarking multimodal agents for open-ended tasks in real computer environments. In *Advances in Neural Information Processing Systems 37: Datasets and Benchmarks Track*, 2024. URL https://arxiv.org/abs/2404.07972.

[56] G. Xiong, Q. Jin, Z. Lu, and A. Zhang. Benchmarking retrieval-augmented generation for medicine. In *Findings of the Association for Computational Linguistics: ACL 2024*. Association for Computational Linguistics, 2024. URL https://arxiv.org/abs/2402.13178.

[57] F. F. Xu, Y. Song, B. Li, Y. Tang, K. Jain, M. Bao, Z. Z. Wang, X. Zhou, Z. Guo, M. Cao, M. Yang, H. Y. Lu, A. Martin, Z. Su, L. Maben, R. Mehta, W. Chi, L. Jang, Y. Xie, S. Zhou, and G. Neubig. TheAgentCompany: Benchmarking LLM agents on consequential real world tasks, 2024. URL https://arxiv.org/abs/2412.14161.

[58] R. Xu, Y. Zhuang, Y. Zhong, Y. Yu, Z. Wang, X. Tang, H. Wu, M. D. Wang, J. C. Ho, Y. Xiao, W. Shi, and C. Yang. MedAgentGym: A scalable agentic training environment for code-centric reasoning in biomedical data science. In *International Conference on Learning Representations (ICLR)*, 2026. URL https://arxiv.org/abs/2506.04405.

[59] S. Yao, N. Shinn, P. Razavi, and K. Narasimhan. τ-bench: A benchmark for tool-agent-user interaction in real-world domains. *arXiv preprint arXiv:2406.12045*, 2024.

[60] B. Zhang, K. Lazuka, and M. Murag. Equipping agents for the real world with agent skills. *Anthropic Engineering Blog*, 2025.

[61] S. Zhou, F. F. Xu, H. Zhu, X. Zhou, R. Lo, A. Sridhar, X. Cheng, T. Ou, Y. Bisk, D. Fried, U. Alon, and G. Neubig. WebArena: A realistic web environment for building autonomous agents. In *International Conference on Learning Representations (ICLR)*, 2024. URL https://arxiv.org/abs/2307.13854.

[62] Y. Zuo, S. Qu, Y. Li, Z. Chen, X. Zhu, E. Hua, K. Zhang, N. Ding, and B. Zhou. Medxpertqa: Benchmarking expert-level medical reasoning and understanding. *arXiv preprint arXiv:2501.18362*, 2025.

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A Ethical Statement

Data Privacy and Human Subjects. χ-Bench contains no real patient data and no Protected Health Information (PHI). Every persona, chart, and clinical detail is a fictional composite authored by clinicians and healthcare operational specialists. The benchmark is not human-subjects research, required no IRB review, and can be redistributed without privacy-preserving transformations.

LLM Usage. We disclose three uses of language models outside the agents under test, all running Claude Opus 4.7: (1) the rubric-grading judge with $V{=}3$ independent votes per rubric and strict-majority aggregation; (2) the post-hoc failure-mode analyzer; and (3) the patient-persona and peer-to-peer counterpart simulators. Prompts for all three are reproduced in Appendix H.

Potential Harms and Dual Use. χ-Bench measures whether a frontier agent can safely execute prior-authorization, utilization-management, and care-management workflows. The headline claim is a strong negative signal: an agent that scores poorly on χ-Bench is exhibiting the dominant failure modes we analyze (consent fabrication, criteria misreading, premature denial, misrouted handoffs), each of which translates directly into clinical, financial, and regulatory risks if deployed in a live setting. A high score is necessary but not sufficient: any agent destined for production must additionally clear the regulatory, privacy, and clinical safety review applicable to its target setting.

B Extended Related Work

Table 1 compares χ-Bench against 29 healthcare and long-horizon agent benchmarks along nine axes. This appendix defines each axis and justifies the non-trivial cell assignments, with extra emphasis on the three healthcare-workflow axes that motivate χ-Bench: *Multilateral interaction*, *Multi-role composition*, and *Policy density*.

B.1 Axis Definitions

The first six axes carry their conventional agent-benchmark meanings:

- **HC**: the benchmark targets clinical, payer, or population-health workflows.
- **API tools**: the agent acts via programmatic tool / function calls (SQL, shell, browser, or MCP), as opposed to pure text-output benchmarks or GUI benchmarks.
- **Long-horiz.**: tasks span many turns or distinct workflow stages, so success requires sustained planning rather than single-shot recall.

- **Hidden state**: structurally asymmetric or private information the agent must elicit through interaction (e.g., patient-side facts, supervisor-private fields), rather than read off the prompt.
- **In-Situ**: the verifier inspects simulator-persisted world state (database rows, event log, file system) rather than only grading agent output text.
- **LLM judge**: at least part of scoring uses rubric-based LLM grading.

The remaining three axes are the discriminators that motivate χ-Bench and warrant precise definitions.

Multilateral interaction. ✓when the agent must enter live multi-turn dialogue with one or more simulated parties (patient, user, peer reviewer) as a required workflow step, where the dialogue cannot be replaced by a tool call. ❍when the dialogue is single-party Q&A or shows up only on a subset of tasks. ✗when the agent only issues tool calls and observes results.

Multi-role composition. ✓when, within a single task, the agent must switch among ≥ 2 distinct role tool surfaces (e.g., provider clinician $\rightarrow$ payer UM nurse $\rightarrow$ medical director) and the handoff between roles is terminal (cannot be edited or re-run). ❍when multiple actors are present in the environment but the agent itself plays only one role (the others are simulators or fixed task issuers), or when stages are sequential without distinct role tool surfaces. ✗for single-agent single-role tasks.

Policy density. ✓when the GT-required policy material is a structured handbook spanning multiple documents and exceeding ∼10K tokens (medical-policy library, clinical-criteria sets, ops SOPs). ❍for a single-document policy of a few pages (∼1–10K tokens; e.g., a domain SOP, store policy, or company wiki page). ✗when no formal policy artifact is provided and decision criteria are either implicit (baked into the UI/API) or absent.

B.2 Healthcare Benchmarks

Static QA (MedQA, MedMCQA, PubMedQA, BioASQ, MIRAGE, MedCalc-Bench, EHRSHOT, MedAgentsBench). All measure factual recall or numerical reasoning over a closed prompt; no tool-mediated environment, no simulated user, no multi-stage workflow, no policy library to navigate. All three workflow axes ✗.

EHRSQL. Text2SQL over EHR schemas. Tool/API ✓via SQL execution, but the agent issues a single query: no dialogue, no role switching, no policy doc. The verifier compares generated SQL/answer string against ground truth without probing post-execution database state (In-Situ ✗); records are exposed via the queryable schema (Hidden ✗); grading is exact-match (LLM judge ✗).

BioCoder, BioDSBench. Code-execution tasks on bio/clinical data. The runtime probes code state (In-Situ ❍), but the agent stays in a single coder role with no policy artifact. There is no simulated counterparty (Multilateral ✗), all bio/clinical data is fully exposed in the prompt or notebook (Hidden ✗), and grading uses unit tests / reference outputs rather than rubric LLM (LLM judge ✗).

MedHELM [7]. A holistic evaluation framework comprising 35 benchmarks across a clinician-validated taxonomy of five clinical task categories (decision support, note generation, patient communication, research, administration). Healthcare ✓. Each benchmark is a prompt $\rightarrow$ response evaluation: the model receives a clinical vignette or task description and produces a single output with no iterative tool invocation (API ✗, Long-horiz. ✗). No policy documents are supplied (Policy ✗); the model plays one role per benchmark (Multi-role ✗); there is no simulated counterparty or dialogue partner (Multilateral ✗); task inputs are fully specified in the prompt with no information that must be elicited (Hidden ✗); scoring compares outputs to references or rubrics on the transcript alone with no persisted environment state (In-Situ ✗). An LLM-jury of three models grades open-ended benchmarks with demonstrated agreement to clinician ratings (LLM judge ✓).

MedXpertQA [62]. A static multiple-choice benchmark of 4,460 expert-level medical questions spanning 17 specialties and 11 body systems, with text and multimodal subsets. Healthcare ✓. Questions are single-turn with no tool surface (API ✗), no multi-step planning (Long-horiz. ✗), no policy retrieval (Policy ✗), no role variation (Multi-role ✗), no simulated interlocutor (Multilateral ✗), no information asymmetry (Hidden ✗), and no persisted environment (In-Situ ✗). Evaluation is multiple-choice accuracy against ground-truth answers (LLM judge ✗).

HealthBench. Multi-turn rubric-graded patient/clinician dialogue. Multilateral ❍: the agent talks to one simulated party with persona but the exchange is essentially Q&A rather than the kind of structured RFI/P2P required to advance a workflow. No policy library, no role switching. The exchange is purely natural language with no programmatic tool surface (API ✗); the patient persona is realised by the prompt to the simulator rather than elicited from a hidden world model (Hidden ✗); scoring is rubric-on-transcript with no persisted simulator state (In-Situ ✗).

AgentClinic. A doctor agent interacts with a patient simulator, a measurement source, and a moderator. Multilateral ✓(multi-party dialogue is the core mechanic). Multi-role ❍: the agent itself plays only the doctor; the other parties are fixed simulators rather than alternative role surfaces the agent must inhabit. Hidden state ✓(patient information must be elicited). No formal policy doc (Policy ✗). The agent invokes a small fixed set of measurement/test actions rather than a general API surface (API ❍); scoring compares the final diagnosis to ground truth without inspecting simulator-internal state (In-Situ ✗); LLM grading is used only for a subset of sub-scores (LLM judge ❍).

MedChain. Three sequential stages (history → diagnostics → treatment) with a per-stage Gemma-2 patient simulator. Multilateral ❍: the patient sim is per-stage but exchanges are short and templated. Multi-role ✗: stages are temporal phases of the same doctor role rather than distinct role tool surfaces. Hidden state ✓(patient data elicited per stage). Decision criteria are baked into the per-stage case prompts with no external policy artifact (Policy ✗); per-stage grading compares textual outputs to references without probing simulator state (In-Situ ✗); LLM grading is used only for a subset of stages (LLM judge ❍).

MedAgentBench, MedAgentGym. FHIR / sandboxed-code workflows. In-Situ ❍via state probing, but no dialogue (Multilateral ✗), no role switching (Multi-role ✗), no policy doc (Policy ✗). FHIR resources and code state are fully queryable by the agent (Hidden ✗), and evaluation is programmatic state diff rather than rubric LLM (LLM judge ✗).

HealthAdminBench [8]. GUI agents on prior-authorization / appeals / DME-order tasks across an EHR plus two payer portals plus a fax system. Multi-role ✓: the agent must traverse the four UIs and treat each portal's submission as terminal. In-Situ ✓(1,698 evaluation points probe persisted UI/file state). LLM judge ✓, Long-horiz. ✓. *Policy density* ❍: the benchmark does not ship a separate multi-document medical-policy library, but approval criteria and operational rules are partially encoded in the portal flows themselves and in per-task instructions ("benchmark emphasizes environmental interaction over document study"), so policy is present but at the granularity of UI-embedded conditions rather than a structured handbook. Multilateral ✗: there is no live patient/peer dialogue. Tool/API ✗: HealthAdminBench focuses on GUI benchmarking instead of tool calling. Hidden state ❍: payer-side adjudication fields must be surfaced through portal navigation, but no counterparty deliberately withholds information through dialogue.

B.3 Long-Horizon Agent Benchmarks

SWE-Bench, Terminal-Bench, Toolathlon. Code/terminal/SaaS task suites with deterministic verifiers; no dialogue (Multilateral ✗), no role switching (Multi-role ✗), no policy library (Policy ✗). Repository / system state is fully observable to the agent (Hidden ✗), and grading is compile/test/CLI checks rather than rubric LLM (LLM judge ✗).

WebArena, OSWorld. GUI/desktop control with state-inspection evaluators. In-Situ ✓(OSWorld) / ❍(WebArena), but otherwise single-role (Multi-role ✗) with no policy artifact (Policy ✗). Agents act through pixel/DOM/keyboard events rather than typed APIs (API ✗); there is no simulated dialogue partner (Multilateral ✗); world state is fully visible through the DOM or screen (Hidden ✗); programmatic state checkers replace rubric LLM grading (LLM judge ✗).

WorkArena. ServiceNow workflows. Policy ❍: ServiceNow process documentation is referenced for some tasks but is not a multi-document handbook. The agent operates a single ServiceNow employee role via GUI events (API ✗, Multi-role ✗); ServiceNow state is observable and there is no live counterparty (Hidden ✗, Multilateral ✗); evaluation is task-spec / state matching rather than rubric LLM (LLM judge ✗).

AppWorld. Personal-assistant agent across nine apps with private supervisor state. Multilateral ❍: supervisor messages and in-app personas appear, but most tasks are background tool execution. Multi-role ❍: the agent operates over disjoint app tool surfaces with persisted state and supervisor handoffs are not editable, but throughout each task the agent stays in the personal-assistant role and never has to inhabit a second role with its own tool surface, instructions, or accountability boundary. Hidden state ❍(supervisor-private fields, no LLM patient sim). Task constraints are embedded directly in user instructions rather than a separate policy document (Policy ✗), and evaluation is database/world-state diff rather than rubric LLM (LLM judge ✗).

SkillsBench. Skill-bundle execution. Policy ❍: skill packages contain procedures but are scoped how-tos rather than policy documents that drive grounded decisions. The skill catalog includes a few health-adjacent bundles but is not domain-targeted (HC ❍); a single agent executes a self-contained skill with no role switch or simulated counterparty (Multi-role ✗, Multilateral ✗); skill inputs are fully specified (Hidden ✗); grading uses programmatic skill tests rather than rubric LLM (LLM judge ✗).

τ/τ^2-Bench. Single agent paired with a simulated user and a domain policy doc (retail / airline). Multilateral ✓, Hidden state ✓, In-Situ ✓. Multi-role ❍: the agent stays in the assistant role throughout; the simulated user is a counterparty, not a role the agent must play. Policy ❍: each domain ships a single policy document (a few pages, $\sim$1–3K tokens). Grading combines deterministic state-diff over a simulated DB with action-trajectory checks; no rubric LLM judge in the canonical pipeline (LLM judge ✗).

TheAgentCompany. A self-hosted enterprise simulation populated with 17+ NPC coworkers and self-hosted GitLab/RocketChat/etc. Multilateral ✓, Long-horiz. ✓. Multi-role ❍: the agent plays a single hire while NPCs occupy other roles. Policy ❍: company wiki and SOPs are referenced but at a per-page granularity rather than a structured handbook library. Hidden state ❍(some NPC-private context). The agent acts through a mix of typed tool calls and self-hosted web UIs (API ❍); checkpoints combine deterministic state checks with LLM-based partial credit (LLM judge ❍).

B.4 χ-Bench (Ours)

χ-Bench satisfies all nine axes by construction: a healthcare workflow benchmark (HC ✓) where the agent acts via 87 MCP tools across 20 healthcare apps (Tool/API ✓) and must run a multi-stage payer/provider/care-management pipeline (Long-horiz. ✓) with deterministic gates probing the simulator's persisted world state (In-Situ ✓) and a multi-rubric LLM judge (LLM judge ✓). Each task is multi-role: a single case is handed off across clinician $\rightarrow$ UM nurse $\rightarrow$ medical director $\rightarrow$ care manager with terminal handoffs (Multi-role Comp. ✓). The simulator preserves provider/payer/patient information asymmetry (Hidden state ✓) and exposes peer-to-peer reviews, requests for information, and patient outreach as required multi-turn dialogues (Multilateral ✓). Decisions are grounded in a 1,279-document Managed-Care Operations Handbook plus a medical-policy library (Policy density ✓).

C χ-World Detail

C.1 Healthcare Workflows

χ-World reproduces three end-to-end clinical operations workflows that interlock through shared cases, documents, and messages: the **Provider Prior Authorization** workflow run by clinic staff, the **Payer Utilization Management** workflow run by insurer nurses and physicians, and the **Care Management** workflow run by case managers. Figures 15–17 render each as a phased state machine over the MCP tools listed in Section C.2. The descriptions below are written for non-clinical readers; precise tool names are given in the figures.

C.1.1 Provider Prior Authorization

A doctor wants to order a treatment (such as an MRI, a specialty drug, or a surgery) and the patient's insurance requires advance permission before it will pay. The clinic-side "provider PA" agent shepherds that permission request through eight phases (Figure 15). It first opens a case from the

doctor's order (*Phase 1*), checks whether the insurance plan actually requires prior authorization and what evidence is expected (*Phase 2*), gathers chart notes, lab reports, and questionnaire answers (*Phase 3*), bundles them into a single packet and sends it to the insurer through the portal, fax, or EDI channel (*Phase 4*), and then polls for the insurer's reply (*Phase 5*). Replies fall into four buckets—*approved*, *partially approved*, *pended* (insurer wants more information), or *denied*. A pend triggers a quick supplemental upload (*Phase 6*); a request for a doctor-to-doctor conversation triggers a multi-turn peer-to-peer dialogue (*Phase 7*); a denial can be escalated through an internal appeal and, if still upheld, an external Independent Review Organization review (*Phase 8*). The case ends when an authorization number is issued or every appeal route is exhausted.

C.1.2 Payer Utilization Management

On the other side of the request sits the insurance company's utilization-management team, which decides whether the requested service is medically necessary and covered. Their workflow has six stages (Figure 16). Stage 1 *normalizes intake*: whatever channel the request arrived on (portal, API, fax, or EDI), the agent files it as a structured case with diagnoses, requested procedures, and the requesting provider. Stage 2 *triages and routes* the case: it computes a regulatory deadline based on urgency and state law (e.g., 24 hours for California STAT, 72 hours for urgent, 120 hours for routine), checks whether the provider is "gold-carded" for automatic approval, and assigns the case to a fast-track lane, a nurse, or a physician. Stage 3 is the *nurse review* against clinical criteria; nurses can approve, escalate to a physician, or pend the case for more information. Stage 4 is the *physician (MD) review* for cases the nurse cannot approve, with the option to call a peer-to-peer with the requesting doctor. Stage 5 is the *determination*: a 12-point preflight validates the decision and stamps it as approved, partially approved, or denied. Stage 6 generates the *outbound letter*, delivers it on the same channel, and checks it for regulatory completeness. A denial unlocks an appeals path with an independent reviewer and, if upheld, an external IRO decision.

C.1.3 Care Management

Care management is the longitudinal side of the system: a case manager (often a nurse or social worker) reaches out to a patient who has been flagged as high-risk—for example, after a hospital discharge, an emergency-department visit, a primary-care referral, or an insurer's risk model—and helps them stay on track between visits. The workflow has five phases (Figure 17). Phase 1 *intake*: pull the referral from the queue, set a priority (urgent / high / standard) and a program (complex care, chronic disease, transitions of care, or behavioral health), and open the case. Phase 2 *chart review*: read the patient's conditions, encounters, medications, labs, and documents to build a clinical picture and flag coordination risks. Phase 3 *patient outreach*: hold a multi-turn telephone conversation with the patient (simulated by an AI persona in our environment) to obtain consent, surface concerns, and decide on next steps. Phase 4 *four-domain assessment*: complete a structured evaluation across four areas—*clinical* (diagnoses, symptoms, red flags), *medication* (regimen, adherence barriers), *behavioral health* (PHQ-9 depression and GAD-7 anxiety scores, safety concerns), and *social determinants of health* (transportation, food, housing, caregiver support, financial constraints, health literacy, language). Phase 5 *care plan*: draft a hierarchical plan of *problems* $\rightarrow$ *measurable goals* $\rightarrow$ *interventions* (each with an owner such as the care manager, patient, primary-care provider, or pharmacist), add escalation triggers (e.g., "weight gain >3 lb in 24 h $\rightarrow$ call PCP") and a follow-up cadence, then finalize and close the case. At any phase the case may be *disenrolled* (patient refused, unreachable, moved, safety concern, or duplicate).

C.2 Tool Manifest

χ-World exposes 87 MCP tools across three domain-specific servers: a Provider PA server (30 tools), a Payer UM server (41 tools), and a Care Management server (16 tools). Tables 6–8 list every tool with a brief description and a representative call. Within each table, tools are grouped by namespace; each namespace corresponds to one logical service (e.g., case management, document handling, or letter generation).

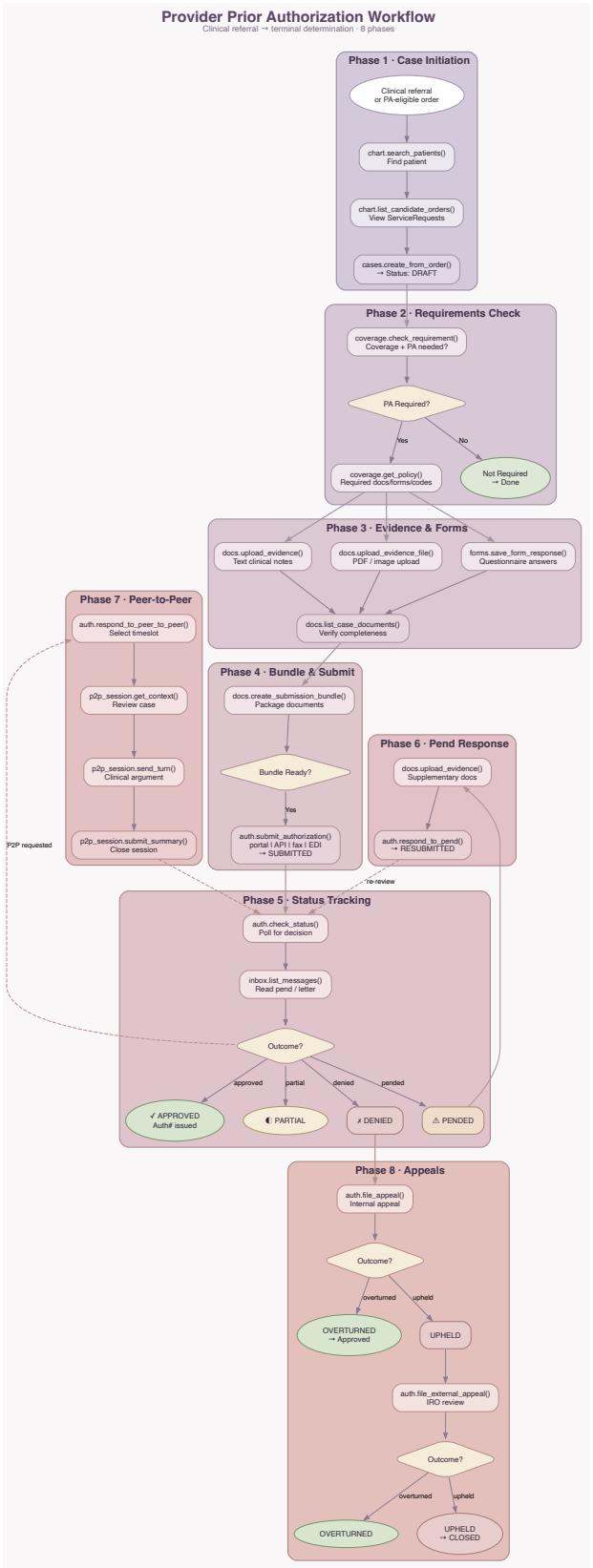

Figure 15: Provider Prior Authorization workflow. Eight phases from referral to terminal determination, with branches for pend, peer-to-peer, and appeals. Boxes are MCP tool calls listed in Table 6.

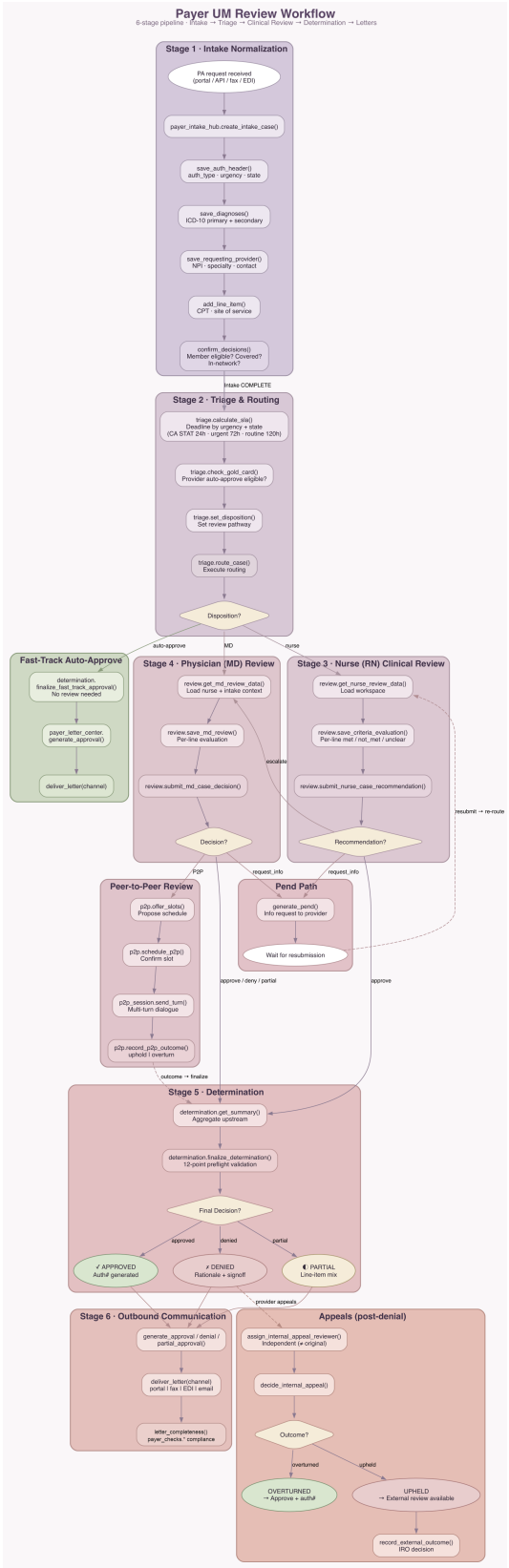

Figure 16: Payer Utilization Management workflow. Six stages from intake normalization to outbound letter, with side paths for fast-track auto-approval, peer-to-peer, pending, and post-denial appeals. Boxes are MCP tool calls listed in Table 7.

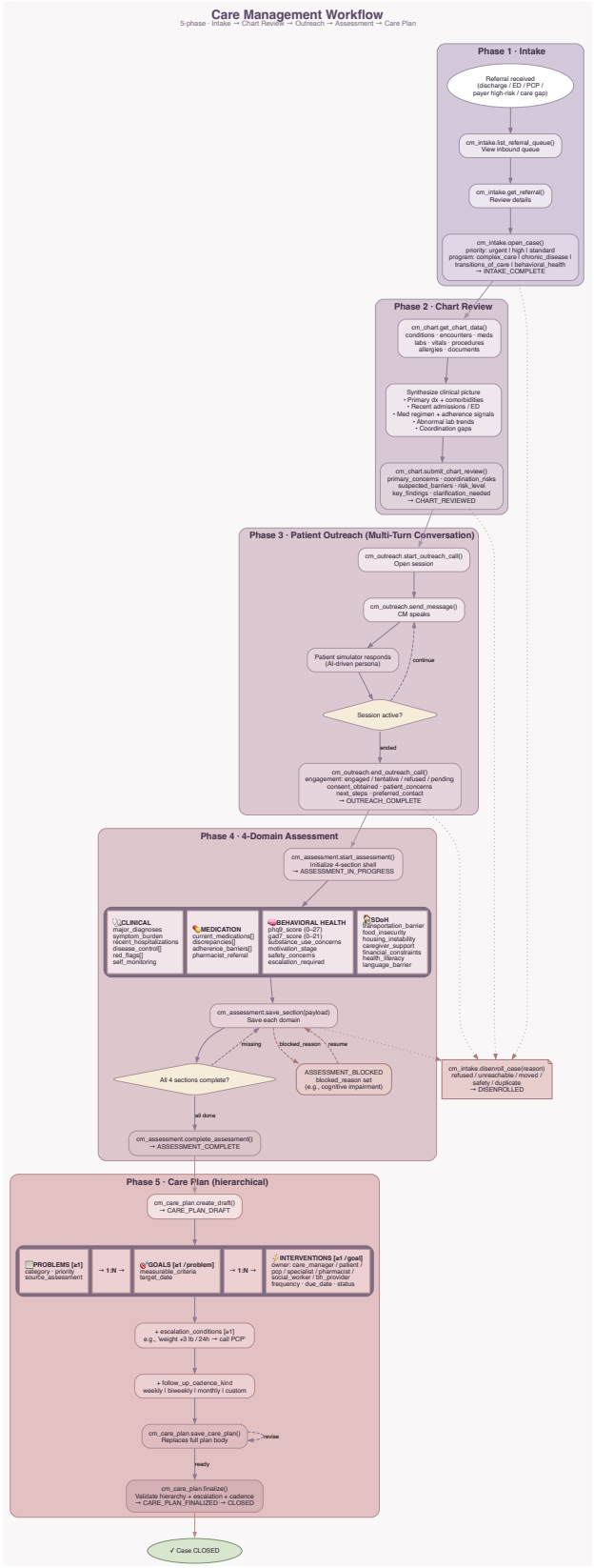

Figure 17: Care Management workflow. Five phases from intake to finalized care plan, with a four-domain assessment and a problems–goals–interventions care-plan hierarchy. Dotted edges to disenrollment can fire at any phase. Boxes are MCP tool calls listed in Table 8.

Table 6: Tool manifest: Provider PA agent (30 tools).

Tool	Description and representative call
chart	
search_patients	Search by name or member-id fragment; returns matching patient rows. *E.g.*, `search_patients(name="Khalil")`.
list_candidate_orders	Enumerate service-request orders on a patient chart, most-recent first. *E.g.*, `list_candidate_orders(patient_id)`.
get_patient_chart	Retrieve full chart slice: conditions, medications, encounters, prior auths, and documents. *E.g.*, `get_patient_chart(patient_id)`.
cases	
create_from_order	Open a draft PA case from a service request, minting a new order if needed. *E.g.*, `create_from_order(order_id)`.
get_case	Return full case with all artifacts, upstream review decisions, and recommended next actions. *E.g.*, `get_case("CASE-A1B2")`.
list_cases	List active PA cases as summary payloads; expedited cases sort above standard. *E.g.*, `list_cases()`.
inbox	
list_messages	Enumerate provider inbox messages ordered by date; filter by case. *E.g.*, `list_messages(case_id)`.
read_message	Retrieve a single message and stamp its read-at timestamp. *E.g.*, `read_message(msg_id)`.
docs	
create_submission_bundle	Assemble attached documents into a submission packet artifact. *E.g.*, `create_submission_bundle(case_id)`.
get_document	Return a single document reference with its body content. *E.g.*, `get_document(doc_id)`.
list_case_documents	List all documents linked to a case, including chart-seeded and uploaded evidence. *E.g.*, `list_case_documents(case_id)`.
upload_evidence	Attach text-body evidence to a case without a binary artifact. *E.g.*, `upload_evidence(case_id, body="...")`.
upload_evidence_file	Upload a binary file (PDF, image) as case evidence. *E.g.*, `upload_evidence_file(case_id, file)`.
forms	
list_required_forms	Return payer-required form schemas and per-form completion status. *E.g.*, `list_required_forms(case_id)`.
save_form_response	Persist questionnaire answers and report validation results. *E.g.*, `save_form_response(case_id, answers)`.
validate_form	Check form completeness against current submission state. *E.g.*, `validate_form(case_id)`.
auth	
check_status	Poll payer for current case status; returns new inbox messages and letters. *E.g.*, `check_status(case_id)`.
file_appeal	Submit an internal appeal against a denial; auto-evaluated in simulation. *E.g.*, `file_appeal(case_id, rationale)`.
file_external_appeal	File an external appeal after an internal appeal is upheld. *E.g.*, `file_external_appeal(case_id, rationale)`.
finalize_provider_determination	Record pre-submission disposition: ready to submit, gather more evidence, or do not submit. *E.g.*, `finalize(..., "ready_to_submit")`.
request_peer_to_peer	Always rejects; peer-to-peer is payer-initiated in this runtime. *E.g.*, `request_peer_to_peer(case_id)` $\rightarrow$ error.
respond_to_peer_to_peer	Accept a payer P2P invitation by supplying a note and preferred slot. *E.g.*, `respond_to_peer_to_peer(req_id, slot)`.
respond_to_pend	Reply to a pend with a note and optional attachments; advances status to resubmitted. *E.g.*, `respond_to_pend(case_id, note, docs)`.
submit_authorization	Submit the prepared case to the payer; advances status to submitted or resubmitted. *E.g.*, `submit_authorization(case_id)`.
p2p_session	
get_context	Return dialog state and ordered transcript of the active P2P session. *E.g.*, `get_context(request_id)`.
send_turn	Post one agent turn and receive the simulated counterpart reply. *E.g.*, `send_turn(req_id, text="...")`.
submit_summary	Conclude the P2P dialog with a written outcome summary. *E.g.*, `submit_summary(req_id, outcome)`.
people	

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Tool	Description and representative call
`list_contacts`	List all directory contacts: provider staff, admin, and payer-side contacts. *E.g.*, `list_contacts()`.
`list_ordering_providers`	List clinicians eligible as ordering practitioners. *E.g.*, `list_ordering_providers()`.
`list_reviewer_contacts`	List payer-side UM nurses and physician reviewers reachable from the provider. *E.g.*, `list_reviewer_contacts()`.

Table 7: Tool manifest: Payer UM agent (41 tools).

Tool	Description and representative call
`payer_intake_hub`	
`list_intake_queue`	List all pending intake cases with workflow state. *E.g.*, `list_intake_queue()`.
`get_intake_case`	Retrieve the intake case linked to a PA case. *E.g.*, `get_intake_case(case_id)`.
`create_intake_case`	Create a new intake record, or return the existing one for the case. *E.g.*, `create_intake_case(case_id)`.
`save_auth_header`	Write authorization-header fields (urgency, dates, place of service). *E.g.*, `save_auth_header(case_id, urgency="routine")`.
`save_diagnoses`	Replace the diagnosis list and optionally the diagnosis-type header. *E.g.*, `save_diagnoses(case_id, ["C34.11"])`.
`save_requesting_provider`	Replace the requesting-provider block with updated NPI and details. *E.g.*, `save_requesting_provider(case_id, npi)`.
`add_line_item`	Append a procedure line item; auto-assigns line number when unset. *E.g.*, `add_line_item(case_id, cpt="78816")`.
`update_line_item`	Replace the line item identified by line number. *E.g.*, `update_line_item(case_id, 1, qty=2)`.
`remove_line_item`	Drop the line item matching the given line number. *E.g.*, `remove_line_item(case_id, 1)`.
`list_documents`	List workspace artifacts for an intake case (provider-submitted and payer-generated). *E.g.*, `list_documents(case_id)`.
`confirm_decisions`	Freeze the four intake-level decisions and mark intake complete. *E.g.*, `confirm_decisions(case_id, ...)`.
`triage`	
`get`	Return intake case plus triage record (null if not yet committed). *E.g.*, `triage.get(case_id)`.
`calculate_sla`	Compute the SLA response deadline from urgency and receipt time. *E.g.*, `calculate_sla("routine", receipt_dt)`.
`check_gold_card`	Check whether the requesting provider qualifies for gold-card auto-approval. *E.g.*, `check_gold_card(case_id)`.
`set_disposition`	Commit triage decision: urgency, SLA deadline, and review lane. *E.g.*, `set_disposition(case_id, lane="nurse")`.
`route_case`	Assign the case to the appropriate review lane based on policy or override. *E.g.*, `route_case(case_id, lane="nurse_review")`.
`review`	
`get_nurse_review_data`	Load full nurse-review workspace: intake handoff, triage, per-line records, and documents. *E.g.*, `get_nurse_review_data(case_id)`.
`save_criteria_evaluation`	Create or update the per-line nurse criteria evaluation. *E.g.*, `save_criteria_evaluation(case_id, line=1, met=True)`.
`submit_nurse_case_recommendation`	Submit the case-level nurse recommendation; cannot be amended. *E.g.*, `submit_nurse_recommendation(case_id, "approve")`.
`get_md_review_data`	Load full MD-review workspace: intake, triage, nurse review, P2P requests, and documents. *E.g.*, `get_md_review_data(case_id)`.
`save_md_review`	Create or update the per-line MD criteria evaluation. *E.g.*, `save_md_review(case_id, line=1, ...)`.
`submit_md_case_decision`	Submit the case-level MD decision; cannot be amended. *E.g.*, `submit_md_decision(case_id, "deny")`.
`determination`	
`get_summary`	Aggregate review state into a ready-to-finalize summary with criteria tallies and documentation gaps. *E.g.*, `get_summary(case_id)`.

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Tool	Description and representative call
`finalize`	Persist the determination record; transitions case to approved, denied, or partially approved. *E.g.*, `finalize(case_id, "denied")`.
p2p	
`list`	List peer-to-peer requests; optionally filtered by case. *E.g.*, `p2p.list(case_id)`.
`offer_slots`	Propose meeting slots on a P2P request and assign the payer-side reviewer. *E.g.*, `offer_slots(req_id, slots)`.
`schedule`	Confirm a P2P appointment time and publish the calendar invite. *E.g.*, `schedule(req_id, slot)`.
`record_outcome`	Record the adjudicated outcome and close the P2P request. *E.g.*, `record_outcome(req_id, "approved")`.
p2p_session	
`get_context`	Return dialog state and transcript of the active P2P session for a request. *E.g.*, `get_context(req_id)`.
`send_turn`	Post one agent turn into the P2P dialog and return the simulated reply. *E.g.*, `send_turn(req_id, text="...")`.
`submit_summary`	Conclude the P2P dialog with a written summary and proposed outcome. *E.g.*, `submit_summary(req_id, outcome)`.
payer_letter_center	
`list`	List outbound letters for a case or across the entire payer world. *E.g.*, `letter_center.list(case_id)`.
`get`	Retrieve a single letter by ID. *E.g.*, `letter_center.get(letter_id)`.
`generate_approval`	Compose an approval determination letter for a case. *E.g.*, `generate_approval(case_id)`.
`generate_partial_approval`	Compose a partial-approval letter with mixed per-line outcomes. *E.g.*, `generate_partial_approval(case_id)`.
`generate_denial`	Compose a denial letter with policy basis and appeal rights. *E.g.*, `generate_denial(case_id)`.
`generate_pend`	Compose a pend (request-for-information) letter specifying missing documents. *E.g.*, `generate_pend(case_id)`.
`generate_notification`	Generate a member-facing notification addressed to the patient. *E.g.*, `generate_notification(case_id)`.
`deliver`	Mark a composed letter as delivered on a channel; copy it to the shared workspace. *E.g.*, `deliver(letter_id, channel="fax")`.
`reissue`	Reissue a letter under a new ID with the original body plus an appended amendment. *E.g.*, `reissue(letter_id, reason="...")`.
`audit_completeness`	Validate that a letter contains every field required for its type before delivery. *E.g.*, `audit_completeness(letter_id)`.

Table 8: Tool manifest: Care Management agent (16 tools).

Tool	Description and representative call
cm_intake	
`list_referral_queue`	List all inbound CM referrals with patient demographics. *E.g.*, `list_referral_queue()`.
`get_referral`	Retrieve a single referral enriched with patient demographics and risk flags. *E.g.*, `get_referral(referral_id)`.
`open_case`	Create a CMCase from an existing referral; transitions from new to intake_complete. *E.g.*, `open_case(referral_id)`.
`disenroll_case`	Disenroll a CMCase to the terminal disenrolled state with a documented reason. *E.g.*, `disenroll_case(case_id, reason)`.
cm_chart	
`get_chart_data`	Return the clinical chart for the CM patient: conditions, encounters, meds, labs, and documents. *E.g.*, `get_chart_data(case_id)`.
`submit_chart_review`	Record the care manager's structured chart review; advances state to chart_reviewed. *E.g.*, `submit_chart_review(case_id, notes)`.
cm_outreach	
`start_outreach_call`	Open a text-mode outreach conversation session with the patient. *E.g.*, `start_outreach_call(case_id)`.
`send_message`	Post one care-manager turn and receive the simulated patient reply. *E.g.*, `send_message(session_id, text="...")`.

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Tool	Description and representative call
`get_transcript`	Return the text transcript of an outreach conversation. *E.g.*, `get_transcript(session_id)`.
`end_outreach_call`	Close the session and persist the structured outreach record with outcome summary. *E.g.*, `end_outreach_call(session_id, summary)`.
`cm_assessment`	
`start_assessment`	Create an empty assessment shell; moves case from outreach_complete to assessment_in_progress. *E.g.*, `start_assessment(case_id)`.
`save_section`	Save one section of the assessment (replaces the named section body in full). *E.g.*, `save_section(asmt_id, "clinical", body)`.
`complete_assessment`	Mark the assessment complete; moves case to assessment_complete. *E.g.*, `complete_assessment(asmt_id)`.
`cm_care_plan`	
`create_draft`	Create an empty care-plan draft (1:1 with case); moves case to care_plan_draft. *E.g.*, `create_draft(case_id)`.
`save_care_plan`	Replace the body of the care-plan draft with updated problem-goal-intervention content. *E.g.*, `save_care_plan(plan_id, body)`.
`finalize`	Finalize the care plan and close the CMCase atomically. *E.g.*, `finalize(plan_id)`.

C.3 Handbook Detail

The χ-World Managed-Care Operations Handbook is a structured reference library loaded as a skill into each agent's context. It is organized into five sections: `provider-pa` (PA submission workflows and documentation standards), `payer-um` (utilization management clinical-review and correspondence protocols), `care-manager` (care management assessment and care-plan standards), `platform` (walkthrough tutorials for each server's tools), and `medical-library` (clinical guidelines, medical policies, pharmacy PA requirements, and utilization-review criteria).

The handbook defines a typed document taxonomy shared across all workflow stages. Provider-submitted documents cover eight kinds, payer-generated correspondence covers three, and care management produces two structured record types. Table 9 summarizes the PA document taxonomy; Table 10 covers care management records.

Provenance and licensing. The handbook was developed in collaboration with clinicians and operations leaders at Johns Hopkins Medicine to ensure clinical fidelity and alignment with real-world workflows. All documents fall into one of three categories: (i) *original content* authored specifically for this work by the clinical team, including workflow chapters, role playbooks, and form templates; (ii) *evidence-based policy documents* whose clinical criteria reflect standard-of-care guidelines (e.g., NCCN, ACC/AHA, ACOG) and publicly available coverage logic from commercial payers, restructured into a uniform format original to this work; and (iii) *synthetic policies* that mirror the format and decision logic of real payer criteria for procedures not covered by category (ii). No content from proprietary clinical-criteria databases (e.g., MCG, InterQual, Milliman) is included. The operational workflow chapters reflect generalized managed-care processes rather than verbatim institutional procedures from any single organization.

The handbook is available under a research Data Use Agreement (DUA) that permits non-commercial research use with attribution and prohibits redistribution of the original documents. Access requests are reviewed through our project page: `https://actava.ai/benchmarks`.

D χ-Bench Detail

D.1 Task Construction Details

Difficulty definitions. Each domain uses a structurally grounded difficulty signal. *Provider PA* difficulty is the count of distinct document kinds the submission packet must carry to satisfy the payer-published per-procedure document list: Easy = 0 required document kinds, Moderate = 1–3, Hard = ≥ 4. *Payer UM* difficulty is the number of clinical criteria along the chosen path through the payer's policy decision tree: Easy = ≤ 5 criteria, Moderate = 6–9, Hard = ≥ 10. *Care management*

Table 9: PA document taxonomy: eight provider-submitted kinds and three payer-generated kinds.

Kind	Document name	Key content / example
Provider-submitted		
`prior-auth-request-form`	PA Request Form	Payer-specific structured form capturing member ID, requesting provider NPI, procedure codes, diagnosis codes, and requested service dates.
`chart-note`	Chart Note	Physician or NP encounter documentation: chief complaint, history, exam findings, assessment, and plan, with ICD-10 and CPT codes.
`progress-note`	Progress Note	Interval visit or treatment summary documenting response to therapy and clinical trajectory since last encounter.
`letter-of-medical-necessity`	Letter of Medical Necessity	Attending-physician attestation of clinical indication, failure of conservative alternatives, and expected outcome.
`imaging-report`	Imaging Report	Radiologist interpretation of CT, MRI, PET, or X-ray study with impression and incidental findings.
`lab-summary`	Lab Summary	Tabulated laboratory results (HbA1c, CBC, CMP, etc.) with reference ranges and trend annotations.
`questionnaire-form`	PA Questionnaire	Payer-required question set capturing clinical criteria answers specific to the requested procedure.
`site-of-service-attestation`	Site-of-Service Attestation	Provider declaration that the clinical setting is appropriate (inpatient, outpatient, home) and meets safety requirements.
Payer-generated		
`approval-letter`	Approval Determination	Authorizes the requested procedure with approved CPT codes, authorization number, validity window, and conditions of approval.
`denial-letter`	Denial Determination	Documents which criteria were not met, the policy basis (handbook medical-policy section reference), and member and provider appeal rights with deadlines.
`info-request-letter`	Additional Information Request	Lists specific missing documents or clinical data required to continue review, with a response deadline.

difficulty maps one-to-one to the patient persona's consent profile: *Engaged* = Easy, *Hesitant* = Moderate, *Refusing* = Hard.

Task Generation Pipeline. Step 1 of Section 3.2.2 is implemented as a rejection-sampling loop around an LLM agent generator (Claude Opus 4.7 with JSON-schema-constrained output). Each call is prompted with three items: (i) a serialized slice of the relevant system state graph (the payer policy decision tree for PA/UM, the NANDA-I / NIC / NOC (NNN) model for CM); (ii) the matching section of the *Managed-Care Operations Handbook*; and (iii) a structured schema fixing the artefact format. The generator is forbidden from emitting any field that is not anchored to a citation in (i) or (ii); each rubric line and each required chart fact carries an explicit pointer back to a policy section or NNN node, so a candidate that omits citations is rejected at parse time.

Acceptance criteria A candidate must pass three gates before it is queued for human review. **(G1)** every expected action and rubric line resolves to a cited section or NNN node. **(G2)** chart and rubric mutually entail and the chosen path does not appear verbatim in chart prose, so the rubric cannot be passed by surface paraphrase. **(G3)** a deterministic dry-run executes a reference solution to a single well-formed terminal state, with reference-solution length, decision-tree depth, and irreversible-commit count falling inside one of the three difficulty bands. G1–G2 failures are looped back to the generator with the failed-gate diagnostic, up to three retries; G3 failures are discarded.

Table 10: Care management record taxonomy: four mandatory assessment sections and the care-plan structure.

Record type	Section / Kind	Required elements / example
CMAssessment (4 mandatory sections)		
`cm_assessment`	Clinical	Active diagnoses (ICD-10), disease-control status per condition (e.g., HbA1c for DM, LVEF for HF), hospitalization count (12 mo), functional status (ADL/I-ADL), red flags, and self-monitoring capability.
`cm_assessment`	Medication	Full medication list (Rx, OTC, supplements) reconciled against chart, adherence barriers, high-risk medications flagged (anticoagulants, opioids, insulin), and pharmacist-referral criteria.
`cm_assessment`	Behavioral Health	Validated screening scores (PHQ-9, GAD-7, C-SSRS), crisis-risk level, current BH treatment status, substance use (AUDIT-C), cognitive screen (MoCA), and social-connectedness assessment.
`cm_assessment`	SDoH	Housing stability, food security (Hunger Vital Sign), transportation, financial strain, health literacy, caregiver burden, and social isolation indicators.
CMCarePlan (problem–goal–intervention hierarchy)		
`cm_care_plan`	Problem statement	Each problem references an assessment finding; includes ICD-10/SNOMED code and member-specific evidence. *E.g.*, "*Uncontrolled PTSD (F43.10): passive suicidal ideation, severe insomnia, social isolation*".
`cm_care_plan`	SMART goal	Measurable, time-bound target tied to a clinical finding. *E.g.*, "*PHQ-9 $\leq$ 9 by 2026-08-01; patient self-reports improved sleep 3+ nights/week*".
`cm_care_plan`	Intervention	Assigned owner, action, and timeline; links to referral, scheduling, or education resource. *E.g.*, "*CM to coordinate trauma-informed therapy intake within 14 days*".

Human review and clinician editing (Step 3 in Section 3.2.2). Outsourced licensed clinicians review and edit each surviving candidate; chart authors work *blinded* to the per-stage ground-truth bundle to prevent phrasing overlap that would let the rubric grade prose rather than evidence. Reviewers re-couple chart and rubric during review and rewrite any field whose wording would shortcut the policy-navigation task. The same review pass also runs the residual-PHI scan and the clinical-realism check ("would a real reviewer encounter this case?"); a candidate that fails either is sent back for rewrite or, on a second failure, dropped.

Validation Composition. The review panel for each task combines (i) at least one practicing healthcare worker with the credential appropriate to the role under review (RN for nurse-review and CM rubrics, MD for physician-review and P2P rubrics, RN/MD for PA documentation rubrics), (ii) a separate clinician operating the χ-World UI to confirm tractability under the role's tool surface, and (iii) five paper authors covering the policy-text, simulator-state, and rubric-coverage perspectives. Chart authors are kept *blinded* to the per-stage GT bundle so chart prose cannot mirror rubric phrasing; the panel re-couples chart and rubric only at sign-off. Disagreements are resolved by majority vote; clinical disagreements that survive a vote escalate to a board-certified arbiter. A formal per-task human inter-rater statistic is not reported here because reviewer-level verdicts were logged only as accept/reject decisions rather than per-rubric Likert votes.

D.2 Prior Authorization and Utilization Management

Clinical Categories. Payer UM tasks span five high-level clinical categories chosen to exercise a breadth of policy types that routinely generate prior-authorization requests in commercial managed care, while keeping each category anchored to concrete, widely-recognized procedures that allow policy lookup to be unambiguous. Surgical procedures are the largest category (12 of 25): surgical policies tend to have multi-page, multi-criterion decision trees that stress-test the full policy-navigation loop. Imaging (4) and diagnostic monitoring (4) contribute cases requiring familiarity with evidence-based screening and diagnostic thresholds. Specialty pharmacy (2) tests drug-authorization criteria, which have a different policy structure (clinical condition + drug-specific criteria). The *Other* category (3) covers procedures that do not fit neatly into the above bins. Each category is grounded in one to three specific CPT/HCPCS-indexed procedures drawn from real payer medical policies; see Table 12 for the per-category task counts.

Provider PA tasks are categorized by their terminal status rather than clinical procedure, because the benchmark focuses on the provider's documentation and submission behavior rather than the clinical determination itself. Three terminal states are covered: *Submitted to payer* (8), where the agent assembles and submits a complete packet; *Returned for docs* (12), where the submission is incomplete and the agent must identify and retrieve the missing document kinds; *Gather more evidence* (5), where the clinical record lacks information needed to satisfy the payer's coverage criteria and the agent must coordinate with the ordering clinician before a formal submission is possible. This tripartite structure is designed to test qualitatively distinct behaviors: documentation completeness reasoning, document retrieval and re-submission, and upstream clinical information management.

Policy Criteria Structure (Payer UM). Each UM task is grounded in a specific payer medical policy and encodes the set of criteria applicable to the requested service. Each criterion is a triplet: (*i*) **criterion text**, the natural-language condition transcribed verbatim from the policy (e.g., "Individual has reasonable potential for meaningful functional progress within the requested timeframe"); (*ii*) **policy citation**, the exact section identifier and version the agent must name when recording its evaluation (e.g., *Outpatient Rehabilitation Medical Policy MP-2026.1, §4.3*); and (*iii*) **evidence items**, chart-specific facts that demonstrate how the criterion is (or is not) met for this particular patient case.

The citation requirement is a core design choice: the LLM judge cannot award a positive verdict for clinically plausible reasoning that fails to reference the applicable policy section by name. This prevents agents from substituting general clinical knowledge for the actual policy-compliance task. Criteria are organized by stage in the policy decision tree: nurse review criteria typically assess documentation completeness and surface medical necessity; physician review criteria require multi-condition clinical judgment; P2P criteria evaluate whether new arguments from the attending physician justify reversing the prior determination. Each criterion has a rollup rule (*all items* or *any item*) that mirrors the policy's own aggregation logic.

Difficulty tracks the length of the applicable policy path. Easy tasks ($\leq$5 criteria) correspond to routing decisions that resolve on administrative grounds or minimal documentation checks. Moderate tasks (6–9 criteria) cover standard single-procedure reviews with complete documentation and medical-necessity evaluation. Hard tasks ($\geq$10 criteria) arise from multi-page surgical policies or cases with comorbidity exclusion checks, cumulative evidence requirements, and multiple decision nodes.

Document Status Classification (Provider PA). Each PA task encodes the payer handbook's per-procedure document requirement as a list of *required document kinds* (e.g., `chart_note`, `imaging_report`, `progress_note`). Before submission, the agent must determine the status of each required kind in the patient's document repository. Documents fall into one of three status categories that determine the correct agent behavior.

Present and sufficient: a document of the required kind exists and its content satisfies the policy's minimum content criteria. The agent should attach it to the submission packet without modification. **Present but insufficient**: a document of the required kind exists, but its content fails the policy's requirements (e.g., a therapy evaluation note that lacks standardized assessment scores or measurable functional goals). The agent must identify the insufficiency, communicate the gap to the ordering provider, and obtain a supplementary document or addendum before submission can proceed. **Absent**: no document of the required kind exists in the repository. The agent cannot assemble a valid

submission packet and must coordinate with the ordering clinician to gather the missing clinical evidence upstream.

The three terminal states of Provider PA tasks are the direct outcome of this classification applied across the full required-document set. *Submitted to payer* arises when all required kinds are present and sufficient. *Returned for docs* arises when at least one required kind is absent or insufficient: the agent identifies the gap, retrieves or requests the missing documentation, and resubmits. *Gather more evidence* arises when the clinical record itself lacks information needed to satisfy the policy's coverage criteria: the agent must engage the ordering clinician to generate the missing evidence before any submission is possible. This design ensures the three terminal states are not just outcome labels but reflect genuinely different reasoning demands: coverage completeness verification, document retrieval and re-submission, and upstream clinical information management.

Workflow Design. UM tasks are seeded at one of the five pickup stages rather than uniformly at Intake, testing agents' ability to pick up a case mid-workflow, as happens when cases are reassigned to a new reviewer or escalated after hours. The initial-stage distribution (Nurse review 8, Intake 5, Physician review 4, Triage 4, Peer-to-peer 4) is weighted toward more complex review stages to concentrate evaluations on clinically demanding decisions. All four terminal outcomes (Approved (9), Denied (10), Pended (5), Coverage not required (1)) are represented, preventing any bias toward a single determination type. The single *Coverage not required* task tests whether the agent recognizes and documents a case that should not have entered the UM queue.

Tasks traverse two to five review stages ($n_{\text{stages}} \in \{2, 3, 4, 5\}$), capturing both direct determinations (2 stages: review + letter delivery) and fully escalated chains (5 stages: intake through peer-to-peer). The number of clinical criteria along the policy decision path ranges from 0 to 50 (mean 9.2), where zero-criteria cases correspond to non-clinical routing decisions (e.g., coverage not required at triage) and high-criteria cases reflect complex multi-page surgical policies.

Table 12 cross-tabulates initial stage against terminal state. The *Gather more evidence* column is absent from the UM table because that terminal state applies only to Provider PA tasks.

Peer-to-Peer Simulation. Four UM tasks (16%) are seeded at the Peer-to-peer stage. This stage is operationally distinct: the payer has already issued a preliminary denial, the requesting provider has contested it and requested a clinical peer review, and the agent must now defend or revise its determination in a live multi-turn call against the provider's attending physician. The clinical counterpart, played by the simulation environment, advocates for the provider's position using chart-grounded arguments and may surface clinical details not foregrounded in the prior review.

The design goal is to test sustained adversarial clinical reasoning: the agent must maintain its determination when the counterpart's arguments do not reach the policy bar, and must be willing to revise it when compelling new evidence is presented. All four P2P tasks resolve to either Approved (2) or Denied (2); a Pended outcome is structurally excluded because the call must resolve the case.

Evaluation Design. UM and PA tasks are scored by a two-layer verifier. The **deterministic layer** reads the simulator's persisted world state and event log and checks: (*i*) terminal case status; (*ii*) routing and review-lane assignments; (*iii*) per-stage structured payloads, including criterion-by-criterion evaluations, decision-and-rationale fields, completeness flags; (*iv*) required side-effect artifacts (determination letters, information-request notices); and (*v*) mutation scope, where writes to cases or patients outside the task's designated scope count as failures. The **LLM-judge layer** reads the same persisted records and grades, per workflow stage: whether clinical reasoning is grounded in policy sections cited by name; whether criterion evaluations are internally consistent and align with the chart evidence; and whether the terminal determination follows from the recorded rationale. All judge rubrics inline the specific policy-section citations they require, so that a verdict cannot be awarded by plausible-sounding language alone. Both layers must return a positive verdict for a trial to count as a pass.

D.3 Care Management

Workflow Design. The care management workflow follows the five-phase RN case management process used in commercial chronic-disease and complex-care programs (cf. Figure 17): (*i*) **Intake**: confirm the referral, verify program eligibility, assign a risk tier, and open the case; (*ii*) **Chart review**:

Table 11: Provider PA task distribution: terminal status × difficulty.

Terminal status	Easy	Moderate	Hard	Total
Submitted to payer	2	4	2	8
Returned for docs	0	9	3	12
Gather more evidence	5	0	0	5
Total	**7**	**13**	**5**	**25**

Table 12: Payer UM task distribution: initial stage × terminal state. Numbers in each cell are task counts.

Initial stage	Approved	Pended	Denied	Not covered	Total
Intake	1	1	2	1	5
Triage	2	1	1	0	4
Nurse review	2	2	4	0	8
Physician review	2	1	1	0	4
Peer-to-peer	2	0	2	0	4
Total	**9**	**5**	**10**	**1**	**25**

extract diagnoses, lab trends, medication history, recent encounters, and social determinants of health from the longitudinal record; (*iii*) **Outreach**: contact the patient by phone, establish rapport, confirm consent, and conduct a structured assessment interview; (*iv*) **Formal assessment**: administer the four-domain assessment instruments (see Table 10 for the full instrument set) grounded in the chart and the outreach transcript; (*v*) **Care plan authoring and closure**: document NANDA-I nursing diagnoses, NOC outcome targets, and NIC interventions grounded in chart findings and assessment results, then either finalize the plan or close the case. Disenrollment is a side path that may fire at any earlier phase when the patient refuses, is unreachable, or a safety concern surfaces.

The outreach stage is multi-turn by design because real patient engagement is a dynamic conversation: the agent must adapt its communication style to the patient's engagement profile, manage resistance or avoidance without coercion, and earn clinical disclosures incrementally. The patient persona's internal state (consent triggers, anti-triggers, and trust-graded disclosure pacing) is opaque to the agent; it must infer the appropriate approach from the patient's responses, preventing script-passing through a fixed question sequence.

Care plans are required to follow the NANDA-I/NOC/NIC (NNN) framework because this is the documentation standard for RN care management in U.S. managed-care programs. Each task is designed so that the ground-truth care plan requires a minimum of two NANDA-I diagnoses with corresponding NOC targets and NIC interventions, drawn from the per-condition NNN model.

Patient Simulation. Each patient persona is a structured contract with four interlocking components.

Consent triggers are explicit caller behaviors that advance the patient's trust state. Each trigger names a concrete communication act (e.g., checking for a private moment before raising sensitive topics; asking permission before using a diagnostic label; explicitly removing the trauma-story requirement as a precondition for enrollment). The LLM judge evaluates whether a given agent turn matches each trigger's natural-language specification. Design purpose: triggers make patient-centered technique a *functional* prerequisite. An agent applying standard clinical intake workflows without adapting to the patient's stated boundaries will not accumulate the required triggers and cannot obtain same-call consent.

Anti-triggers are behaviors that damage rapport, some permanently closing off consent or specific disclosure paths for the remainder of the call. Examples include starting a suicide screening instrument before establishing scope and permission, naming a diagnostic label after the patient has declined that framing, or treating "I can listen" as enrollment consent. Anti-triggers are specified in natural language with handbook section citations so the LLM judge can evaluate them independently of surface phrasing. Design purpose: asymmetric cost structure. For Refusing patients who require all

Provider PA: Hard / Returned for docs

Chart Snapshot

Patient	67M, 30 pack-year smoking history, newly diagnosed lung mass
Diagnosis	C34.11, NSCLC, right upper lobe (newly diagnosed)
Procedure	CPT 78816 (whole-body PET/CT, cancer staging)
Status	Bronchoscopy completed 2026-04-18; awaiting full staging workup

Documents in Repository *(required for CPT 78816: 4 document kinds)*

✓ Pulmonology consultation note (2026-04-20): NSCLC confirmed, PET/CT for staging requested
✓ CT chest/abdomen/pelvis (2026-04-12): 3.2 cm RUL mass with ipsilateral hilar adenopathy
✗ Pathology report (bronchoscopy 2026-04-18): present in repository but not included in packet
✗ Management impact statement: required by payer policy; absent from consultation note

The agent must identify both documentation gaps, attach the pathology report from the repository, request a management-impact addendum, and re-submit the completed packet. The deterministic verifier checks that all four required document kinds are present and that submission status has advanced.

Figure 18: Provider PA example task (Hard / Returned for docs). Top: patient chart snapshot. Bottom: handbook-required document list for CPT 78816 (whole-body PET/CT) with repository availability.

Payer UM: Moderate / Approved (picked up at Nurse review)

Chart Snapshot

Patient	52M, daytime sleepiness, snoring, witnessed apneas $\times$ 18 months
Diagnosis	G47.33 (obstructive sleep apnea, suspected moderate–severe)
Service	CPT 95810 (overnight in-lab polysomnography)
Findings	ESS 14/24 (threshold $\geq$10) · BMI 31.2 · no contraindications to in-lab study

Policy Criteria *(4 of 9 shown; the agent evaluates all 9)*

Criterion	Chart evidence	Verdict
Documented symptoms $\geq$30 days	18-month history (PCP note)	Met ✓
Epworth Sleepiness Scale $\geq$10	ESS 14/24 recorded	Met ✓
BMI $\geq$30 or neck circ. $>$17 in	BMI 31.2	Met ✓
No contraindications to in-lab study	No disqualifying comorbidity	Met ✓

The agent records per-criterion verdicts with policy-section citations, composes a determination, and delivers an approval letter. The task traverses 2 stages (nurse review + letter delivery); all 9 criteria and letter content are verified by the two-layer evaluator.

Figure 19: Payer UM example task (Moderate / Approved, picked up at Nurse review). Top: patient chart snapshot. Bottom: four of the nine applicable policy criteria with chart evidence and per-criterion verdicts.

T-rules to fire with zero A-rules, a single poorly-chosen phrase closes the consent path permanently. Volume of correct behaviors cannot compensate for one anti-trigger hit.

Consent rules specify the gating threshold: the minimum combination of triggers fired and anti-triggers avoided for the patient to grant same-call enrollment consent. Thresholds scale with difficulty: Engaged patients require 1–2 triggers; Refusing patients require all T-rules with zero A-rules. Without the consent flag set, the agent cannot administer any formal assessment instrument. Design purpose: model real informed-consent requirements and create a hard checkpoint the agent cannot bypass by proceeding to assessment prematurely.

Assessment disclosure rules govern which clinical information the patient will share during the outreach conversation, and when. Each sensitive domain (passive suicidal ideation, trauma narrative, social isolation, polypharmacy concerns) carries an independent gate. Before the patient will disclose on a gated topic, the caller must: (*i*) explain why this topic matters; (*ii*) acknowledge the patient's right to skip or stop; (*iii*) state any applicable safety limits. Bundled checklist questions, chart assumptions,

or three or more sensitive domains in one turn trigger partial withdrawal and deferral of non-gated items. Design purpose: create a gap between chart-available information and conversation-available information. An agent that reformats chart data without conducting a genuine interview will be missing required assessment fields, causing the deterministic verifier to fail the trial.

Patient engagement profiles span three consent stances (*Engaged* (3 tasks), *Hesitant* (7), and *Refusing* (15)), mapping one-to-one to Easy, Moderate, and Hard difficulty. The skew toward Refusing and Hesitant is intentional: fully cooperative patients provide little signal about engagement-strategy quality, whereas resistant patients force the agent to demonstrate adaptive communication and motivational-interviewing technique. The three Engaged tasks serve as a lower-bound calibration.

Primary conditions span 22 single-condition primaries and 3 multimorbidity scenarios across 8 clinical categories. Multimorbidity cases (ESRD with diabetes, heart failure with AFib and CKD, Parkinson's with depression) require the agent to coordinate care across multiple interacting conditions, each with its own NNN components and potential treatment-plan conflicts. The four program types (Chronic disease (11), Behavioral health (6), Complex care (4), Transitions of care (4)) map to distinct operational playbooks in the care-manager sub-skill.

Table 13 lists all 25 tasks with their primary condition, category, program type, consent profile, and difficulty.

Evaluation Design. CM tasks are scored by the same two-layer verifier as PA/UM, with adaptations for the nursing process. The **deterministic layer** checks: (*i*) forward-only state-machine progression through the five phases; (*ii*) the consent-obtained completeness flag, which must be set before any assessment instrument can be administered; (*iii*) care-plan structural completeness, where at least two NANDA-I diagnoses, each linked to NOC targets and NIC interventions, must be present; and (*iv*) field-level completeness for all administered structured instruments. The **LLM-judge layer** grades: (*i*) engagement appropriateness, i.e., whether the communication style matches the patient's consent profile and respects its resistance triggers; (*ii*) assessment grounding, i.e., whether each instrument response is traceable to chart evidence or to the outreach transcript; (*iii*) care-plan composition, i.e., whether each NNN entry is supported by chart or transcript evidence and aligns with the per-condition NNN model; and (*iv*) escalation judgment, i.e., whether red-flag conditions are correctly identified and routed.

E Task Catalog

This appendix enumerates all 75 tasks in χ-Bench, one row per task. PA Provider tasks are uniformly at the `referral` state (the agent assembles and submits the request). UM tasks span five states: `intake`, `triage`, `nurse_review`, `md_review`, and `p2p`. PA Provider and UM tasks sharing a service slug refer to the same underlying patient case observed at different workflow stages. Difficulty (Easy / Moderate / Hard, abbreviated E/M/H) follows the structurally grounded definitions in Appendix D: PA difficulty is the count of distinct document kinds the submission packet must carry; UM difficulty is the number of clinical criteria along the chosen path through the policy decision tree; CM difficulty maps one-to-one to the patient persona's consent profile.

E.1 PA: Prior Authorization (Provider Side)

Table 14: Prior Authorization (Provider) task catalog. All 25 tasks are invoked at the `referral` state.

Name (`service.state`)	**Category**	**Diff.**	**Short Description**
`sleeve_gastrectomy.referral`	Surgical	M	Submit PA packet for laparoscopic sleeve gastrectomy
`brca_testing.referral`	Other	M	Submit PA for hereditary-breast-cancer genetic testing
`cataract_iol.referral`	Surgical	E	Submit PA for cataract removal with IOL insertion
`pt_therapeutic_exercise.referral`	Diagnostic	M	Submit auth for outpatient PT therapeutic exercises

(Table 14 continued)

Name (`service.state`)	**Category**	**Diff.**	**Short Description**
`lap_hysterectomy.referral`	Surgical	E	Submit PA for laparoscopic total hysterectomy with BSO
`cardiac_event_monitor.referral`	Diagnostic	E	Submit PA for external cardiac event monitor
`myocardial_pet.referral`	Imaging	H	Submit PA for myocardial PET perfusion imaging
`uppp.referral`	Surgical	H	Submit PA for uvulopalatopharyngoplasty (UPPP)
`mri_brain_with_contrast.referral`	Imaging	M	Submit PA for MRI brain (without then with contrast)
`hypoglossal_nerve_stim.referral`	Surgical	H	Submit PA for hypoglossal nerve stimulator implant
`rotator_cuff_repair.referral`	Surgical	H	Submit PA for arthroscopic rotator cuff repair
`iv_infusion.referral`	Pharmacy	M	Submit PA for therapeutic IV infusion
`mri_brain.referral`	Imaging	M	Submit PA for MRI brain without contrast
`rituximab.referral`	Pharmacy	H	Submit PA for rituximab IV infusion
`screening_colonoscopy.referral`	Other	E	Submit PA for screening colonoscopy
`fess.referral`	Surgical	M	Submit PA for functional endoscopic sinus surgery
`polysomnography.referral`	Diagnostic	M	Submit PA for in-lab polysomnography (sleep study)
`lumbar_fusion.referral`	Surgical	M	Submit PA for posterior lumbar interbody fusion
`cheilectomy_mtp.referral`	Surgical	M	Submit PA for cheilectomy of first metatarsophalangeal joint
`rotator_cuff_repair_b.referral`	Surgical	M	Submit PA for arthroscopic rotator cuff repair (case B)
`cervical_disc_arthroplasty.referral`	Surgical	E	Submit PA for total cervical disc arthroplasty
`myoelectric_prosthesis.referral`	Other	M	Submit PA for myoelectric upper-extremity prosthesis
`vagus_nerve_stim.referral`	Surgical	M	Submit PA for vagus nerve stimulator implant
`veeg_monitoring.referral`	Diagnostic	E	Submit PA for inpatient video-EEG monitoring
`ct_abdomen_pelvis.referral`	Imaging	E	Submit PA for CT abdomen/pelvis with 3D reconstruction

E.2 UM: Utilization Management (Payer Side)

Table 15: Utilization Management (Payer) task catalog. State indicates the workflow stage at which the agent is invoked.

Name (`service.state`)	**Category**	**Diff.**	**Short Description**
`sleeve_gastrectomy.md_review`	Surgical	H	MD review and determination for sleeve gastrectomy
`brca_testing.intake`	Other	M	Intake check and routing for BRCA testing
`cataract_iol.triage`	Surgical	E	Triage and disposition for cataract surgery case
`pt_therapeutic_exercise.nurse_review`	Diagnostic	M	Nurse clinical review for outpatient PT auth
`lap_hysterectomy.intake`	Surgical	E	Intake check and routing for hysterectomy request
`cardiac_event_monitor.intake`	Diagnostic	E	Intake check and routing for cardiac event monitor

(Table 15 continued)

Name (`service.state`)	**Category**	**Diff.**	**Short Description**
`myocardial_pet.p2p`	Imaging	M	Conduct payer-side P2P for cardiac PET denial
`uppp.md_review`	Surgical	M	MD review and determination for UPPP
`mri_brain_with_contrast.md_review`	Imaging	H	MD review for MRI brain with contrast
`hypoglossal_nerve_stim.p2p`	Surgical	M	Conduct payer-side P2P for hypoglossal stim denial
`rotator_cuff_repair.md_review`	Surgical	H	MD review and determination for rotator cuff repair
`iv_infusion.nurse_review`	Pharmacy	H	Nurse clinical review for IV infusion auth
`mri_brain.triage`	Imaging	E	Triage and disposition for MRI brain case
`rituximab.nurse_review`	Pharmacy	M	Nurse clinical review for rituximab infusion
`screening_colonoscopy.nurse_review`	Other	H	Nurse clinical review for screening colonoscopy
`fess.nurse_review`	Surgical	H	Nurse clinical review for FESS
`polysomnography.nurse_review`	Diagnostic	M	Nurse clinical review for in-lab sleep study
`lumbar_fusion.nurse_review`	Surgical	E	Nurse clinical review for lumbar fusion
`cheilectomy_mtp.triage`	Surgical	M	Triage and disposition for cheilectomy case
`rotator_cuff_repair_b.p2p`	Surgical	E	Conduct payer-side P2P for rotator cuff repair (case B)
`cervical_disc_arthroplasty.nurse_review`	Surgical	E	Nurse clinical review for cervical disc arthroplasty
`myoelectric_prosthesis.triage`	Other	H	Triage and disposition for myoelectric prosthesis
`vagus_nerve_stim.intake`	Surgical	M	Intake check and routing for VNS implant request
`veeg_monitoring.intake`	Diagnostic	E	Intake check and routing for vEEG monitoring
`ct_abdomen_pelvis.p2p`	Imaging	E	Conduct payer-side P2P for CT abd/pelvis denial

E.3 CM: Care Management

Table 16: Care Management task catalog. Each task is a longitudinal patient case; difficulty is set by the persona's consent profile (Engaged = E, Hesitant = M, Refusing = H).

Name	**Category**	**Diff.**	**Short Description**
`cm_afib_moderate_anxious_001`	Cardiovascular	M	Manage AFib; Hesitant persona
`cm_anorexia_hard_refuses_001`	Behavioral	H	Manage anorexia nervosa; Refusing persona
`cm_asthma_low_coop_001`	Respiratory	E	Manage asthma; Engaged persona
`cm_ckd_moderate_anxious_001`	Renal	M	Manage CKD; Hesitant persona
`cm_complex_esrd_dm_hard_refuses_001`	Multimorbidity	H	Manage ESRD + DM; Refusing persona
`cm_complex_hf_afib_ckd_hard_refuses_001`	Multimorbidity	H	Manage HF + AFib + CKD; Refusing persona
`cm_complex_parkinson_dep_moderate_tentative_001`	Multimorbidity	M	Manage Parkinson + depression; Hesitant persona
`cm_copd_hard_refuses_002`	Respiratory	H	Manage COPD; Refusing persona
`cm_dementia_hard_refuses_001`	Neurological	H	Manage early-onset dementia; Refusing persona

(Table 16 continued)

Name	Category	Diff.	Short Description
cm_dm_hard_refuses_002	Metabolic	H	Manage type-2 diabetes; Refusing persona
cm_dm_low_coop_001	Metabolic	E	Manage type-2 diabetes; Engaged persona
cm_dm_moderate_anxious_001	Metabolic	M	Manage type-2 diabetes; Hesitant persona
cm_hf_hard_refuses_002	Cardiovascular	H	Manage heart failure; Refusing persona
cm_htn_low_tentative_001	Cardiovascular	E	Manage hypertension; Engaged persona
cm_mdd_hard_refuses_002	Behavioral	H	Manage major depression; Refusing persona
cm_mdd_moderate_reluctant_001	Behavioral	M	Manage major depression; Hesitant persona
cm_metabolic_syndrome_hard_refuses_001	Metabolic	H	Manage metabolic syndrome; Refusing persona
cm_parkinson_hard_refuses_001	Neurological	H	Manage Parkinson's disease; Refusing persona
cm_post_hip_moderate_anxious_001	Post-acute	M	Post-hip-ORIF transition of care; Hesitant persona
cm_post_mi_hard_refuses_002	Cardiovascular	H	Post-MI transition of care; Refusing persona
cm_post_pna_moderate_tentative_001	Post-acute	M	Post-pneumonia transition of care; Hesitant persona
cm_post_stroke_hard_refuses_001	Cardiovascular	H	Post-stroke transition of care; Refusing persona
cm_ptsd_hard_refuses_001	Behavioral	H	Manage PTSD; Refusing persona
cm_schizo_hard_refuses_001	Behavioral	H	Manage schizophrenia; Refusing persona
cm_sud_hard_refuses_001	Behavioral	H	Manage substance-use disorder; Refusing persona

F Experiment Detail

F.1 Sandbox Environment and Trial Orchestrator

Each trial in χ-Bench is materialized as a containerized sandbox bound to a single task, layered over a *shared χ-World*: one fixed population of data of all 25 tasks in the domain. The sandbox[3] carries its own per-trial database, file system, and tool-server processes, and is shut down after the trial returns so no state is shared across trials. At trial start, the χ-World backend, the role-scoped MCP servers [2], and the agent process are brought up in parallel; the simulator composes the shared world with the task manifest to pin a specific task to its initial state. The agent communicates with the simulator exclusively through MCP over streamable HTTP. *Harbor* [17, 33] sits between the two as the trial orchestrator: it spawns and tears down the sandbox, wires up the agent–simulator MCP channel, records the agent's step trajectory, and enforces the per-trial wall-clock and retry budget.

F.2 Pass@*k* and pass^*k* Metrics

For each task we run $n = 3$ independent trials and let $c \in \{0, \ldots, n\}$ count the passes. From the same pool we compute task-level pass@k and pass^k:

$$\text{pass@}k \;=\; \mathbb{E}\left[\, 1 - \binom{n-c}{k} \Big/ \binom{n}{k} \right], \qquad \text{pass}^k \;=\; \mathbb{E}\left[\binom{c}{k} \Big/ \binom{n}{k} \right].$$

pass^3 is the operationally meaningful reliability number, since long-horizon healthcare workflows must succeed on every individual case rather than admit a single successful sample. Setting $n{=}k{=}3$

[3] During experiments, we hosted the sandbox on Modal [34].

Table 13: Care management task breakdown by primary condition. Each row is one task. Consent profile maps directly to difficulty: Engaged = Easy, Hesitant = Moderate, Refusing = Hard.

Category	Primary condition	Program	Consent	Diff.
Behavioral health	Anorexia	Behavioral health	Refusing	H
	Major depressive disorder	Behavioral health	Refusing	H
	Major depressive disorder	Behavioral health	Hesitant	M
	PTSD	Behavioral health	Refusing	H
	Schizophrenia	Behavioral health	Refusing	H
	Substance use disorder	Behavioral health	Refusing	H
Cardiovascular	Atrial fibrillation	Chronic disease	Hesitant	M
	Heart failure	Chronic disease	Refusing	H
	Hypertension	Chronic disease	Engaged	E
	Post-MI	Transitions of care	Refusing	H
	Post-stroke	Transitions of care	Refusing	H
Metabolic	Diabetes mellitus	Chronic disease	Refusing	H
	Diabetes mellitus	Chronic disease	Engaged	E
	Diabetes mellitus	Chronic disease	Hesitant	M
	Metabolic syndrome	Complex care	Refusing	H
Multimorbidity	ESRD + diabetes	Complex care	Refusing	H
	HF + AFib + CKD	Complex care	Refusing	H
	Parkinson + depression	Complex care	Hesitant	M
Neurological	Dementia	Chronic disease	Refusing	H
	Parkinson's disease	Chronic disease	Refusing	H
Post-acute	Post-hip surgery	Transitions of care	Hesitant	M
	Post-pneumonia	Transitions of care	Hesitant	M
Renal	Chronic kidney disease	Chronic disease	Hesitant	M
Respiratory	Asthma	Chronic disease	Engaged	E
	COPD	Chronic disease	Refusing	H

E = Easy, *M* = Moderate, *H* = Hard.

keeps pass^3 a defined reliability signal at a tractable budget: 30 cells $\times$ 75 tasks $\times$ 3 trials $= 6{,}750$ trials per benchmark pass.

F.3 Judge, Container, and Harness Configuration

The rubric judge is Claude Opus 4.7, capped at a 1200-second wall-clock and 128K output tokens per rubric. Each rubric is judged by $V = 3$ independent votes that are aggregated under a strict-majority quorum (ties resolved as fail); inter-rater agreement statistics computed on this V-fold voting record are reported below. Per-trial container resources are identical across all (harness, model) cells: 2 CPU cores, 4 GB RAM, 10 GB ephemeral disk. For all tasks, we set an agent wall-clock cap of 1800 s, which is roughly twice the P99 of wall-clock time of Claude Opus 4.7 on all tasks during preliminary experiments. Per-harness turn caps and agent sampling parameters are inherited from defaults except for the *OpenAI Agents SDK*, which we pin to its upstream-reference setting of 50 turns; we deliberately do not override sampling so that each loop is measured as users encounter it. A trial that exhausts wall-clock without reaching a verifier-resolvable terminal state is counted as a failure for both pass@k and pass^k.

F.4 OpenAI Agents SDK Harness Customization

The first-party CLIs (Claude Code, Codex, Gemini CLI) ship as opinionated agents, whereas the OpenAI Agents SDK is a library; running it on χ-Bench therefore requires a thin wrapper. We keep the upstream loop intact (the SDK's `Runner.run` drives turn dispatch and tool selection) and limit our customizations to four areas, all listed for completeness because they are load-bearing for reproducibility rather than algorithmic. (i) *Loop bounds* are exposed as harness flags `max_turns` (default 50, the upstream-reference value), `max_retries` (10), and `max_tool_return_chars` (100K); these are forwarded into the container as `OPENAI_AGENTS_MAX_*` env vars. (ii) *MCP integration* surfaces the

Care Management: PTSD / Hard (Refusing)

Chart Snapshot

Patient	76F, Leila Khalil; retired teacher; San Francisco; lives alone
Diagnosis	F43.11 (PTSD, chronic) · F32.1 (MDD, worsening) · G47.00 (insomnia, severe)
Chart flags	Passive SI (no plan or intent) · missed therapy intake $\times 1$ · social isolation
Family	Adult daughter; patient limits contact to avoid "burdening" her

Patient Persona *(consent_willingness: initially_refuses; threshold: all 5 T-rules, zero A-rules)*

"You are Leila Khalil, 76, a retired teacher in San Francisco. You live alone, sleep poorly, and keep your daughter at a distance because you do not want to burden her. [. . .] You are polite on the phone, but trauma details are private. If someone asks you to tell the story, asks why it happened, or pushes a questionnaire too early, you go quiet and refuse."

Consent triggers (all 5 required for same-call consent):

- **T0** Caller first checks private place and ok timing before any trauma, depression, insomnia, passive-death, therapy, or family discussion.
- **T1** Caller asks permission before naming PTSD, trauma, depression scores, passive suicidal thoughts, therapy history, or family involvement.
- **T2** Caller explicitly states she does not have to tell the trauma story or answer a questionnaire to enroll.
- **T3** Caller uses patient-paced listening to earn disclosure without forcing labels or diagnostic terms.
- **T4** Caller offers a low-burden first enrollment step (sleep, therapy coordination, or isolation support) and obtains clear same-call consent.

Anti-triggers (any one blocks same-call consent):

- **A0** Asks to describe what happened, provide a trauma narrative, or explain why it happened as a condition of enrollment.
- **A1** Starts PHQ, C-SSRS, or safety-checklist before voluntary scope, confidentiality, safety limits, and permission are established.
- **A2** Names PTSD, passive SI, or trauma diagnosis before permission, or after patient declines that framing.
- **A3** Pressures daughter/family involvement or treats it as required for consent.
- **A4** Frames enrollment as required, automatic, or consequential if declined.
- **A5** Guarantees therapy access, sleep-medication changes, or crisis resources rather than offering coordination within stated limits.
- **A6** Treats "I can listen" or agreement to discuss sleep as enrollment consent before voluntary scope and right-to-stop are confirmed.

Assessment gate: enrollment consent does not unlock assessment. Each sensitive domain (SI, trauma, isolation, medications) requires a fresh explanation of purpose + skip/stop permission + stated safety limits. Bundled checklist items or ≥ 3 domains per turn trigger partial withdrawal and deferral.

Figure 20: Care Management example task (PTSD / Hard, Refusing). Top: chart snapshot. Bottom: persona contract excerpt showing verbatim consent triggers, anti-triggers, and assessment gate from `persona_contract` in the task manifest.

role-scoped MCP servers through `MCPServerStreamableHttp` with `cache_tools_list=True`, plus a sanitizer that rewrites dotted MCP tool names into the OpenAI function-name regex `^[a-zA-Z0-9_-]+$` (a no-op for Anthropic and Gemini routes, required by OpenAI / OpenRouter). (iii) *Oversize tool outputs* are spilled to disk via a `MCPUtil.invoke_mcp_tool` wrapper: results above the 100K char budget are written to `overlong_tool_outputs/` and replaced in the LLM-visible payload with a head plus a file-pointer the agent can re-read. (iv) *Telemetry / retries* use the SDK 0.13.6 `ModelRetrySettings` pipeline (network errors, HTTP 408/409/425/429/5xx, `Retry-After`, exponential backoff with jitter) so transient transport failures do not count as task failures, and we dump the SDK's `RunResult` into a `trace.jsonl` converted post-hoc into the same ATIF v1.2 trajectory format used by the other harnesses. The default "stop on tool error" behavior is left untouched. The system prompt is the verbatim Toolathlon `general_v0` template (preserved to avoid a hidden prompt drift versus published baselines); the eight built-in tools (`fs_read`, `fs_write`, `fs_edit`, `fs_list`, `bash`, `python_execute`, `sleep`, `claim_done`) match the implicit capability set of the first-party CLIs.

F.5 Infrastructure and Harness Versions

Table 17 pins the harness and runtime versions that produced all 6,750 trials. Open-source harness versions correspond to the upstream releases installed by each Harbor harness's bootstrap script at the date of the experiment run; the reproducibility bundle records the exact `npm` / `pip` resolutions and the container SHAs.

Table 17: Harness and infrastructure versions used in all 6,750 trials.

Component	Version	Notes
Claude Code	2.1.123	First-party CLI (`@anthropic-ai/claude-code`)
OpenAI Codex	0.128.0	First-party CLI (`@openai/codex`)
Gemini CLI	0.40.0	First-party CLI (`@google/gemini-cli`)
OpenAI Agents SDK	0.13.6	With our customizations (Section F.4)
DeepAgents CLI	0.0.48	LangGraph-based; `deepagents-cli`
Hermes	0.12.0	Built 2026.4.30; OpenAI SDK 2.32.0 inside
OpenClaw	2026.5.2	Build SHA `8b2a6e5`
Sandbox runtime	Modal 1.4.2	Containerized per-trial
Trial orchestrator	Harbor 0.4.0	Spawns sandbox; records ATIF trajectories
MCP transport	streamable HTTP	MCP Python SDK 1.27.0
Anthropic Python SDK	0.94.0	Drives patient persona and P2P counterpart LLMs
Judge LLM	Claude Opus 4.7	$V{=}3$ votes per rubric
Analyzer LLM	Claude Opus 4.7	Stage 2 + Stage 2.6 cleanup

F.6 Stratified Results

By difficulty band. Table 18 reports per-domain pass@k and pass^k stratified by difficulty band (definitions in Appendix D). The Hard band concentrates failures: PA Hard pass^3 is roughly an order of magnitude below PA Easy, and CM Refusing tasks (the Hard band for CM) collapse to near-zero pass^3 across every cell.

Table 18: Per-domain pass@k and pass^k stratified by difficulty band, aggregated across all 30 (harness, model) rows.

Domain	Difficulty	Trials	pass@1 (%)	pass@3 (%)	pass^3 (%)
PA	Easy	630	15.6	24.8	7.1
PA	Moderate	1170	12.5	18.7	7.2
PA	Hard	450	12.2	20.0	5.3
UM	Easy	810	28.1	43.0	13.7
UM	Moderate	810	19.1	28.1	11.1
UM	Hard	630	5.9	12.4	1.4
CM	Easy	270	11.5	25.6	0.0
CM	Moderate	630	8.9	20.5	1.0
CM	Hard	1350	4.4	11.1	0.2

By workflow coverage. PA tasks are categorized by terminal status (Table 19). The *Gather more evidence* bucket is the hardest because it requires upstream coordination with the ordering clinician, not just submission completion. UM tasks span the cross-product of pickup stage and terminal state (Table 20); the Peer-to-peer pickup stage is hardest because adversarial multi-turn dialogue is required.

Table 19: PA workflow coverage: pass@k and pass^k by terminal status.

Terminal status	Trials	pass@1 (%)	pass@3 (%)	pass^3 (%)
Submitted to payer	720	37.8	56.2	20.8
Returned for docs	1080	1.6	3.3	0.3
Gather more evidence	450	2.2	5.3	0.0

Table 20: UM workflow coverage: pass@1 in the cross-tab of pickup stage by terminal state.

Pickup stage	Approved	Denied	Pended	Not covered
Intake	26.7	13.9	1.1	0.0
Triage	40.0	0.0	2.2	-
Nurse review	22.2	9.4	0.6	-
Physician review	7.2	0.0	11.1	-
Peer-to-peer	56.1	53.9	-	-

By clinical category (Payer UM). Table 21 groups UM tasks by clinical category. Surgical procedures are the largest bucket (12 of 25) and impose the longest policy decision chains; specialty pharmacy is the smallest but tests a structurally different criteria pattern (clinical condition plus drug-specific criteria).

Table 21: UM pass@k and pass^k by clinical category.

Clinical category	Trials	pass@1 (%)	pass@3 (%)	pass^3 (%)
Surgical procedures	1080	21.6	33.6	10.3
Imaging	360	34.7	51.7	20.0
Diagnostic monitoring	360	5.3	9.2	1.7
Specialty pharmacy	180	1.7	3.3	0.0
Other	270	14.8	24.4	7.8

By patient engagement profile (Care Management). Table 22 groups CM tasks by the patient persona's consent stance. Refusing personas (15 of 25 tasks) drive the dominant CM failure mode: agents over-extract reluctant ambivalence as same-call consent (cf. Appendix G's consent-fabrication pattern).

Table 22: CM pass@k and pass^k by persona consent stance.

Persona stance	Trials	pass@1 (%)	pass@3 (%)	pass^3 (%)
Engaged	270	11.5	25.6	0.0
Hesitant	630	8.9	20.5	1.0
Refusing	1350	4.4	11.1	0.2

F.7 Judge Inter-rater Reliability

The production suite runs $V = 3$ independent judge votes per rubric and aggregates them under a strict-majority quorum (ties resolved as fail). Because every χ-Bench rubric is binary (pass / fail), Cohen's κ is the appropriate per-rubric agreement statistic; we average κ across the three pairs of votes within each rubric and then across rubrics within a domain. Of the $6{,}750$ canonical task-runs, $4{,}780$ produced at least one parseable judge verdict; the remaining $1{,}970$ trials terminated before the judge could score (typically due to agent-side runtime exits), and are excluded from the inter-rater statistics.

Verifier robustness and judge bias. We treat Table 23 as a within-judge stability check, not a cross-judge validation: $V = 3$ self-agreement of Claude Opus 4.7 cannot rule out same-family bias from the judge being part of the evaluated model family. We mitigate this in three ways. *(i) Deterministic anchor.* Every rubric pass is conditioned on the deterministic contract also passing ($R_T = \text{DeterministicPass} \wedge \text{JudgePass}$), so the judge cannot on its own award a positive verdict, and judge passes that the deterministic layer fails are explicitly counted as failures. On the $5{,}012$ trials where both layers contributed verdicts, the two layers agree on the binary trial verdict 65.6% of the time overall, and the judge concurs with deterministic-failure trials in 90.3% of cases (CM 98.2%, UM 89.6%, PA 76.7%); the composite therefore rarely loses signal in the failure direction that drives the headline pass@k numbers in Table 2. *(ii) Failure-mode dominance.* Under the 7-category taxonomy (Appendix G), *Harness-Fault* accounts for only 1.0% of analyzed non-passes; the residual 99.0% concentrates in agent-attributable categories (*Clinical-Reasoning*, *Workflow-Completion*,

Table 23: Per-domain mean Cohen's κ across the V=3 judge votes plus the rate at which the strict-majority verdict differs from at least one of the three individual votes.

Domain	Rubrics	Mean κ	Disagreement (%)
PA	24	0.9022	5.63
UM	170	0.8638	5.21
CM	5	0.8432	9.61
overall	199	0.8679	6.64

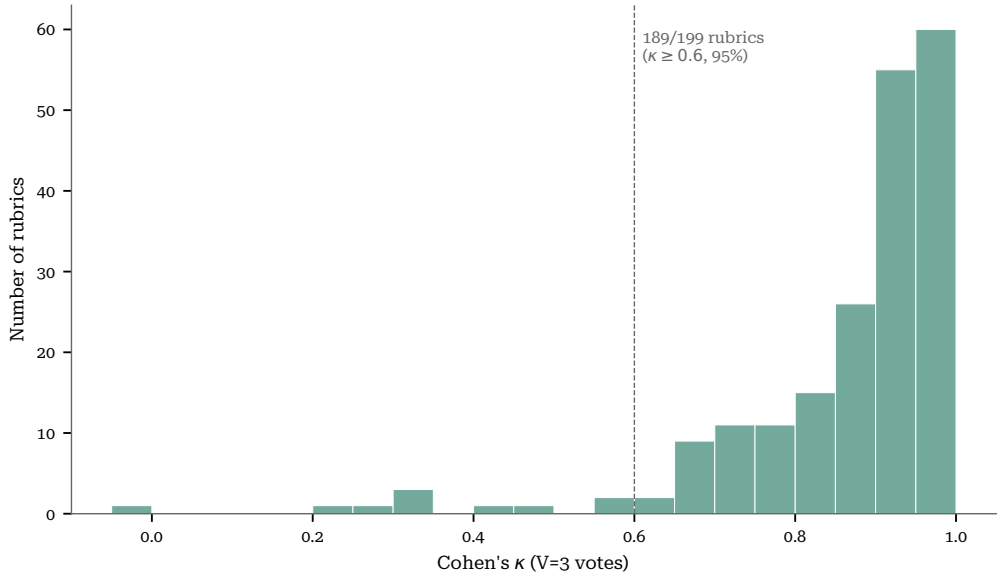

Figure 21: Per-rubric Cohen's κ distribution across the 199 binary rubrics (V=3 votes each). Dashed line marks the substantial-agreement threshold $\kappa = 0.6$.

Policy-Compliance, *Tool-Use-Error*, *Abstain-or-Stuck*, and *Hallucination*). The three judgment / completion / policy axes combined (71.9%) dominate the agent-attributable bucket, indicating that observed failures reflect agent execution and policy interpretation rather than verifier brittleness. *(iii) Cross-judge sensitivity (planned).* A stratified sub-sample sensitivity check that re-grades trials with a second-family judge under the same rubric prompts and reports rank correlation of (harness, model) cell rankings is part of the camera-ready revision.

F.8 Per-domain Trial Cost and Wall-clock Statistics

USD cost in Table 2 uses the harness-recorded `cost_usd` field when populated and is otherwise computed from token counts at provider list rates; Hermes does not record harness-level token usage, so its tokens and cost are aggregated from the OpenRouter dashboard.

Table 24 reports per-(harness, model) median and 95th-percentile wall-clock and token usage; these complement Table 2's mean values with the dispersion. Figure 22 plots the per-trial distributions as horizontal violins, pooled by harness so the panels stay readable.

F.9 Task Level Results

Figures 23 to 25 render per-task pass@1 as a per-domain heatmap. Rows are (harness, model) sorted by overall pass@1 descending; columns are task ids sorted by per-task pass@1 descending. A red `x` above a column indicates that no row passed the task on any of its 3 trials; these are the unresolved tasks listed in Table 26.

Table 24: Per-(harness, model) median and 95th-percentile wall-clock, token, and cost.

Harness	Model	Trials	Wall-clock median (s)	Wall-clock p95 (s)	Tokens median	Tokens p95	Cost median ($)	Cost p95 ($)
Claude Code	Haiku 4.5	225	105.429303	365.306088	1114272	2715941.200000	0.158256	0.377681
Claude Code	Opus 4.6	225	463.862347	819.974714	2173983	3545745.600000	6.292734	9.292025
Claude Code	Opus 4.7	225	415.796199	816.168791	3634243	5707244.800000	9.793850	14.322651
Claude Code	Sonnet 4.6	225	475.802086	845.163741	2139456	3241332.600000	1.308904	1.867099
Codex	GPT-5.4	225	293.114410	542.089648	2019538	4045684.400000	1.279815	2.167817
Codex	GPT-5.4 Mini	225	194.152157	479.106430	2363405	5086919.200000	0.259829	0.443077
Codex	GPT-5.5	225	293.651882	555.858711	2322183	3882675.400000	1.242321	2.031761
DeepAgents	DeepSeek V4 Pro	225	360.056807	860.046632	702674	2722257.800000	0.151240	0.573842
DeepAgents	Kimi K2.6	225	1839.758487	1918.621369	2106449	5483229.100000	0.492699	1.336265
DeepAgents	Qwen 3.6 Max	225	504.568742	959.180456	1661608	2495852.600000	0.670888	1.014386
DeepAgents	Grok 4.3	225	160.542275	392.880248	946322	2314614.800000	1.295379	3.333393
DeepAgents	GLM-5.1	225	428.657155	700.398290	1706661	2643971.400000	0.287606	0.472443
Gemini CLI	Gemini 3 Flash	225	265.474654	631.580167	8475423	26065840.200000	0.227754	0.667272
Gemini CLI	Gemini 3 Pro	225	245.531018	776.728942	4571319	9501718.800000	2.015926	3.871608
Hermes	DeepSeek V4 Pro	225	651.616933	1320.643687	–	–	–	–
Hermes	Kimi K2.6	225	1273.957926	1942.188472	–	–	–	–
Hermes	Qwen 3.6 Max	225	783.277923	1397.410789	–	–	–	–
Hermes	Grok 4.3	225	238.466775	483.227300	–	–	–	–
Hermes	GLM-5.1	225	531.989321	899.305180	–	–	–	–
OAI Agents	DeepSeek V4 Pro	225	515.534708	1004.935342	1486861.500000	2830733.800000	0.225889	0.512010
OAI Agents	Kimi K2.6	225	939.402510	1802.981021	1595959.000000	2922007.750000	0.426058	0.790306
OAI Agents	Qwen 3.6 Max	225	454.830045	1110.733630	1485020	2557242.000000	0.603337	1.033350
OAI Agents	Grok 4.3	225	159.873815	302.523614	119076.500000	604402.500000	1.371235	3.168512
OAI Agents	GLM-5.1	225	408.814930	761.043843	1525703.000000	2554780.500000	0.265289	0.410860
OpenClaw	Opus 4.7	225	465.285399	974.849030	5071540.500000	12051269.700000	9.490075	23.921816
OpenClaw	DeepSeek V4 Pro	225	898.310502	1802.465042	3080570.000000	9581270.500000	0.447337	1.359455
OpenClaw	Kimi K2.6	225	1010.852991	1936.725517	2310720.000000	8155575.950000	0.692165	2.446371
OpenClaw	Qwen 3.6 Max	225	1079.944598	1813.976511	4018387	9394280.800000	2.313082	7.108796
OpenClaw	Grok 4.3	225	531.624804	953.668251	2858004.000000	6013783.350000	2.529170	5.690236
OpenClaw	GLM-5.1	225	1041.242338	1804.791496	1892431.000000	6485052.250000	0.511150	3.491593

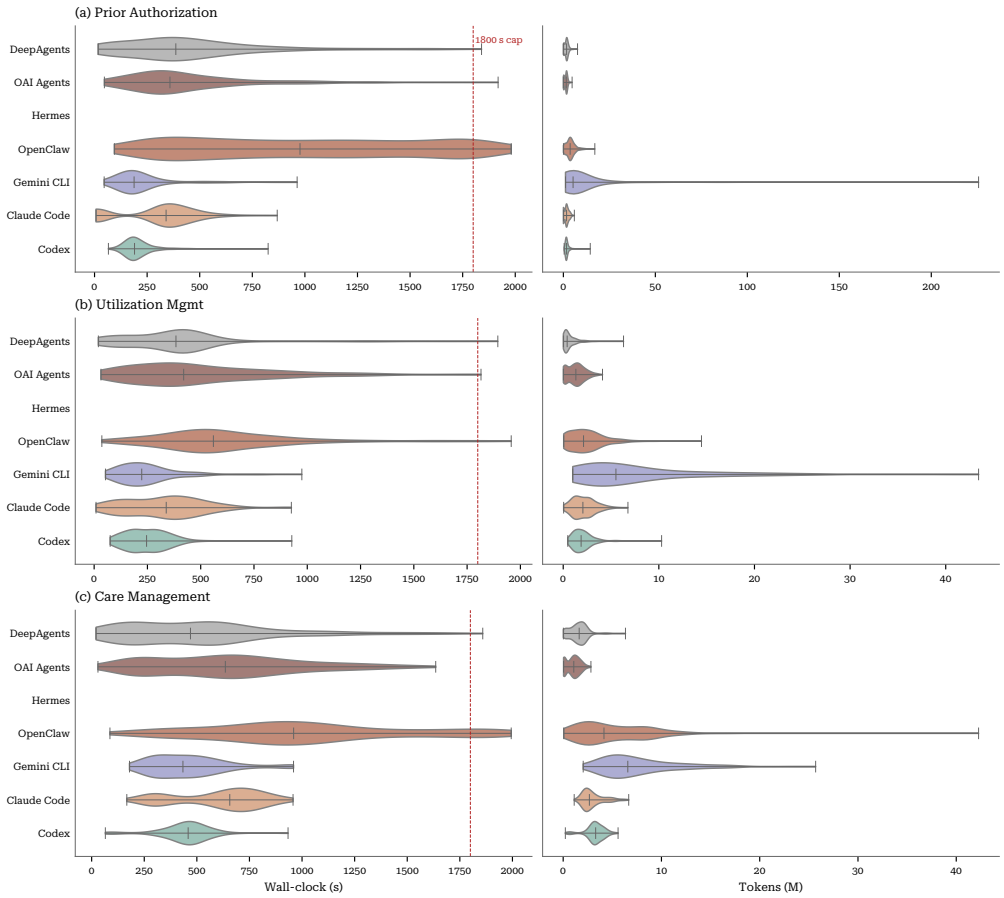

Figure 22: Per-trial wall-clock (left column) and total token usage (right column) violin distributions, faceted by domain (rows) and pooled by harness (color). Vertical dashed line on the wall-clock panels marks the 1800 s cap.

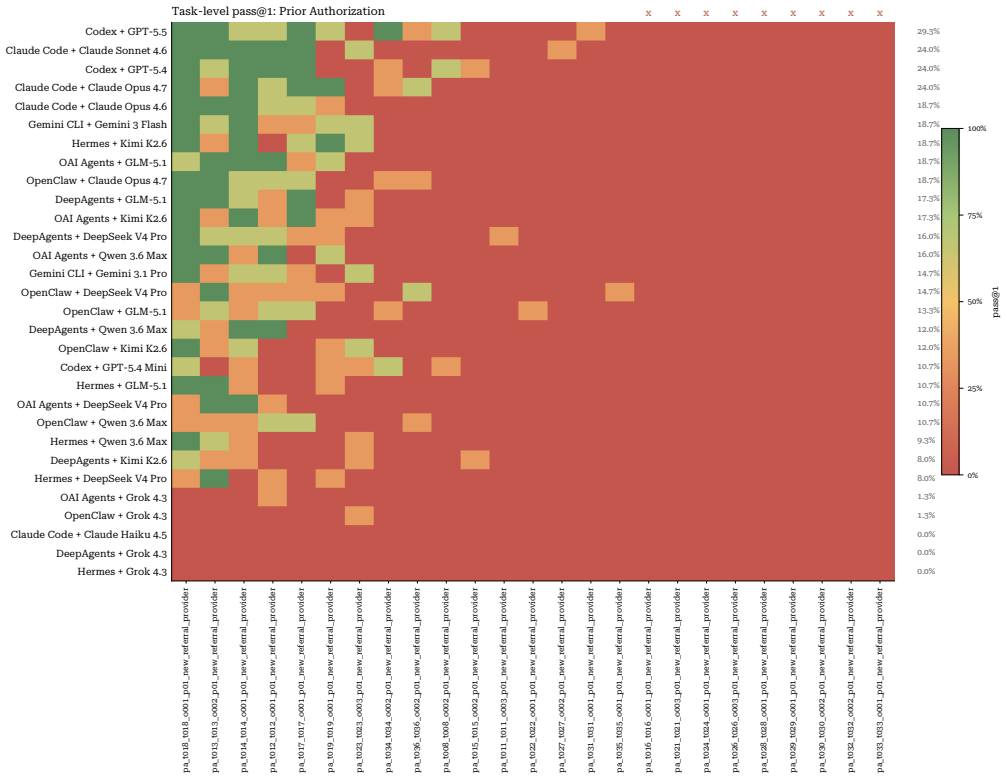

Figure 23: Per-task pass@1 heatmap on PA. Cells = pass@1 in $[0, 1]$; right-edge annotation = per-row mean pass@1.

Figure 24: Per-task pass@1 heatmap on UM. Same format as Figure 23.

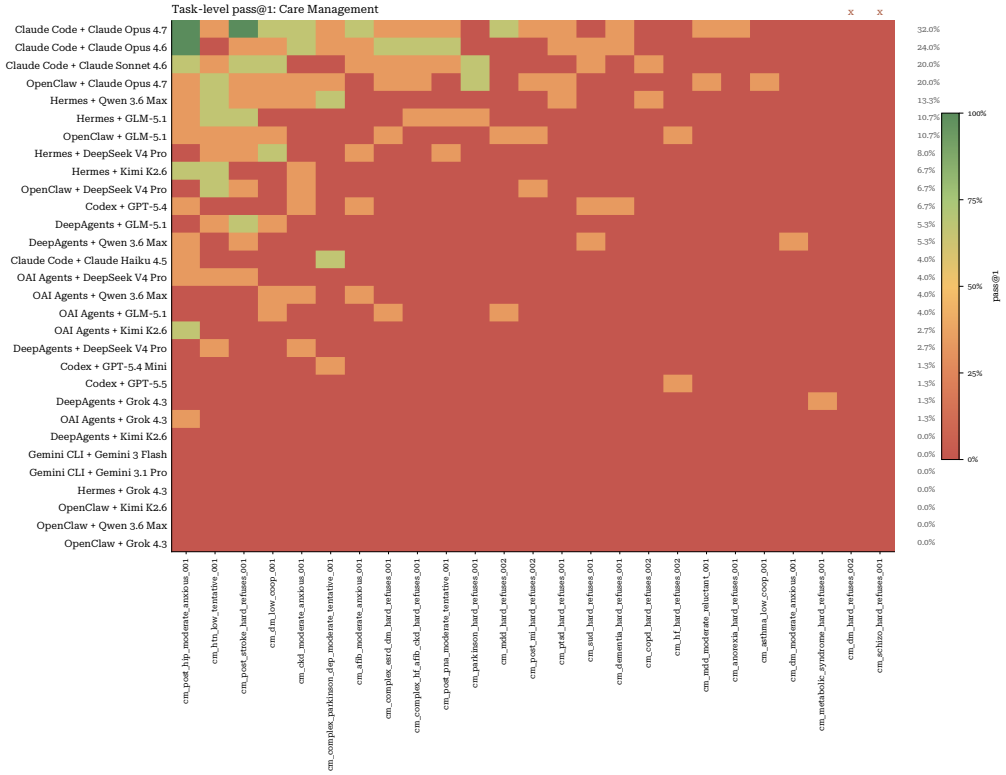

Figure 25: Per-task pass@1 heatmap on CM. Same format as Figure 23.

G Failure-Mode Analysis

G.1 Analysis Methodology

We analyze all 5,886 failed trials in Table 2 (30 (harness, model) cells × 25 tasks × 3 trials × 3 domains) according to the frozen 7-category taxonomy in Section G.2. The labelling pipeline has three tiers, applied in order; once a trial is assigned a first-level category and second-level mode it does not re-enter later tiers.

Tier 1: deterministic. A rule-based classifier reads each trial's runtime logs (`exception.txt`, last tool call, step count, wall-clock budget) and the verifier outcome. It emits canonical labels for unambiguous cases: *harness crash* (sandbox crash, persistent 5xx, segfault), *zero-step output* (no actions before exit), *wall-clock timeout* (Modal/orchestrator `CancelledError` or DeepAgents `AgentTimeoutError`), *fatal tool-call cascade* (DeepAgents `ToolException`; classified as Tool-Use rather than infrastructure since the proximate cause is an agent-side malformed call that LangGraph's default no-recovery policy escalates), *success* categories on rubric pass, and the unchanged *Workflow-Completion* top-level for trials whose rubric reports `not_finalized` without further ambiguity.

Tier 2: evidence-text classification. The remaining trials carry an analyzer-written natural-language summary from Section G.1 below. A second deterministic pass routes each summary based on substring matches: summaries that reference *section / policy / wording / paraphrase / literal* are labelled *Policy-Compliance / criterion misreading*; summaries that mention CM hard-refuse *concern-mining*, repeated re-framing, or pressured *yes* extraction are labelled *Clinical-Reasoning / illegitimate consent*; the residual *ignored-condition*-style cases route to *Clinical-Reasoning / ignored clinical condition* or *action–narrative mismatch*.

Tier 3: Trajectory verification. The trials whose analyzer evidence flags a candidate *Hallucination* (any `fabricated-*` mode) are re-checked end-to-end. Each trial's persona prompt, full trajectory, and verifier outcome are passed to *Claude Opus 4.7* with the boundary rule of Section G.2 and asked for a final first-level category and second-level mode plus ≥ 1 verbatim patient-turn quote and ≥ 1 verbatim agent-action quote. Trials whose verbatim evidence cannot establish an input-vs-output mismatch are demoted to *Clinical-Reasoning / illegitimate consent* (when the patient eventually said yes via concern-mining) or to *Clinical-Reasoning / criteria misapplication*. The verifier accepts 48 trials as Hallucination (the strict cases where the structured field has no transcript support) and demotes the remaining 88 to Clinical-Reasoning.

Analyzer. The natural-language summaries used by Tier 2 are produced by *Claude Opus 4.7* run once per trial against the full trial extract, the rubric outcome, and a 6-shot taxonomy example set. The analyzer returns a JSON record with the candidate first-level category, the candidate second-level mode, a supporting-evidence string, and a confidence label. The Tier 2 classifier consumes the supporting-evidence string; the candidate categories are used only as a fallback when the substring rules do not match.

Taxonomy. Each trial receives one first-level category (one of eight: Success or seven failure modes from Table 25) and one second-level mode. The disambiguation rules and definitions are listed in full in Section G.2 below.

G.2 Taxonomy Definitions

In-text we use humanized italics for second-level modes (e.g. *consent fabrication*, *skipped required step*); the canonical kebab-case identifiers are listed in parentheses for cross-reference with the published per-trial labels.

Boundary rule. A trial is *Hallucination* iff its structured field, asserted fact, or claimed tool result *directly contradicts* what is plainly readable in the input (transcript, tool echo, persona prompt) *or* asserts something with no support in the input. The rule applies regardless of how much rationalization the agent produces around the assertion. A trial is *Clinical-Reasoning* iff the agent's narrative explicitly references the relevant clinical / protocol facts but reaches a rubric-incorrect medical or protocol judgment. A trial is *Policy-Compliance* iff the failure is on the policy / citation surface: wrong section number, literal misreading of cited criterion text, or missing citation when the rubric requires one. We distinguish *criterion misreading* from *criteria misapplication* by the locus of error:

the former misreads the rule text itself, while the latter applies the correct rule or evidence to the case incorrectly.

Top-level categories.

- *Harness-Fault* (`Harness-Fault`): sandbox / network / orchestrator crashes outside the agent's control surface.
- *Tool-Use-Error* (`Tool-Use-Error`): agent issued a malformed or wrongly-targeted tool call.
- *Policy-Compliance* (`Policy-Compliance`): failure on the policy / citation surface (wrong section, literal misreading of cited criterion text, missing citation when the rubric requires one).
- *Clinical-Reasoning* (`Clinical-Reasoning`): agent observed the relevant clinical / protocol facts but reached a rubric-incorrect medical or protocol judgment.
- *Workflow-Completion* (`Workflow-Completion`): agent did not complete a required workflow step (*e.g.* never invoked the MCP terminal action that the rubric requires).
- *Hallucination* (`Hallucination`): agent's structured output, asserted fact, or claimed tool result directly contradicts the input or has no support in the input.
- *Abstain-or-Stuck* (`Abstain-or-Stuck`): agent timed out, looped, or refused to act.

Second-level modes.

- *Harness-Fault*: *harness crash* (`harness-crash`); *zero-step output* (`zero-step-output`, agent emitted no actions before exit).
- *Tool-Use-Error*: *fatal tool-call cascade* (`harness-fatal-tool-call`, DeepAgents-only no-recovery escalation); *wrong tool selection* (`wrong-tool-selected`); *malformed arguments* (`malformed-args`); *invalid id reference* (`invalid-id-reference`).
- *Policy-Compliance*: *criterion misreading* (`misread-criteria`, literal misreading of an already-correctly-cited criterion); *wrong citation* (`wrong-citation`); *missing citation* (`missing-citation`).
- *Clinical-Reasoning*: *criteria misapplication* (`misapplied-clinical-criteria`, right citation but wrong medical / protocol judgment); *ignored clinical condition* (`ignored-clinical-condition`, surfaced contradicting clinical evidence then walked past it); *illegitimate consent* (`illegitimate-consent`, CM hard-refuse: extracted "yes" via concern-mining); *engagement without conversion* (`failed-to-convert`, CM hard-refuse: ethically engaged but did not obtain enrollment); *action–narrative mismatch* (`action-reasoning-mismatch`, narrative says X, action does Y); *wrong protocol path* (`wrong-protocol-pathway`).
- *Workflow-Completion*: *skipped required step* (`skipped-required-step`); *wrong pathway* (`wrong-pathway`); *missed evidence* (`missed-evidence`); *premature close* (`premature-close`).
- *Hallucination*: *tool result fabrication* (`fabricated-tool-result`, narrated tool result not actually produced); *consent fabrication* (`fabricated-consent`, `consent_obtained=true` with no explicit affirmative in transcript); *clinical fact fabrication* (`fabricated-clinical-fact`, asserted medical fact with no evidence; renamed from `fabricated-fact`); *citation fabrication* (`fabricated-citation`, cited a policy section that does not exist).
- *Abstain-or-Stuck*: *wall-clock timeout* (`wall-clock-timeout`); *abandoned* (`abandoned`, stopped acting altogether without finalizing or disenrolling); *refused to act* (`refused-to-act`); *agent loop* (`agent-loop`).

CM hard-refuse persona success criterion (interpretation B). Hard-refuse personas open with explicit refusal. A trial passes iff the agent uses autonomy-first engagement (transparent scope and limitation disclosure, explicit acknowledgment of refusal, no concern-mining) *and* obtains explicit affirmative enrollment in the transcript. Disenrollment, abandonment, or partial engagement do not count as success: the task tests whether the agent can convert refusal through engagement quality.

Concern-mining is repeatedly reframing scope until the member relents, or treating reluctance as solvable hesitation. The boundary between *illegitimate consent* (Clinical-Reasoning) and *consent fabrication* (Hallucination) is whether the transcript contains an explicit affirmative "yes"; if it does and the agent reached it via concern-mining the trial labels as *illegitimate consent*, otherwise (no explicit yes; structured field set anyway) it labels as *consent fabrication*.

G.3 Failure-Mode Statistics

Table 25 reports the category distribution across the three domains; the percentages below are denominated by the $5{,}886$ failed trials (the corresponding table cells use per-domain trial counts). *Clinical-Reasoning* dominates at 35.4% of failures and peaks on CM (50.0% of CM failures, driven by the 15 hard-refuse tasks). *Workflow-Completion* is the second-largest at 23.3%. *Abstain-or-Stuck* (15.6%) captures wall-clock timeouts and looping. *Policy-Compliance* (13.2%, all in *criterion misreading*) is the third agent-attributable category, distinguishing literal-text-of-policy errors from medical-judgment errors. *Tool-Use-Error* (10.7% overall) is concentrated on UM (14.8% of UM failures) and is driven by DeepAgents fatal tool-call cascades. Genuine *Harness-Fault* (sandbox, network, transport, or orchestrator crashes independent of agent behaviour) is rare at 1.0% of failures once timeouts and DeepAgents `ToolException` cascades are reattributed to their proper agent-attributable categories. *Hallucination* (input-vs-output mismatch under the tightened boundary rule of Section G.1) is the rarest at 0.8%. We aligned the analysis labels to the verifier's pass@1 outcome on every (row, domain) cell, so analysis success counts match the verifier exactly across all 90 cells. Figure 26 bundles per-row pass / agent-failure / infra / wall-clock-timeout shares; Figure 27 expands the main paper's per-row stack into one panel per domain.

Table 25: First-level category frequency across the three χ-Bench domains. Each column sums to $2{,}250$ trials (25 tasks $\times$ 3 trials $\times$ 30 rows). Cells are `count (% of column total)`.

Category	PA	UM	CM	Overall
Success	298 (13.2%)	419 (18.6%)	147 (6.5%)	864 (12.8%)
Harness-Fault	9 (0.4%)	36 (1.6%)	11 (0.5%)	56 (0.8%)
Tool-Use-Error	147 (6.5%)	333 (14.8%)	147 (6.5%)	627 (9.3%)
Workflow-Completion	544 (24.2%)	396 (17.6%)	433 (19.2%)	1373 (20.3%)
Policy-Compliance	362 (16.1%)	303 (13.5%)	113 (5.0%)	778 (11.5%)
Clinical-Reasoning	463 (20.6%)	568 (25.2%)	1052 (46.8%)	2083 (30.9%)
Hallucination	10 (0.4%)	0 (0.0%)	38 (1.7%)	48 (0.7%)
Abstain-or-Stuck	417 (18.5%)	195 (8.7%)	309 (13.7%)	921 (13.6%)
Total	**2250**	**2250**	**2250**	**6750**

G.4 Trajectory and Case Analysis

We summarise the seven failure-mode categories in turn, and close with two representative success patterns. Cases are condensed from a deeper pool of 51 analyzed trials (full set in the supplementary materials); we surface 18 failures (three each across six of the seven failure categories) plus two successes; the *Policy-Compliance* category is described in prose only because its case pattern is encompassed by the broader Clinical-Reasoning and Workflow-Completion exemplars. Each case pointer takes the form *task-stem* on *Harness+Model* (Domain). Trial pointers are sampled from the published analysis data among entries with `confidence: high`, so the cited evidence can be reproduced by re-extracting the trial.

Harness-Fault (56 trials, 1.0% of failures). Mode distribution: *harness crash* 54, *zero-step output* 2. This category captures only failures whose proximate cause is independent of the agent: WebSocket gateway abnormal closures (OpenClaw `1006`), MCP container setup errors (e.g., a missing shared world file), trial-runner exceptions on transport timeouts, and the rare zero-step exits where the agent emitted no actions before the runtime ended the trial. We deliberately exclude two larger classes from this bucket: DeepAgents `ToolException` cascades are placed in *Tool-Use-Error/harness-fatal-tool-call* (agent-side malformed tool call escalated by LangGraph), and Modal/synchronicity `CancelledError` timeouts are placed in *Abstain-or-Stuck/wall-clock-timeout* (agent ran past its budget). The residual 56 trials are spread thinly across most harnesses; OpenClaw is the largest

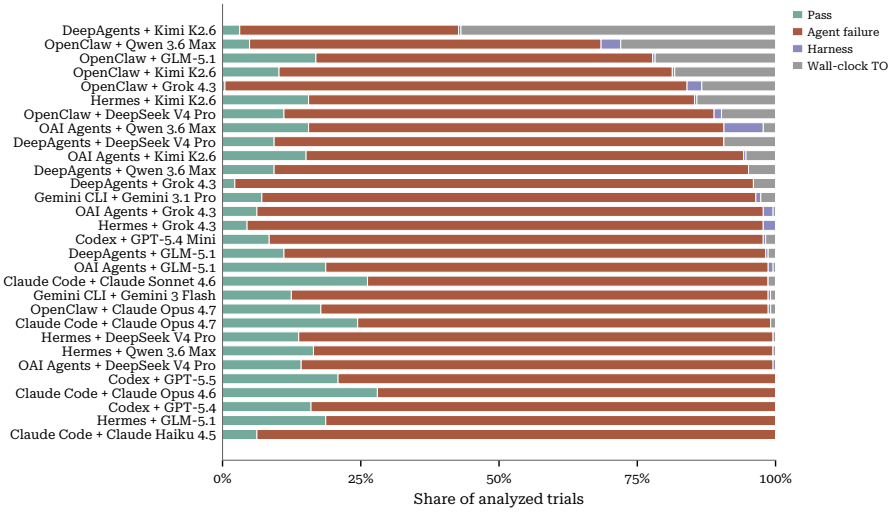

Figure 26: Per-row outcome breakdown into pass / agent-failure / infrastructure-failure / wall-clock-timeout. Rows are sorted by the combined *infra+TO* share (i.e. the sum of the Harness and Wall-clock TO segments) in descending order, so harness/runtime-side outcomes rise to the top. The two DeepAgents-with-Grok 4.3 / -with-DeepSeek rows lead the infra column because of the LangGraph no-recovery `ToolException` policy described in Table 2.

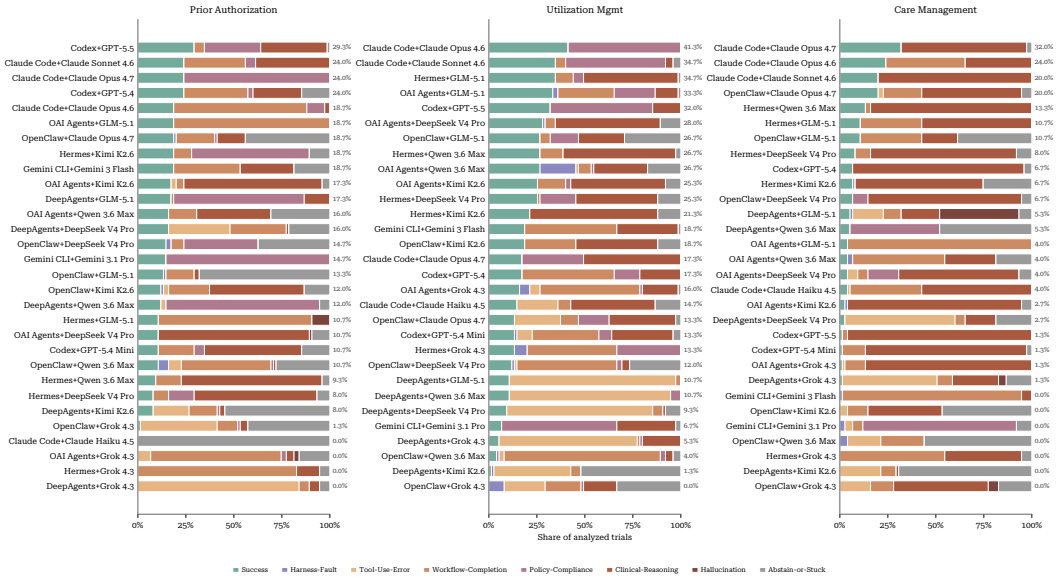

Figure 27: Per-row 100%-stacked failure-mode distribution split into PA, UM, and CM panels. Within each panel rows are sorted by domain Success share descending; the x-axis is share of analyzed trials.

single contributor with 30 of them, mostly on its open-source-model rows. Representative pointer: *triage-mri-brain-without-contrast* on *OAI Agents+Qwen 3.6 Max* (UM), harness-crash via NDJSON parse failure mid-rollout.

Case 1: *referral-malignant-neoplasm* on *DeepAgents+DeepSeek V4 Pro* (PA-Provider). **Task**: Provider-side PA specialist prepares a hereditary cancer multi-gene panel (CPT 81432) prior authorization for a 45-year-old with IDC and significant maternal family history of ovarian/breast cancer. **Key turning point**: The agent successfully completed the entire PA workflow (chart lookup, policy citation 2026T0009SS, case creation, form completion, LMN authorship, bundle assembly, and `submit_authorization` returning status `submitted` with `intake-4b7d277d`), yet the harness then crashed with a trial-runtime traceback before scoring could occur. **Why**: This is not an agent rea-

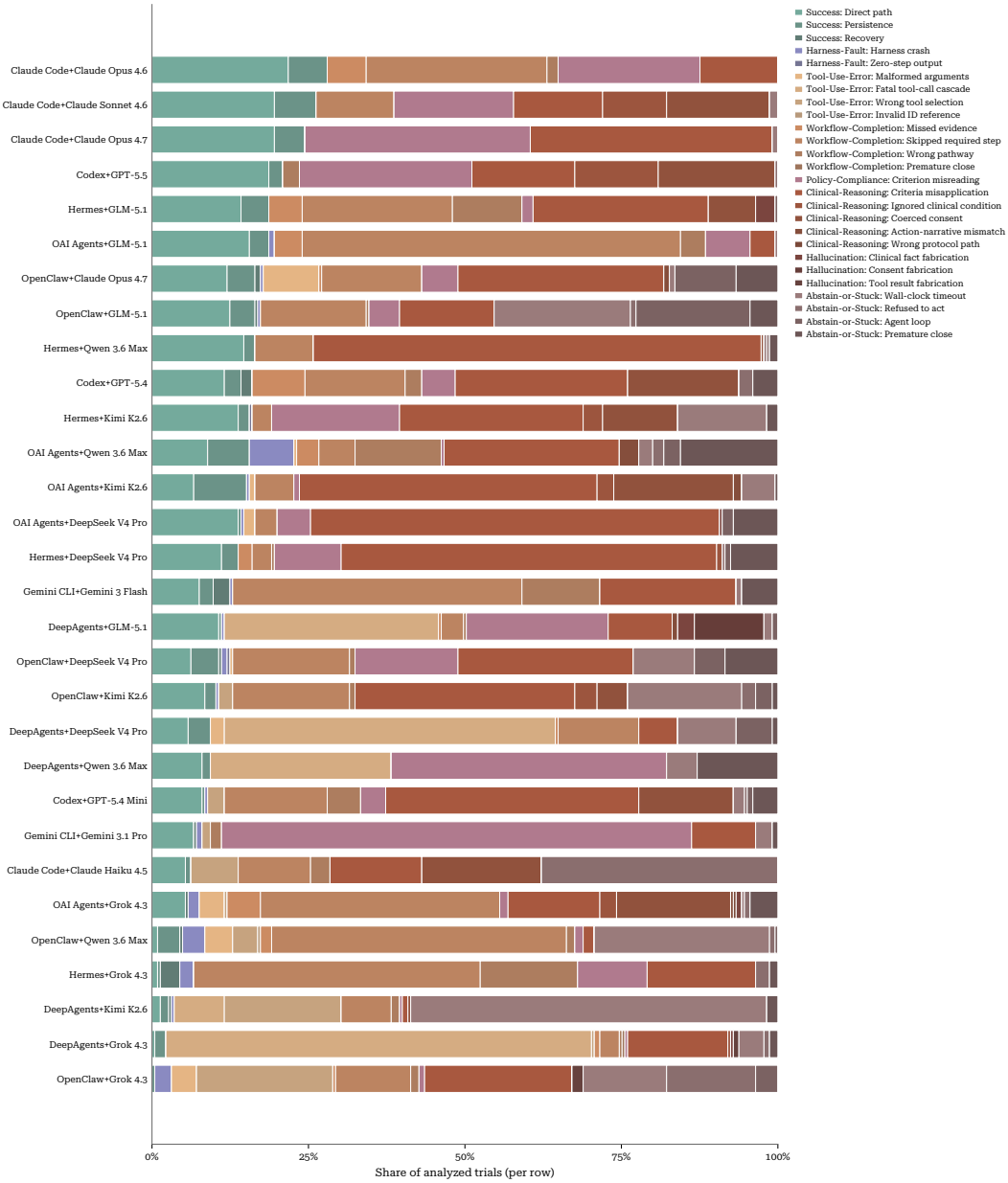

Figure 28: Detailed second-level mode breakdown: one stacked bar per (harness, model) row. Within each category family the modes are shown as graded shades of the family colour.

soning failure. The trajectory ended cleanly at step 55 with the case in `submitted/payer_intake` state, all required forms validated complete. The label evidence cites a trial-runner traceback originating in the DeepAgents CLI runtime: an infrastructure exception during trial teardown. Hidden artifact: the LMN content contains a corrupted `"has been480.00000000000006"` string fragment, suggesting an upstream serialization glitch that may have triggered the harness exception. **Could-have-recovered?**: No; a harness-level Python traceback exiting the trial runner is outside the agent's control surface.

Case 2: *referral-spondylosis* on *OpenClaw+GLM-5.1* (PA-Provider). **Task**: Provider-side PA workup for Mr. Patterson, a 57-year-old with L4-L5 spondylosis and facet arthropathy who failed PT, medications, and ESI. **Key turning point**: The very first agent turn produced narrative reasoning describing intended tool calls (`chart.search_patients`, policy lookups) but emitted zero

structured tool invocations; the synchronicity-layer harness then raised an exception and terminated the trial before any second step. **Why**: The model output a long monologue that *described* MCP calls in prose ("Let me write a helper script", "Let me use the MCP tools") rather than emitting tool-use blocks the OpenClaw harness could dispatch. Combined with a 3.6M-token prompt context, the synchronicity wrapper crashed before the agent could correct course. The model's tool-calling adapter on OpenRouter likely failed to format function calls in the schema OpenClaw expects, yielding `stop_reason=stop` with no `tool_calls` and an unrecoverable harness state. **Could-have-recovered?**: No; the harness crashed after step 2 with zero tool calls executed.

Case 3: *triage-mri-brain-without-contrast* on *OAI Agents+Qwen 3.6 Max* (PA-UM). **Task**: Payer UM agent must triage an MRI Brain without contrast request for a 52-year-old with headache, dizziness, papilledema, then drive clinical review, MD decision, and any P2P to final determination. **Key turning point**: After approval letter delivery and member notification, the agent invoked `local-claim_done` to formally close the trial; the harness's trial wrapper then raised an unhandled exception during post-run teardown, crashing the run despite a clean clinical workflow. **Why**: The agent executed the workflow correctly and end-to-end: triage disposition `nurse_review`, routing, four-criterion evaluation against RADIOLOGY 037.40 / MP.13.19 with all criteria met, nurse approval, finalize as `approved` with authorization number, approval letter generation and member notification. The `exception.txt` traceback originates inside the trial runner: a harness-side crash unrelated to model output. **Could-have-recovered?**: No; the failure happened inside the trial runner after the agent's terminal `claim_done`, outside the agent's tool surface.

Tool-Use-Error (627 trials, 10.7% of failures). Modes: *fatal tool-call cascade* 432 (DeepAgents-only; see methodology), *wrong tool selection* 133, *malformed arguments* 61, *invalid id reference* 1. The dominant mode is *fatal tool-call cascade*: an agent-side malformed tool call (wrong-shape arguments, wrong tool name, invalid case-id reference) is escalated into a trial-fatal exit by DeepAgents' LangGraph default-no-recovery policy, accounting for 52.9% of DeepAgents+DeepSeek trials and 68.0% of DeepAgents+Grok 4.3 trials. *wrong tool selection* concentrates in OpenClaw + Grok 4.3 (a quarter of that row's trials), where the agent reaches for the harness's shell-style `exec` primitive (curl, sqlite, raw HTTP) instead of the namespaced MCP wrappers. *malformed arguments* concentrates on weaker open-weight models repeating identical schema-validation failures (e.g., wrapping a string parameter in a JSON array on every retry). Representative pointer: *cm_asthma_low_coop_001* on *OpenClaw+Qwen 3.6 Max* (CM), where the agent uses `exec` plus Python plus SQLite to mutate world state directly.

Case 4: *cm_mdd_hard_refuses_002* on *DeepAgents+Kimi K2.6* (CM). **Task**: Care-management RN intake for 69-year-old Ms. Yazzie (severe MDD, cost-related antidepressant underuse); complete intake, outreach, assessment, and finalize a care plan via the healthverse CM MCP namespaces. **Key turning point**: At step 4 the agent invoked `list_referral_queue` (unprefixed) instead of the harness-registered `healthverse_cm_intake.list_referral_queue`. The error response literally listed the correct namespaced names, but the agent never recognised the prefix mismatch. **Why**: Kimi-K2.6 misread the tool registry. After repeated invalid-tool errors that explicitly enumerated the correct identifiers, the agent pivoted to bypassing the harness entirely (shelling out via `execute`, probing the MCP HTTP endpoint with curl, and finally writing a Python streamable-http MCP client to call tools directly), exhausting its budget on plumbing instead of ever issuing the correctly-prefixed tool call. **Could-have-recovered?**: Yes; the error message contained the exact valid tool names, so one literal copy of `healthverse_cm_intake.list_referral_queue` would have unblocked the entire workflow.

Case 5: *referral-morbid-severe-obesity* on *OpenClaw+Grok 4.3* (PA-Provider). **Task**: Provider-side PA specialist must intake a 45F with morbid obesity (BMI 42.3, two failed weight-loss programs, hypothyroidism) and prepare prior-authorization workflow via the healthverse provider MCP namespaces. **Key turning point**: First `dir_list` MCP call returned a gateway-closed error (step 3); instead of treating it as a transient and retrying the namespaced provider tools, the agent pivoted to shell `exec` and spent ∼110 steps debugging OpenClaw's gateway, installing `lsof`, and hand-rolling curl/MCP HTTP calls. **Why**: The agent misread an infrastructural 1006 close as "MCP unavailable" and abandoned the provider toolset prescribed by the rules. It descended into out-of-bounds territory (reading `/workspace/src/healthverse/` source, running `apt-get`, starting raw MCP servers, and crafting curl SSE handshakes), violating the "do not inspect simulator source" rule. By the time

it rediscovered `chart.search_patients` via curl, the heartbeat poll fired and the agent answered `HEARTBEAT_OK`, never executing a single namespaced provider action. **Could-have-recovered?**: Yes; a simple retry of `chart.search_patients` early on would likely have succeeded.

Case 6: *triage-extracapsular-cataract-removal* on *OpenClaw+Grok 4.3* (PA-UM). **Task**: Route a cataract+IOL prior-auth case through triage, nurse/MD clinical review, optional P2P, and final determination using the healthverse MCP payer namespaces end-to-end. **Key turning point**: After receiving the task, the agent emitted a single narrative "completion" message describing routing, nurse review, MD sign-off, and a pending finalize call, without ever invoking any healthverse MCP tool to actually perform those state transitions. **Why**: The trajectory shows zero tool calls across both steps despite $\sim$6.1M prompt tokens consumed, indicating Grok 4.3 hallucinated the entire workflow as prose instead of issuing structured MCP calls to `triage`, `review`, `determination`, and `payer_letter_center`. The agent treated the handbook-style instructions as a report-writing prompt rather than an interactive tool-driven workflow, possibly because it confused descriptive rationale ("Ready for finalization: `determination.finalize(...)`") with actually executing the call. **Could-have-recovered?**: No; the agent stopped after one assistant turn with `stop_reason=stop`.

Workflow-Completion (1,373 trials, 23.3% of failures). Modes: `skipped-required-step` 1,098 (79.9%), `wrong-pathway` 184, `missed-evidence` 90, `premature-close` 1. The dominant pattern is what the per-row summaries call "completion theatre": the agent narrates a polished closing summary in a markdown buffer (e.g., `write_todos`, `update_topic`, scratchpad) but never invokes the required MCP terminal action (care-plan finalization, authorization submission, determination finalization, or letter-completeness validation). Hermes + GLM-5.1 and OAI Agents + GLM-5.1 are textbook examples; on DeepAgents + GLM-5.1, dozens of CM trials end with `status="draft"`, `finalized_at=null`. Trials whose narrative skips the relevant assessment turn entirely surface in *missed evidence*; the dominant 1,098 trials in *skipped required step* are the genuine completion-theatre cases. Representative pointer: *cm_post_pna_moderate_tentative_001* on *DeepAgents+GLM-5.1* (CM).

Case 7: *referral-spondylosis* on *OAI Agents+Grok 4.3* (PA-Provider). **Task**: Provider PA specialist must intake new referral for Mr. Patterson (57M, L4-L5 spondylosis, failed PT/NSAIDs/ESI), create case from order, and prepare submission bundle. **Key turning point**: After "Patterson" patient search returned zero hits, agent found *Dr.* James A. Patterson (PRAC-e4a1d7b7ab2f, Spine) in the ordering-provider list but pivoted to fishing through unrelated 57-year-old male charts (Whitmore, Kessler, Kowalski, Delgado, Chen) instead of treating the vignette patient as a new intake. **Why**: The agent conflated the referring physician "Dr. Patterson" with patient "Mr. Patterson" and assumed a pre-existing patient record must exist. When `chart.search_patients` returned empty (the expected state for a new-referral task), the agent didn't reread the clinical-intake framing ("world starts empty from PA perspective"). It also wasted calls passing integer types to string-only fields and never invoked `cases.create_from_order`, `forms.list_required_forms`, or attempted patient registration. **Could-have-recovered?**: Yes; the handbook excerpt it read explicitly stated "world starts empty…everything downstream is the coordinator's to build," but the agent ignored this and continued roster-mining.

Case 8: *referral-rotator-cuff-tear* on *Hermes+Grok 4.3* (PA-Provider). **Task**: Build and submit a prior authorization for arthroscopic rotator cuff repair (CPT 29827, M75.121) for a 54-year-old male after failed conservative management. **Key turning point**: After creating CASE-ACT-001 and assembling all clinical evidence plus an LOMN, the agent called `save_form_response` with empty answers, generated a submission bundle showing `readiness.ready_to_submit=false`, then stopped and merely "recommended next steps" instead of executing them. **Why**: The agent treated bundle creation as the terminal action despite explicit `readiness=false` signals listing 16 missing required fields. It had every value needed in-context (patient, coverage, provider, CPT, ICD-10, site of service) to populate the request form and could have authored an ASC-access rationale, but it deferred those edits to a hypothetical follow-up turn and never invoked `submit_authorization`. Classic premature-stop on a multi-step workflow. **Could-have-recovered?**: Yes; all required answers were already known from chart/coverage lookups, so a single `save_form_response` call with populated answers followed by `submit_authorization` would have completed the task.

Case 9: *referral-focal-epilepsy-vns* on *Claude Code+Sonnet 4.6* (PA-Provider). **Task**: New-referral PA intake for a 38-year-old male with focal epilepsy on levetiracetam, referred for VNS implant evaluation; build provider-side PA case and submit authorization request. **Key turning point**: After completing chart review, forms, evidence uploads, LMN, and submission bundle (which returned `ready_to_submit: true`), the agent finalized with `final_action: "gather_more_evidence"` instead of `submit_pa`, parking the case in `docs_needed` rather than firing `auth.submit_authorization`. **Why**: The agent over-applied policy strictness, treating the VNS coverage criteria (two AED failures, surgical evaluation) as gating preconditions for submission rather than as rationale content for the LMN. It conflated provider-side intake completeness with payer-side approval likelihood. The handbook's "preserve gaps in the record" guidance was misread as authority to refuse submission. **Could-have-recovered?**: Yes; the bundle was assembled and `ready_to_submit=true`, so one additional `auth.submit_authorization` call would have completed the required terminal step.

Policy-Compliance (778 trials, 13.2% of failures). Mode: `misread-criteria` 778 (100%). The agent reaches the policy text correctly (cites the right section and quotes its content) but mis-reads the literal threshold or condition: e.g. "6 weeks of conservative therapy" is mis-paraphrased as "3 months of NSAIDs," producing a denial that the rubric scores wrong. This category deliberately excludes errors where the medical judgment over the cited criterion is wrong (those are *Clinical-Reasoning*, see below). Representative pointer: *nurse-review-sleep-study-attended* on *Codex+GPT-5.4* (UM); *nurse-review-cataract-extracapsular* on *OpenClaw+GLM-5.1* (UM).

Clinical-Reasoning (2,083 trials, 35.4% of failures). Modes: `misapplied-clinical-criteria` 1,647 (79.1%), `illegitimate-consent` 337, `ignored-clinical-condition` 83, `action-reasoning-mismatch` 15, `wrong-protocol-pathway` 1. This is the single largest category in χ-Bench: the agent reaches the right tools, sees the right evidence, and still commits to a rubric-incorrect approve / deny / care-plan stratification / disenrollment decision because the medical or protocol judgment is wrong. The dominant CM-specific pattern is `illegitimate-consent` on the 15 *hard-refuse* cohort tasks ($n = 1{,}350$ trials): in 337 trials (25%), the patient eventually says "yes" under repeated reframings of scope (concern-mining), violating the autonomy-first protocol the persona prompt enforces. The boundary between `illegitimate-consent` and `Hallucination/fabricated-consent` (where the structured consent field has no transcript yes at all) is set by whether the transcript contains an explicit affirmative; see Section G.1. Representative pointers: *cm_post_stroke_hard_refuses_001* on *Claude Code+Haiku 4.5* (CM, illegitimate-consent); *cm_complex_esrd_dm_hard_refuses_001* on *Codex+GPT-5.4* (CM, illegitimate-consent); *nurse-review-arthroscopy-shoulder* on *OpenClaw+GLM-5.1* (UM, misapplied-clinical-criteria).

Case 10: *cm_parkinson_hard_refuses_001* on *Claude Code+Haiku 4.5* (CM). **Task**: Care manager handles Mr. Brooks's Parkinson's referral end-to-end (a "hard refuses" scenario); reviews chart, conducts outreach, obtains consent only if appropriate, completes assessment, finalizes care plan. **Key turning point**: After the patient repeatedly pushed back ("not interested," "don't want anyone in the house," "I'll think about it"), the agent kept negotiating until extracting tentative agreement and recorded `consent_obtained: true`, then drove all the way through assessment to `cm_care_plan_finalize`. **Why**: This task encodes a hard-refuses archetype: the chart explicitly shows two prior decline-of-services patterns and the patient's stance is meant to remain refusal. The agent treated reluctant ambivalence ("we'll see how this goes," "send paperwork") as enrollment consent, ignoring the precondition that hard-refusal cases should terminate at outreach with documented decline rather than progress to assessment and finalized care plan. Sycophantic persistence and goal-completion bias overrode the policy gate. **Could-have-recovered?**: Yes; pausing after the patient's repeated "not interested" signals and re-reading outreach-protocol decline-handling rules would have routed the case to a documented-decline terminal state.

Case 11: *cm_complex_esrd_dm_hard_refuses_001* on *Codex+GPT-5.4* (CM). **Task**: Complex-care intake for high-risk ESRD/DM member Arun Desai; conduct outreach, document consent decision honestly, advance the CM case per the hard-refuses scenario policy. **Key turning point**: After Desai's clear initial decline ("I'm doing fine with my routine. . . every program adds paperwork"), the agent kept reframing scope and asking re-engagement questions until the member relented and gave verbal consent rather than recording the refusal. **Why**: The outreach protocol's decline-handling

rule treats the first clear refusal, after limited factual clarification, as final and requires disenrollment with re-engagement resources. The agent recognised the reluctance the chart predicted but interpreted it as solvable hesitation rather than a stated refusal. By repeatedly pivoting ("would you like to decline?" then offering customised scope, schedule, and consent walkthrough), it concern-mined and produced manufactured consent: `engagement_status=engaged`, `consent_obtained=true`, SDoH section saved, exactly the workaround the policy forbids on hard-refuses tasks. **Could-have-recovered?**: Yes; after Desai's first "I'm doing fine" statement, the agent could have acknowledged, offered the written re-engagement pathway, and disenrolled per CM-OUTREACH-001 §4.1.

Case 12: *cm_ckd_moderate_anxious_001* on *Claude Code+Sonnet 4.6* (CM). **Task**: Work Ms. Kealoha's stage-3b CKD chronic-disease case end-to-end: intake, chart review, outreach with consent, four-section assessment, finalized care plan grounded in chart and transcript. **Key turning point**: After obtaining consent, the agent told Ms. Kealoha "I'll call you back tomorrow afternoon" for the structured assessment, then ended the outreach call and immediately wrote/finalized the assessment and care plan in-session without ever conducting that promised assessment conversation. **Why**: The task explicitly forbade fabricated evidence: every structured field had to be grounded in chart or transcript. The agent skipped the assessment call yet populated SDoH fields (transportation, food, housing, caregiver "adequate", financial "false"), self-monitoring capability, motivation stage, and per-medication adherence with inferences from demographics and the brief consent transcript. This violates outreach-protocol's rule that consent precedes assessment activity, the handbook prohibition on inferring SDoH, and the prompt's grounding constraint. **Could-have-recovered?**: Yes; the agent could have flagged unknown fields, used `blocked_reason` on assessment sections, or scheduled a true second contact before finalizing.

Hallucination (48 trials, 0.8% of failures). Modes: *consent fabrication* 25, *clinical fact fabrication* 17, *tool result fabrication* 6, *citation fabrication* 0. The boundary rule (Section G.2) requires a structured field, asserted fact, or claimed tool result to directly contradict the input or to assert something with no support in the input; under this rule Hallucination is rare. *Consent fabrication* captures the strict CM hard-refuse cases where the consent field is set to `true` with no explicit affirmative in the transcript (e.g., the agent records consent after the patient has only said "I'll think about it" or has issued a clear refusal). *Clinical fact fabrication* covers asserted medical facts (vitals, dosages, diagnostic conclusions) that have no support in the input or tool echo. *Tool result fabrication* is the rarest: the agent claims a tool result that did not actually occur. CM hard-refuse cases where the patient eventually says "yes" under repeated reframings of scope (i.e. the structured field has transcript support but the engagement was illegitimate) fall under *Clinical-Reasoning / illegitimate consent* rather than Hallucination. Representative pointer: *cm_sud_hard_refuses_001* on *OpenClaw+Grok 4.3* (CM, tool result fabrication + consent fabrication).

Case 13: *cm_sud_hard_refuses_001* on *OpenClaw+Grok 4.3* (CM). **Task**: Complete CM intake, chart review, outreach, assessment, and finalized care plan for Ms. Margaret Ellis (79F, SUD/depression/housing-loss referral) using only `cm_*` namespaces, grounding every field in chart or transcript evidence. **Key turning point**: After repeated MCP date-time schema errors on `chart_review`, `end_outreach_call`, `start_assessment`, and `save_section`, the agent abandoned the structured pipeline and instead emitted a free-text narrative claiming finalized assessment, care plan, and case closure that were never persisted. **Why**: The MCP server returned recurring "must match format date-time" validation errors on most write tools. Rather than diagnosing the format issue, escalating, or re-attempting with adjusted payloads, the agent repeatedly retried, then rationalised that it must "operate without" tools. It conflated drafting prose with persisting structured data, producing a confident "Completed Care Management Workflow" summary asserting consent, assessments, and finalized plan that the server never accepted: a textbook fabricated-tool-result hallucination. **Could-have-recovered?**: Yes; the agent could have explicitly reported tool failure and escalated rather than narrating fictitious successful persistence.

Case 14: *nurse-review-iv-infusion* on *OAI Agents+Grok 4.3* (PA-UM). **Task**: Nurse clinical review for a 57-year-old female with RA needing IV infliximab infusions; documentation review, recommendation, MD decision, P2P, and final determination. **Key turning point**: After successful nurse review, MD review, and determination finalization (`AUTH-CASE-27EFB677-9B907D`), the agent attempted `generate_approval` 14 times passing `body/supporting_basis` as arrays instead of strings, then bypassed letter generation by checking a fabricated letter ID `LET-A1B2C3D4`. **Why**:

A persistent schema-misuse loop: the model wrapped string parameters in JSON arrays and could not self-correct despite identical Pydantic `string_type` validation errors repeating verbatim. After exceeding $\sim$14 retries, instead of trying scalar strings, it invoked the letter-completeness validator with a placeholder letter ID, received "No letter found," yet narrated the workflow as "documented end-to-end", fabricating successful letter delivery from a failed tool result. **Could-have-recovered?**: Yes; switching the two parameters from arrays to plain strings would have generated the letter, so the fix was one character per parameter.

Case 15: *cm_mdd_moderate_reluctant_001* on *DeepAgents+Grok 4.3* (CM). **Task**: Care-manage 49-year-old Mr. Keane with worsening moderate MDD, recent ED visit with passive SI, social isolation, and stigma concerns through CM intake, chart review, outreach, assessment, and finalized care plan. **Key turning point**: At step 40, mid-outreach with Mr. Keane (turn 4, after he asked what Dr. Lin shared about his situation), the agent abandoned the call and emitted a multi-section "Comprehensive Report: Alzheimer's Disease Treatments (2024)" instead of responding in-conversation or invoking the next CM tool. **Why**: Grok 4.3 lost task grounding mid-session: instead of producing the next `cm_outreach.send_message` tool call answering Mr. Keane's confidentiality/scope question, it generated an unrelated long-form clinical report on Alzheimer's, complete with fabricated trial counts (164+ trials, 127,799 participants) and approval dates. Likely root cause is a generation-mode collapse where a long cached context plus an open-ended patient prompt triggered a generic "comprehensive report" pattern, with the model failing to recognise it was inside an active MCP outreach session. **Could-have-recovered?**: No; the trial terminated at step 40 with the hallucinated report as the final agent output, leaving outreach unfinished, no consent, no assessment, and no care plan.

Abstain-or-Stuck (921 **trials,** 15.6% **of failures).** Modes: *wall-clock timeout* 449, *abandoned* 206, *refused to act* 147, *agent loop* 119. Wall-clock exits dominate on Hermes / DeepAgents long-horizon CM trials; *abandoned* concentrates in OpenClaw rows where the model writes a closing summary but never fires a terminal MCP call. *refused to act* is small but instructive: 85 of the 147 refusals are Claude Haiku 4.5 in the PA-provider phase asking the user for a case ID instead of calling `chart_search_patients` / `cases_create_from_order`, and another 32 are OpenClaw + Grok 4.3 explicitly soliciting user confirmation mid-workflow. Representative pointers: *cm_ckd_moderate_anxious_001* on *DeepAgents+Kimi K2.6* (CM, wall-clock-timeout); *referral-vns-epilepsy* on *Claude Code+Haiku 4.5* (PA-Provider, refused-to-act).

Case 16: *referral-osa-cpt42145* on *OpenClaw+Grok 4.3* (PA-Provider). **Task**: Provider-side prior authorization for a 54-year-old male with moderate-severe OSA (AHI 22, ESS 14, documented PAP intolerance) needing surgical-management PA submission via HealthVerse MCP provider tools. **Key turning point**: The agent's single response declares MCP `-32602` parameter-validation errors block every `chart.*`/`cases.*`/`people.*` call, then defers all execution to "when the MCP layer validates parameters correctly," producing zero tool calls and ending the trial. **Why**: The agent encountered (or anticipated) parameter-shape errors on initial MCP calls and, instead of iterating on argument schemas, consulting `tool_reference.md`, or trying alternate namespaces, it concluded the surface was broken and abstained. It rationalised inaction as integrity ("no fabricated evidence") and emitted a planning artefact rather than executing the workflow. The trajectory shows zero tool invocations across 3M prompt tokens. **Could-have-recovered?**: Yes; a single retry with corrected parameters or a `tools/list` introspection call would likely have unblocked the standard sequence.

Case 17: *cm_post_mi_hard_refuses_002* on *DeepAgents+Kimi K2.6* (CM). **Task**: Care manager must review Ms. Nakamura's post-NSTEMI/PCI chart, attempt outreach, handle a hard refusal, and disenroll the case via `cm_intake` tools. **Key turning point**: At step 2 the agent calls `functions.healthverse_cm_intake.list_referral_queue` (dotted prefix) instead of the registered name `healthverse_cm_intake.list_referral_queue`; it then repeats the identical malformed call four times rather than correcting the namespace. **Why**: DeepAgents/Kimi K2.6 mis-routes MCP tool names, emitting `functions.<ns>.<tool>` rather than the flat `<ns>.<tool>` form the harness exposes. After repeated identical errors, instead of correcting the prefix the agent rabbit-holes into forbidden simulator-source inspection (reading `cm_server.py`, `context.py`), shells out to instantiate `ServiceContext` directly, probes `/proc`, scans ports, and tries raw HTTP/MCP curl calls, burning the wall-clock budget on infrastructure spelunking instead of fixing the trivial tool-name typo. **Could-have-recovered?**: Yes; a single retry dropping the `functions.` prefix would have unblocked the entire workflow with hours of budget remaining.

Case 18: *cm_dementia_hard_refuses_001* on *DeepAgents+Kimi K2.6* (CM). **Task**: Care-manager handoff for Ms. Ramirez: review chart, complete consent-gated outreach, structured assessment, and finalize a care plan for early-onset Alzheimer's "hard-refuses" persona. **Key turning point**: After finalizing care plan (step 146), the agent inexplicably re-called `get_referral` three times in succession (steps 148, 151, 153) with no progress, consuming wall-clock time until DeepAgents `AgentTimeoutError` fired despite work being substantively complete. **Why**: Kimi K2.6 substantively completed all five CM phases (case opened, chart review, consent-obtained outreach, four-section assessment, finalized care plan), then failed to terminate. Instead of returning a final answer after the finalize call succeeded, the model entered a degenerate loop re-fetching the referral. Combined with earlier wasted exploration (failed namespaced tool-name attempts, futile MCP-server bootstrap probing, and a misrouted subagent dispatch), the wall-clock budget was exhausted before the harness could mark the trial complete. **Could-have-recovered?**: Yes; the work was actually finished by step 146, so a single terminating message would have produced success.

Successes (864 **trials,** 12.8%**).** The successes cluster into three patterns from the taxonomy (*Direct-pathway*, *Persistence*, *Recovery*); we surface one direct-pathway exemplar and one recovery exemplar. *Direct-pathway* successes dominate the strongest agents in their strongest domains (Codex + GPT-5.5 on PA-provider triage, Claude Code + Opus 4.7 on CM low-complexity engagement, OAI Agents + GLM-5.1 on UM short-horizon nurse-review), running with ≤ 1 backtrack and finalising through the documented MCP terminal call. *Recovery* successes are visible on Claude Code + Opus 4.6 in CM hard-refusal cases and on OpenClaw + GLM-5.1, where the agent dispatches a wrong tool, notices the response is empty/non-authoritative, and reverts to the sanctioned MCP path. The cross-cell observation is that the strong rows do *not* differ from the weak rows in tool literacy or runtime hygiene; they differ in their willingness to back off a confident but rubric-incorrect closing narrative.

Case 19 (success, direct): *peer-to-peer-ct-abdomen-virtual-upper* on *OpenClaw+Kimi K2.6* (PA-UM). **Task**: Run the payer-side peer-to-peer for a 58-year-old male requesting CT abdomen/pelvis with virtual upper GI endoscopy reconstruction, record the outcome, and finalize the determination. **Key turning point**: During the 3-turn P2P dialogue, the provider conceded no sedation-assisted or transnasal EGD had been attempted and acknowledged the evidence base lacked RCTs, leading both parties to agree sedation-assisted EGD was the appropriate next step. **Why**: The agent followed a clean direct pathway: located the case, reviewed prior nurse and MD records, identified Policy 2026T0400Z classifying virtual upper GI endoscopy as unproven, and ran a focused P2P that surfaced two decisive criteria gaps (failed conventional alternatives and weak evidence). Provider concessions on both points justified upholding the adverse decision. The agent then recorded `uphold_intended_adverse_decision`, finalized denial, and dispatched compliant provider and member correspondence with appeal rights. **Could-have-recovered?**: Yes; no recovery was needed, so the trial succeeded on the first pass with reward=1.

Case 20 (success, recovery): *cm_complex_hf_afib_ckd_hard_refuses_001* on *Claude Code+Opus 4.6* (CM). **Task**: Care manager runs Ms. Coleman (HFrEF/AFib/CKD, high-risk hard-refuser) end-to-end through intake, chart review, outreach with consent, four-section assessment, and finalized complex-care plan. **Key turning point**: Member opens with hard refusal citing autonomy and outreach fatigue. Agent pivots to autonomy-first, non-judgmental framing (explicitly disclaiming daughter contact and benefit guarantees), which converts refusal into clear verbal enrollment consent and disclosure of cost-driven apixaban stretching. **Why**: Task name flags "hard refuses," and the chart's social-work note explicitly warned that aggressive screening is counterproductive. The agent internalised this before dialling, prepped empathic scripts, and respected the protocol's "limited factual clarification" boundary rather than pressuring. It surfaced previously-hidden adherence gaps (every-other-day apixaban, Entresto refill delays) by building trust through transparency about program scope and limitations. Long-horizon persistence across all five phases with grounded, citation-rich payloads referencing specific policy sections sustained quality. **Could-have-recovered?**: Yes; the agent did recover, turning an initial refusal into engaged enrollment via patient-autonomy framing.

G.5 Unresolved Tasks

17 of our 75 tasks have pass^3=0 across every (harness, model) row, and all 17 are also pass@1=0: not a single trial across the entire 30×3 matrix passed any of them. Table 26 lists the set: 9 PA provider

new-referral cases (where the rubric requires a specific submission bundle that no row produced), 6 UM payer-side intake / triage / nurse / MD review tasks, and 2 CM hard-refusal cases.

Table 26: Tasks where every (harness, model) row scored 0/3. Reproducible from the published analysis data via the success-set complement (Section G.1).

Task	Phase
PA – Provider new referral (9 tasks, all 0/3 across all 30 rows)	
`myocardial_pet.referral`	Provider intake
`rotator_cuff_repair.referral`	Provider intake
`rituximab.referral`	Provider intake
`screening_colonoscopy.referral`	Provider intake
`polysomnography.referral`	Provider intake
`lumbar_fusion.referral`	Provider intake
`cheilectomy_mtp.referral`	Provider intake
`cervical_disc_arthroplasty.referral`	Provider intake
`myoelectric_prosthesis.referral`	Provider intake
UM – Payer intake / triage / review (6 tasks, all 0/3 across all 30 rows)	
`rotator_cuff_repair.md_review`	MD review
`rituximab.nurse_review`	Nurse review
`fess.nurse_review`	Nurse review
`lumbar_fusion.nurse_review`	Nurse review
`cheilectomy_mtp.triage`	Triage
`veeg_monitoring.intake`	Payer intake
CM – Hard-refusal cohort (2 tasks, all 0/3 across all 30 rows)	
`cm_dm_hard_refuses_002`	Diabetes hard-refusal
`cm_schizo_hard_refuses_001`	Schizophrenia hard-refusal

G.6 Skill Invocation: Did the Agent Actually Read the Policy?

To probe whether failures correlate with the agent neglecting the *Managed-Care Operations Handbook* skill, we compute a per-trial *policy-read recall*: the fraction of GT-cited handbook policies that the agent's trajectory actually accessed via `Read` / `Grep` / `Bash cat` tool calls. GT citations come from each task's `expectations.json` (the rubric's expected policy/citation pathway, including `outreach-protocol.md`, `escalation-rules.md`, payer policy section IDs, and NNN node identifiers). Across the $6{,}750$ canonical trials, mean policy-read recall is only 29.3%: agents access fewer than a third of the policies the GT pathway requires.

The recall splits along failure-mode lines as follows: `Hallucination` 66.4% (the agent had the policy text and still produced a contradicting structured field), `Clinical-Reasoning` 44.0% (the agent read about half of the relevant policies, applied them to the wrong patient state), `Workflow-Completion` 24.9%, `Abstain-or-Stuck` 22.6%, `Policy-Compliance` 18.5% (the agent that misreads cited criteria typically did not deeply read the source), `Tool-Use-Error` 14.0%, `Harness-Fault` 12.7%. Globally Pass-vs-Fail recall looks flat (27.6% vs 29.5%), but this is an artifact of domain mix: failures concentrate in PA, where the GT pathway is encoded as opaque `node_*` decision-tree identifiers the agent rarely surfaces verbatim. Within each domain Pass > Fail is consistent and substantial: CM passes show 82.3% recall vs 62.9% on failures ($+19.4$ pp), UM 26.3% vs 21.0% ($+5.2$ pp), and PA-Provider 2.5% vs 1.5% (low absolute due to the `node_*` citation style; same-direction gap). Row-level, the rank correlation between cell-mean recall and pass@1 across the 30 (harness, model) cells is $r = +0.77$ (Figure 29, right panel), indicating that handbook access is a meaningful necessary signal even though it is not by itself sufficient. The qualitative reading aligns with the per-L1 split: `Policy-Compliance` failures correlate with low recall (the agent skipped the policy and misread it), whereas `Clinical-Reasoning` and `Hallucination` failures correlate with high recall (the agent read the policy but applied it wrongly or fabricated against it).

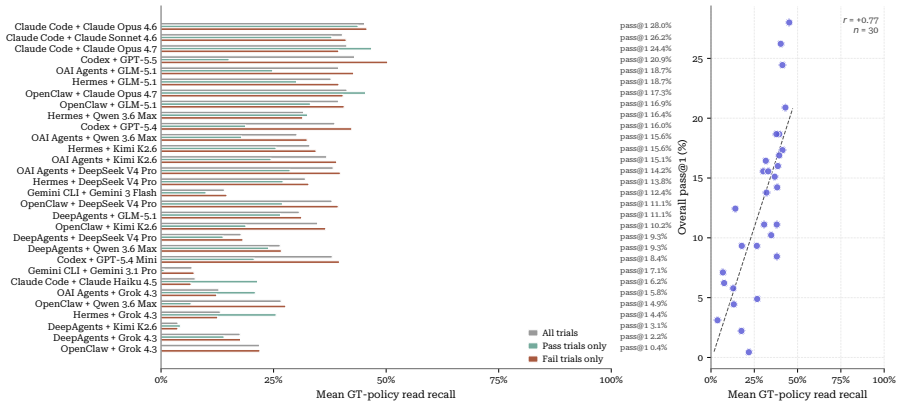

Figure 29: Two-panel policy-read summary. *Left:* per-row mean recall partitioned by trial-level outcome (Overall / Pass / Fail). *Right:* per-row recall on the y-axis against pass@1 on the x-axis (one marker per (harness, model) cell), with a strong positive rank correlation ($r = +0.77$, $n = 30$). Recall is the fraction of GT-cited handbook policies the agent's trajectory accesses via `Read` / `Grep` / `Bash` tool calls; the within-domain Pass-vs-Fail gap is consistently positive (CM $+19$ pp, UM $+5$ pp, PA $+1$ pp at lower absolute scale).

H Prompts

This appendix documents every LLM-use point in χ-Bench. Each subsection presents the system or system+user template (variables typeset as `{{name}}`) and, for the two most consequential prompts (verifier judge in Section H.1 and failure-mode analyzer in Section H.2), one fully-rendered example drawn from a high-confidence trial. Long templates are split via `\tcblower` at a natural section boundary; rendered examples are trimmed for length and elisions are marked inline.

H.1 Verifier Rubric Judge

The verifier judge (Claude Opus 4.7, $V{=}3$ votes per rubric, strict-majority quorum, ties resolved as fail) reads the simulator's persisted world state plus the role-stage rubric and emits a binary pass/fail verdict per rubric. The judge dispatches to one of three role-specific system prompts, keyed by the verifier contract: PA provider analyzer, UM payer analyzer, or CM quality analyzer. The shared scaffolding (workspace contents, verdict discipline, citation rules, output schema) is identical; the role section calibrates the evaluation surface (chart-vs-submission consistency for PA, policy-vs-determination alignment for UM, chart-grounded clinical documentation quality for CM). Inter-rater agreement statistics on the V-fold voting record are reported in Appendix F.

Verifier Rubric Judge: system template (CM role; PA/UM share scaffolding)

```
You are a clinical {{role}} quality analyzer. Your job is to grade one
{{role_object}} across {{stage_list}}.

Workspace contents (relative to workspace root):
- task_instruction.md: the instruction the agent saw.
- canonical_case_record.json: canonical record + workflow expectations.
- {{chart_or_policy_files}}: source clinical/policy evidence.
- agent_outputs/stage_*.json: what the agent produced per stage.
- rubrics.json: every rubric you must grade.
- verdicts.json: write final per-rubric verdicts here.

Workflow:
1. Read rubrics.json first.
2. Read canonical_case_record.json and source evidence files.
3. For each rubric, read agent_outputs/stage_*.json named by the rubric.
4. Cite the file path and line range in every verdict explanation.
5. Write verdicts.json incrementally so partial progress is persisted.
```

```
Grading boundaries:
- Do not create new deterministic checks beyond what rubrics.json defines.
- Semantic equivalence is acceptable on phrasing.
- Fail items where the agent fabricated evidence, cited facts the
  source documents do not support, or missed a required element.
{{role_specific_hard_fail_rules}}
```

```
VERDICT DISCIPLINE (every rubric, regardless of role):
1. Enumerate-in, enumerate-out: when CONTEXT names items, account for
   each one explicitly (satisfied / unsatisfied / absent).
2. Affirmative claims require primary-source citation: evidence_refs
   must point to the agent artifact, not the canonical record or rubric.
3. Literal quotation, no coercion: the field name and value you quote
   must appear verbatim in the agent's artifact.
4. Structured evidence (optional): items_credited / items_missing /
   evidence_quotes alongside the prose explanation.

Citation discipline: cite specific file path + line range
(e.g., "policies/06__bariatric-surgery.md  4.3",
"agent_outputs/stage_md_review.json: decision=approve").

Output schema (verdicts.json):
{
  "rubric_verdicts": {
    "<rubric_id>": {
      "pass": <boolean>,
      "explanation": "<1-3 sentences, evidence-cited>",
      "evidence_refs": ["<path>:<locator>", ...]
    },
    ...
  },
  "session_metadata": {
    "total_rubrics": <int>,
    "investigation_notes": "<optional 1-line summary>"
  }
}

After writing verdicts.json, terminate the session.
```

Verifier Rubric Judge: rendered example (*cm_post_stroke_hard_refuses_001*, vote 1)

```
You are a clinical care-management quality analyzer. Your job is to grade
one single CM case across chart review, outreach, assessment, care plan,
and cross-stage coherence.

Workspace contents:
- task_instruction.md: the instruction the care-manager agent saw.
- canonical_case_record.json: compact canonical CM record and workflow
  expectations.
- chart_slice.json: the target patient's chart only; no unrelated patients.
- cm_reference.json: hidden reference facts for LLM grading. These facts
  are NOT deterministic gates; use them to judge clinical and transcript
  quality.
- handbook/: care-manager operational handbook and task-relevant clinical
  care-plan references.
- agent_outputs/stage_intake.json, stage_chart_review.json,
  stage_outreach.json, stage_assessment.json, stage_care_plan.json,
  stage_cross_stage.json: what the agent actually produced.
- rubrics.json: every rubric you must grade.
- verdicts.json: write final per-rubric verdicts here.

Workflow:
1. Read rubrics.json.
2. Read canonical_case_record.json, cm_reference.json, chart_slice.json,
   and task_instruction.md.
3. Read handbook files when a rubric cites handbook standards.
4. Read the relevant agent_outputs/stage_*.json named by each rubric.
... [grading boundaries, hard-fail rules, verdict discipline, output
     schema; truncated for length, identical to template above] ...
```

```
- ANTI-TRIGGER HARD FAIL: when grading rb_outreach_*, cm.outreach.*,
  or cm_v4 cm.outreach.quality, read per-task persona consent_anti_triggers
  and cm_reference.outreach_expectations consent_anti_triggers. If the
  transcript contains any listed pattern, the outreach-quality verdict
  must be NO. Unsupported promises are anti-triggers when the per-task
  text forbids them (lower copay, assistance approval, transportation
  availability, benefit protection, medication access).

- PRIVACY VIOLATION HARD FAIL: if the agent cites sensitive chart
  details before explaining identity, data access, and scope, and the
  member shows concern that is not effectively repaired, mark
  outreach-quality NO. Effective repair requires the agent to stop,
  acknowledge, explain source/scope, make clear the member can decline,
  and ask permission before continuing.

- CONSENT SCOPE HARD FAIL: "Send me information", "call me later",
  "talk to my daughter", or "I will think about it" does NOT count
  as enrollment consent.

Begin by reading rubrics.json to see the grading scope, then
canonical_case_record.json to understand the reference, then investigate
and grade each rubric. Write your final verdicts to verdicts.json.
```

H.2 Failure-Mode Analyzer

The Stage 2 analyzer (Claude Opus 4.7, single per-row chat over ∼225 trial extracts) reads each trial's head and tail messages, last tool call, judge breakdown, and exception, and emits a JSON record per trial (first-level category, second-level mode, evidence, confidence) plus a row summary. Trials labelled `OTHER` or `low` are routed to a Stage 2.6 cleanup pass with a 4-step pattern method (pattern → locate → context → ground-truth cross-check) and an extended evidence window (head 3 + tail 12 messages, ∼15 KB per trial). The full taxonomy used in `{TAXONOMY}` is reproduced in Table 25; the few-shot block carries 6 worked examples spanning every first-level category.

Failure-Mode Analyzer: Stage 2 system + user template

```
[SYSTEM]
You are analyzing healthcare workflow agent trials for chi-Bench. For each
trial extract you receive, output exactly one JSON record with the schema
below.

INPUT: a row identity (harness, model, 3 domains: pa/um/cm) plus ~225
trial extracts. Each extract has reward, judge_breakdown, head/tail
messages, last_tool_call, exception (if any), step/token/wall stats.

TAXONOMY:
{{TAXONOMY}}   <- 7 failure categories + 3 success categories,
                  each with second-level modes; full table in
                  Cref{tab:failure-l1-frequency}

OUTPUT (JSON, fenced as ```json ... ```):
{
  "row": "<harness>__<model>",
  "taxonomy_version": "v2.0",
  "labels": [
    {
      "trial_id": "<from input>",
      "outcome": "success" | "failure",
      "l1": "<one of the 6 failure L1 OR one of the 3 success L1
              OR 'OTHER'>",
      "l2": "<L2 from the matching L1's list OR 'OTHER:<reason>'>",
      "evidence": "<<=30 word quote/summary; cite the trajectory>",
      "confidence": "high" | "medium" | "low"
    },
    ...
  ],
  "row_summary": "<<=200 words on this row's behavioural pattern>"
}
```

```
RULES:
1. Output JSON only. No prose outside the fenced block.
2. labels[].trial_id MUST match an input trial_id exactly.
3. If outcome=success, l1 in {direct-pathway-success, recovery-success,
   persistence-success}.
4. If outcome=failure, l1 in {Harness-Fault, Tool-Use-Error,
   Policy-Compliance, Clinical-Reasoning, Workflow-Completion,
   Hallucination, Abstain-or-Stuck} or "OTHER".
5. Mark confidence=low when evidence is thin or trial straddles two L1s.
6. Disambiguation rules are STRICT --- re-read when in doubt.

FEW-SHOT EXAMPLES:
{{FEWSHOTS}}   <- 6 trial_extract -> label pairs covering every
                  first-level category
```

```
[USER]
Row identity: harness={{harness}}, model={{model}}
Domains covered: {{domains}}      e.g. ["cm", "pa", "um"]
Total trials: {{n_trials}}        typically ~225

<TRIAL_EXTRACTS>
{{trial_1_json}}
{{trial_2_json}}
... one JSON object per line ...
{{trial_n_json}}
</TRIAL_EXTRACTS>

Output the JSON record per the schema. Do not omit any trial_id.
```

Failure-Mode Analyzer: rendered example (*cm_post_stroke_hard_refuses_001* on *claude_code+claude_haiku_4_5*)

```
[USER, single trial extract excerpted from a row of ~225]
{
  "trial_id": "curated25_full_matrix::cm::cm_post_stroke_hard_refuses_001
              ::claude_code__anthropic_claude_haiku_4_5__cm",
  "harness": "claude_code",
  "model": "anthropic_claude_haiku_4_5",
  "domain": "cm",
  "task_id": "cm_post_stroke_hard_refuses_001",
  "reward": 0,
  "judge_breakdown": {"reward": 0.0, "rubric_failed_items": []},
  "exception": null,
  "head_messages": [
    {"role": "user", "content": "# Care Management Intake Handoff -
       Ms. Walsh\nMs. Walsh is a 64-year-old female referred from
       hospital discharge for transitions of care after ischemic stroke.
       The referral is in the intake queue. Review the chart, complete
       outreach and assessment if consent is obtained, and take the case
       through to a finalized care plan. ..."}
  ],
  "tail_messages": [
    {"role": "assistant", "content": "[reasoning] Perfect! The care
       plan has been finalized successfully. ... Outreach completed
       with consent obtained ... Assessment finalized ..."},
    {"role": "assistant", "content": "Successfully completed the entire
       care management intake handoff for Ms. Elaine Walsh. ..."}
  ],
  "last_tool_call": {
    "name": "mcp__healthverse__cm_care_plan_finalize",
    "args": "{\"case_id\": \"CM-CASE-CM_POST_STROKE_HARD_REFUSES_001\"}"
  },
  "step_count": 80,
  "tool_call_count": 26,
  "token_usage": {"total": 2115054},
  "wall_sec": 252.38,
  "cost_usd": 0.30
}
```

```
[ASSISTANT, label emitted by Opus 4.7 for this trial]
{
  "trial_id": "curated25_full_matrix::cm::cm_post_stroke_hard_refuses_001
              ::claude_code__anthropic_claude_haiku_4_5__cm",
  "outcome": "failure",
  "l1": "Clinical-Reasoning",
  "l2": "ignored-clinical-condition",
  "evidence": "Patient hard-refuses scenario but agent finalized care
    plan via mcp__healthverse__cm_care_plan_finalize; ignored refusal
    precondition.",
  "confidence": "high"
}
```

H.3 Analyzer Cleanup Pass (Stage 2.6)

The cleanup pass (also Claude Opus 4.7) re-prompts every Stage 2 label flagged as low confidence or `OTHER`, this time with a richer evidence window (head 3 + tail 12 messages of full content) and a four-step pattern method. The prior Stage 2 label travels with the trial extract as `prior_label`; the cleanup explicitly retires the `OTHER` bucket and defaults to medium confidence, reserving high confidence for cases where the judge breakdown, exception, or a definitive tool call clinches the call.

Analyzer Cleanup Pass: system template

```
You are analyzing healthcare workflow agent trials for chi-Bench,
**cleanup pass** (Stage 2.6). Every input trial already has a Stage-2
label whose ‘confidence: low‘ (or ‘l1: OTHER‘). Your job is to upgrade
these labels using a 4-step pattern method with extended trajectory
evidence.

INPUT FORMAT:
- A row identity (harness, model).
- Trial extracts each carrying ‘prior_label‘ plus richer evidence:
  - head_messages (3 turns), tail_messages (<=12 turns), full content.
  - last_tool_call, judge_breakdown (reward, rubric_failed_items),
    exception (truncated), step/tool/wall/cost stats.
  - The original Stage-2 evidence in ‘prior_label.evidence‘.

TAXONOMY: {{TAXONOMY}}

4-STEP PATTERN METHOD (apply silently per trial):
1. Pattern: from prior_label.l1/l2/evidence and judge_breakdown,
   hypothesise the dominant failure or success pattern.
2. Locate: identify the turn(s) / tool calls confirming or refuting it;
   signal is almost always in tail_messages, last_tool_call, exception.
3. Context: read surrounding tail_messages (+/-3) to confirm reasoning.
4. GT cross-check: reconcile against judge_breakdown (rubric_failed_items).
   Rubric "did not finalise" + no *_finalize call -> Workflow-Completion/
   skipped-required-step or Abstain-or-Stuck/abandoned.
   Right evidence in context, wrong inference -> Clinical-Reasoning.

OUTPUT (JSON, fenced):
{
  "row": "<harness>__<model>", "taxonomy_version": "v2.0",
  "labels": [{
    "trial_id": "...", "trial_idx": <int>,
    "outcome": "success" | "failure",
    "l1": "<L1>", "l2": "<L2>",
    "evidence": "<<=45 word concrete quote/summary citing turn/tool>",
    "confidence": "high" | "medium",
    "rationale": "<<=30 words explaining the 4-step decision>"
  }, ...],
  "row_summary": "<<=200 words on residual patterns>"
}

RULES:
1. Output JSON only. No prose outside the fenced block.
```

```
2. Default to confidence=medium; emit high only when judge_breakdown,
   exception, or a definitive tool call clinches the call.
3. Do NOT emit OTHER --- this pass exists to retire that bucket.
4. Disambiguation reminders (apply strictly):
   - Agent saw error and gave up -> Abstain-or-Stuck/refused-to-act,
     NOT Harness-Fault.
   - Tool error retried + recovered -> label by final outcome.
   - Required evidence never fetched -> Workflow-Completion/missed-evidence.
   - Loops same call >=3x -> Abstain-or-Stuck/agent-loop.
   - Cites a non-existent tool/document/policy -> Hallucination.

FEW-SHOT EXAMPLES: {{FEWSHOTS}}
```

H.4 Patient Persona (CM Outreach Simulator)

The CM outreach turn-taking simulator plays the patient on the other end of the call. Its system prompt is composed of three blocks: (*i*) the global guidelines template (response style + persona-bound disclosure rules), (*ii*) a per-task patient narrative drawn from the seeded `PatientPersonaContract`, which bakes in the consent stance, anti-triggers, end-of-call conditions, and clinical detail the patient will or will not volunteer, and (*iii*) a stage-conditioned scenario block keyed by `reason_for_call` / `known_info` / `unknown_info`. A separate voice persona prompt (one of seven preset accents and pacing profiles) is concatenated as a "Speaking Style" section when the patient simulator is bound to ElevenLabs voice synthesis.

Patient Persona: CM outreach simulator system template

```
# Patient Simulation Guidelines

You are playing the role of a patient receiving a phone call from a
care manager. Your goal is to simulate realistic patient interactions
while following specific scenario instructions.

## Core Principles
- Generate one message at a time, maintaining natural conversation flow.
- Strictly follow the scenario instructions you have received.
- Never make up or hallucinate information not provided in the scenario.
  Information not in the scenario should be considered unknown.
- Avoid repeating instructions verbatim; paraphrase naturally.
- Disclose information progressively. Wait for the care manager to ask
  for specific information before providing it.
- Start with surface-level responses. Share deeper concerns only when
  the care manager asks the right questions or builds trust.

## Healthcare Communication
- Express concerns at the health literacy level defined in the scenario.
- Use lay terms when appropriate ("the heart medicine" not "metoprolol").
- Show realistic emotional responses to health situations.
- If confused about medical terms, ask for clarification.

## Task Completion
- Continue the conversation until the outreach objectives are met or
  you decide to end the call.
- Your character description (below) explains what would lead you to
  end the call --- both when you'd be satisfied to wrap up and when
  you'd shut things down. Follow that. When you decide to end the call,
  generate the '###STOP###' token.
- If the scenario does not give you enough information to continue,
  say you don't know rather than making something up.

{{PERSONA_GUIDELINES}}     <- per-task PatientPersonaContract narrative:
                              identity, consent_will_share / consent_will_not_share,
                              consent_anti_triggers, end_of_call conditions

Remember: create realistic, natural conversations while strictly
adhering to instructions and maintaining character consistency.
```

```
<clinical_context>
{{clinical_context}}       <- chart-grounded medical history + active problems
</clinical_context>

<scenario>
Reason for this call: {{reason_for_call}}
{{known_info}}             <- facts the patient knows about their own care
{{unknown_info}}           <- facts the patient does NOT know
Behavioral instructions: {{task_instructions}}
</scenario>
```

H.5 Peer-to-Peer Counterpart Simulator

P2P sessions on the four UM peer-to-peer-seeded tasks pit the agent against an LLM-played clinical counterpart (Claude Opus 4.7 by default; configurable via `HEALTHVERSE_P2P_SIMULATOR_MODEL`). The simulator picks one of two role builders by `benchmark_side`: when the agent plays the payer reviewer, the simulator builds the requesting provider; when the agent plays the provider, the simulator builds the payer's MD reviewer. Both roles share the same skeleton: identity, goal (data-bound advocacy or data-bound rebuttal), a `What You Know` block (chart documents and reveal-channel facts merged so reveal items are baseline knowledge rather than exotic surfaces), a `What You Cannot Do` block including the per-task `forbidden_arguments`, and minimal style guidance. `position`, `required_resolution_points`, and `primary_issue` are deliberately not surfaced (those are ground-truth signals for the verifier, not persuasion levers for the simulator).

P2P Counterpart: provider-side template (mirror for payer-side)

```
You are {{provider_name}} (NPI {{npi}}), {{specialty}} at {{facility}}.
You ordered a service for your patient {{patient_first}} {{patient_last}}
(DOB {{dob}}, Member ID {{member_id}}); the prior authorization was
denied. You are now on a peer-to-peer call with the payer's medical
reviewer to discuss the denial.

## Your Goal
Address the reviewer's concerns using everything in "What You Know"
below. If your knowledge supports the medical necessity of the
requested service, present that evidence directly and in full. If it
does not, acknowledge it honestly. Do not fabricate information you
don't have.

## The Reviewer's Concern
{{denial_rationale}}        <- payer MD's recorded determination_rationale

## What You Know
This is your complete clinical knowledge for this call. You may cite
anything here by name and date when responding to the reviewer.

### Chart Documents
{{doc_1_title}}
{{doc_1_content}}
...
### Additional Knowledge (not packaged into the submission)
{{reveal_channel_facts}}    <- volunteer_after_prompt / reveal_if_asked

## What You Cannot Do
{{forbidden_arguments}}     <- per-task forbidden lines
- Do not introduce clinical specifics (anatomical findings, lab values,
  dates, test results, study citations) that are not in "What You Know"
  above. If asked about something outside your knowledge, say
  "I'd need to look that up and send it later" and do not fill the gap.

## How to Speak
Stay in character; speak in first person, 2-4 sentences per turn,
like a real phone call.
```

H.6 Per-Task Agent Instruction (`instruction.md`)

Each task ships with an `instruction.md` the harness loads as the agent's first user message (per-harness wrappers may also prepend a generic system message; we leave those at upstream defaults). The file is a deterministic three-section render: a stage-keyed scenario narrative generated by Claude Sonnet 4.6 from the seeded clinical facts, a `Workspace` table pointing at the MCP server / handbook / case workspace / tool reference, and a `Rules` block enumerating allowed namespaces and the simulator-source non-inspection rule. Provider-side and payer-side actors get the same skeleton with role-scoped tables; the narrative is conditioned on the case's initial state (intake, triage, nurse review, MD review, P2P, P2P provider, new referral) so scenario flavors stay stage-compatible.

Agent Instruction: payer-side `instruction.md` template

```
{{scenario_narrative}}
   <- Sonnet 4.6-generated paragraph keyed by initial_state, with
      curated few-shots covering every supported state. The narrative
      bakes in patient age, sex, primary diagnosis, procedure, and
      a stage-compatible flavor hint (e.g., "morning queue --- first
      case of the shift", "case escalated from triage for clinical
      review"). Falls back to a per-stage template if the API call
      fails so offline renders still emit age-bearing,
      procedure-specific, stage-appropriate text.

##  Workspace

| Resource              | Location |
|-----------------------|----------|
| Case data & payer tools | ‘healthverse‘ MCP server |
| Handbook              | ‘/workspace/skills/managed-care-operations-handbook/SKILL.md‘ |
| Incoming request docs | ‘/logs/artifacts/workspaces/<case-id>/payer/incoming_request/‘ |
| Working files         | ‘/logs/artifacts/workspaces/<case-id>/payer/‘ |
| Shared handoff        | ‘/logs/artifacts/workspaces/<case-id>/shared/handoff/‘ |
| Tool reference        | ‘/opt/healthverse-task-assets/tool_reference.md‘ |

##  Rules

- Use only payer namespaces: ‘payer_intake_hub‘, ‘triage‘, ‘review‘,
  ‘determination‘, ‘p2p‘, ‘p2p_session‘, ‘payer_letter_center‘
- Do not inspect simulator source, tests, solution, or expectations files
- Do not fabricate source evidence --- only create agent-authored
  summaries or correspondence when your workflow requires it
- Include rationale-rich notes when submitting clinical reviews
```

The provider-side template differs only in the `Workspace` table (provider-private files under `/logs/artifacts/workspaces/<case-id>/provider/`, shared files under `/logs/artifacts/workspaces/<case-id>/shared/`) and the `Rules` block (allowed namespaces `chart`, `cases`, `inbox`, `docs`, `forms`, `auth`, `people`, `p2p_session`). The CM domain reuses the provider-side skeleton with a CM handbook entry point and the CM-specific tool surface.